

When Humanitarian Assistance Fades: Experimental Evidence from a Nationwide Program in Lebanon*

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Abstract

Aid agencies regularly scale down their assistance programs, yet little causal evidence exists on how recipients respond to the loss of support and which strategies might mitigate any adverse consequences. We study the scaling down of a nationwide humanitarian food assistance program in Lebanon, which reduced the caseload from about 60,000 to 25,000 households in 2024. In collaboration with the World Food Programme, we randomized the full beneficiary population across four alternative prioritization methods. Because the methods disagreed for 61 percent of households, the random assignment generated experimental variation in whether marginal households continued receiving assistance or not. A random subset of discontinued households also received a six-month transition period out of assistance. Combining population-wide administrative records with two survey rounds from 6,400 households, we find that: i) discontinued households compensate for the loss of assistance by increasing food expenditure, but still suffer a persistent drop in food security, psychological well-being, and trust towards aid agencies; ii) the longer transition shifted food expenditure before exit and attenuated short-run trust losses, but did not prevent later losses in food security or psychological well-being; iii) alternative targeting methods selected different households and induced different acceptance patterns, yet none dominated across welfare outcomes; iv) combining the different methods improves the identification of the households most in need of assistance. Overall, our findings shed new light on the consequences of scaling-down assistance programs and provide some concrete indications on how to limit disruption.

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1 Introduction

Humanitarian agencies routinely revise, downscale, or terminate their programs in response to funding shortfalls and shifting priorities. Over the past decade, humanitarian assistance has faced persistently large and widening funding gaps.¹ In 2024, the United Nations raised only 46% of the \$49.6 billion required to meet global humanitarian needs (OCHA, 2024). Funding is not only insufficient but also highly volatile, subject to abrupt reallocation triggered by emerging crises or political shifts, as illustrated by recent large-scale cuts from major donor governments.² Scale-down decisions are therefore critical to ensure that scarce resources are allocated to households most in need while preserving the legitimacy of future operations. Yet the evidence base is thin. A large literature studies the launch and design of assistance programs, often experimentally varying targeting rules, transfer modalities, and implementation strategies. Much less is known about the effects of withdrawing support from existing recipients, or about whether alternative exit strategies can mitigate the resulting disruption. Program exits are thus both a practical challenge for humanitarian agencies and an under-studied margin of social protection policy.

The end of assistance is likely to differ from the start of assistance in several ways. Humanitarian programs often begin in response to acute shocks, when transfers can play a critical role in sustaining consumption and well-being. Exit is instead expected to occur once the emergency has passed, households might have rebuilt their livelihoods, and continued support may therefore become inframarginal. Although this need not be true. When programs last for years, households may organize consumption, savings, labor supply, and expectations around recurring transfers. Discontinuation may also be perceived not simply as the absence of a new benefit, but as a broken commitment or unfair exclusion, with implications for trust in aid providers. Moreover, operational scale-downs often reflect budget constraints or competing emergencies rather than improvements in recipients' circumstances. These features make the consequences of exit theoretically ambiguous and empirically important.

In this paper, we provide the first experimental evidence of the consequences of the phasing out of a large-scale humanitarian assistance program. We partner with the largest humanitarian organization in the world, the World Food Programme (WFP), and leverage the cuts to its nationwide food assistance program in Lebanon. The program, which is currently still running, began in 2020 to assist Lebanese households in mitigating the effects of the severe socio-economic crisis, which had eroded their ability to meet basic needs. The program was progressively expanded, and by 2023, it was supporting about 60,000 Lebanese households, corresponding to more than 5% of the Lebanese population. Beneficiary households receive monthly food parcels, valued at \$32 (January 2023 market prices), corresponding to roughly 30% of the median household's monthly food expenditure and 13% of its total expenditure. Due to funding constraints and shifts in priorities, WFP decided to reduce the number of beneficiaries to 25,000 households, or 43% of the previous caseload, by January 2024.

¹Figure A1 in Appendix combines data from Global Humanitarian Overview and UNOCHA financial tracking service to show the evolution of humanitarian needs and unmet funding requirements over the period 2016-2023.

²E.g. "Diminished UK aid budget is 'new normal', says development minister" (The Guardian, 17 May 2025); "Trump administration says it's cutting 90% of USAID foreign aid contracts" (AP, 27 February 2025).

In collaboration with WFP’s Lebanon Country Office, we designed a randomized controlled trial across the entire beneficiary population to shed light on different aspects of the scaling-down process. First, we want to understand the consequences of losing assistance for excluded households. Second, we want to understand the coping strategies households might adopt in anticipation of assistance termination. Third, we want to study whether alternative choices, such as the provision of longer advance notice to discontinued households or different prioritization rules can reduce the adverse consequences.

In order to answer these questions, we build the study around two randomizations. First, the full beneficiary population was randomly divided into four equal-sized arms. Within each arm, we rank households based on their vulnerability, using one of four alternative rules and select the 43 percent that would continue to receive assistance. Three rules were proxy-means tests (PMTs), each predicting a different vulnerability object: per-capita expenditure, food consumption, or a multidimensional vulnerability measure that combines expenditure, food consumption, and coping strategies. The fourth was a categorical score based on expert and practitioner criteria. These methods reflect distinct theories of need: expenditure-based PMT uses a data-driven approach to target low resources; food-consumption PMT uses a data-driven approach to target direct food deprivation; multidimensional PMT uses a data-driven approach to target multidimensional vulnerability and coping capacity; categorical targeting uses a simpler human-driven definition of vulnerability.

The key feature of the design is that the four targeting methods often disagree. Simulating all four methods to the full beneficiary population maps every household into one of 16 “eligibility spaces”, defined by the set of methods under which that household would be selected. For households selected by all methods or by none, assignment to a targeting arm does not change assistance status. For the remaining 61 percent of the caseload, assignment to a targeting arm determines whether the household continues or loses assistance. A household in the E1E2 eligibility space, for example, continues to receive assistance if assigned to method 1 or 2 and stops receiving it if assigned to method 3 or 4. This means that the random allocation into the four targeting arms generates random allocation into the discontinuation of assistance within a given eligibility space. This also means that we can use experimental variation to study the consequences of the end of assistance on the 61% marginal households that are eligible under some targeting methods, but not all.

Second, among households assigned to lose assistance, a subset was randomly selected to receive a longer notice window. The regular process used by WFP informed discontinued households at the end of 2023 that their food assistance would not continue in 2024. The long-notice group was instead informed at the end of 2023 that assistance would continue until June 2024 and then end. We timed our data collection to take advantage of this longer warning window and answer two separate questions. First, we can study whether households change behavior while they still receive assistance but know it will end soon. Second, we can study whether the longer warning plus additional assistance manage to cushion later losses after exit.

Finally, once we have established the consequences of the end of assistance and of the longer warning window, we use the original experimental variation into the four targeting methods to study whether any of them is more successful than the others in identifying the most vulnerable

households, defined as those that would suffer the most from the end of assistance, and in leading to overall better wellbeing among the target population.

The study combines rich administrative data from WFP, covering the entire population of beneficiaries, with two rounds of original survey data collected from a random stratified sample of approximately 6,400 households. The study sample was surveyed twice: in May-June 2024, approximately six months after households whose assistance was discontinued stopped receiving it (while long-notice households were still receiving assistance), and in January-February 2025, approximately seven months later, when only continued assistance households were still receiving assistance. Our primary outcomes of interest cover different dimensions of household wellbeing: food expenditure, food security, psychological wellbeing, and trust.

While a rich battery of balance, implementation, and attrition checks reassures us of the original study design and integrity, the last part of the study had to cope with an unexpected, dramatic event that affected Lebanon. In September 2024, tensions with Israel escalated along the southern border and the conflict that followed led to widespread fatalities, injuries, and displacement. In response to this new crisis, WFP provided temporary emergency cash to registered households in priority areas that were not already receiving WFP support. This unexpected programmatic change exemplifies the challenges inherent in humanitarian settings, where emergencies regularly reshuffle priorities. But it also affects our original design, reducing the contrast between continued and discontinued households in our second-round data. We will therefore interpret second-round discontinuation estimates as the consequences of assigned food-assistance discontinuation in a setting where a subsequent emergency response partially compensated many discontinued households, and we will use non-priority-areas and an instrumental variable approach as supporting evidence on the consequences of losing all WFP support.

Our results can be summarised in three main points.

First, we find that households cut out from assistance (“discontinued households”) compensate through higher food expenditure, but still suffer significant welfare losses. Discontinued households increase their food expenditure by about USD 12.5 in the short run and USD 10 in the longer run, especially on those food categories that were included in the assistance basket. The increase in expenditure is however not enough to compensate for the end of assistance. In the short run, food consumption drops and the adoption of risky coping strategies increases, leading to an .1 SD drop in our food security index. In the medium run, food consumption improves, but the adoption of risky coping strategies increases even further, resulting in a .08 SD drop in the food security index. The drop in food security is also accompanied by a worsening in psychological wellbeing of similar magnitude in the short run (.1 SD), that persists over time (.12 SD by the second round). Discontinued households also experience a large drop in trust (.13 SD in the short run), driven by a dramatic drop in trust towards aid agencies (.25 SD), which shrinks but persists over time (.07 by the second round), despite the emergency support later introduced by WFP that reached more than 70% of our discontinued sample. We find instead little systematic evidence of effects on social cohesion, financial, and labour outcomes. Overall, the effects on food security, psychological wellbeing, and trust are substantial, considering that assistance corresponded to only about 13% of the median household expenditure and that our analysis excludes the most vulnerable households (deemed eligible under all targeting methods).

Second, longer notice of discontinuation changed food consumption behavior before exit but did not eliminate later harm. We find some evidence that households that received longer notice spent less on WFP-basket items and more on several non-basket items than continued assistance households, suggesting some adjustment in food composition. We do not find evidence of other adjustments in their financial or labour behaviour. In the first round, their food security, psychological wellbeing, and trust appears very similar to the ones of continued households. By the second round, however, long-notice households still experienced a drop in food security and psychological wellbeing relative to continued households. The most encouraging signal is on trust, where we do not see any drop for long notice households. This pattern suggests that in-advance communication and a smoother exit process may protect institutional trust, even when they are insufficient to protect material wellbeing.

Third, we find overall little difference from selecting one targeting method over another. The four methods select different households. The categorical rule selects less asset-poor households on average and produces more formal appeals against the exclusion decision than the PMT-based rules, consistent with concerns about coarse targeting criteria and perceived legitimacy. Yet these differences do not translate into any substantial difference in households' wellbeing across the target population, neither in the short, nor in the medium run. This is true both when considering general measures of wellbeing and when considering the specific dimensions targeted by each method (e.g. food consumption or expenditure). Results also hold when restricting focus on marginal households.

Fourth, although there is no evident gain in the choice of one specific targeting method over another, we find evidence of improvements in the ability to identify the most vulnerable households from combining multiple methods and prioritising households that are deemed eligible under more of them. Households eligible under more methods have indeed lower baseline assets and are more likely to appeal when discontinued. We also find suggestive evidence that these households tend to be less able to increase expenditure in response to the end of assistance and experience a larger drop in food security. These results can be explained in two main ways. First, to the extent that different targeting methods focus on different definitions of vulnerability, they might suggest that assistance is relatively more important for households presenting multiple vulnerabilities. The fact that targeting based on a multidimensional score that combined two of the other rules does not perform any better than the others, seems to make this story unlikely. Second, it is possible that by combining the different methods one can lower the noise that is inherent to each model. The fact that the combination of methods leads to the exclusion of asset-rich households that were being selected by some specific methods, suggests this might be the most relevant channel.

Our study relates to multiple strands of literature. First, it complements evidence on cash and in-kind transfers by studying withdrawal from an existing assistance program rather than enrollment into a new one. A rich body of studies have investigated the effectiveness of cash and in-kind assistance across various low-income settings. Many of these studies are based on experimental designs that randomise enrollment into assistance programmes across locations and/or beneficiaries (see [Crosta et al., 2024](#); [Hidrobo et al., 2014](#); [Hoddinott et al., 2018](#); [Hidrobo et al., 2023](#)). There is also some (albeit more limited) evidence from assistance

programs in humanitarian settings, with a recent review identifying a total of nine experimental and twelve non-experimental studies (Jeong and Trako, 2022). Taken together, these studies provide rich evidence on the expected benefits of transfer programs, and on the consequences of different program design choices. We study the opposite margin: what happens when a recurrent transfer is removed from existing recipients and how different scaling-down decisions might affect outcomes.

In doing so, we add to the limited but emerging work on program cuts. To the best of our knowledge, there are only two studies (non-experimental) that assess the impact of program cuts in low-income settings. The first study, by Sterck and Bruni (2025) assesses the consequences of a 20-percent cut in cash assistance to refugees in Kenya, by taking advantage of variation in the timing of the household surveys. The study finds a large drop in food security and expenditure following the reduction in assistance.^{3,4} The second study from Iqbal et al. (2026) uses a regression discontinuity design around a revised eligibility threshold to study the impact of unanticipated program exit from a cash transfer program for women in Pakistan, finding in this case no negative effects. The overall lack of evidence on the consequences of cutting assistance programs is both surprising and concerning, particularly given that organisations are regularly scaling down or ending their programmes in response to changes in priorities and funding. This study contributes to filling this gap by providing the first experimental evidence of the consequences of ending a humanitarian food assistance program on beneficiaries’ well-being.

Our study also contributes to the long-standing literature on targeting assistance and, more broadly, social programs. Over recent decades, PMT has risen as the most popular targeting approach across low-capacity and fragile settings where direct measures of poverty and vulnerability are typically unavailable (Follett and Henderson, 2023; Aiken et al., 2023; Cameron and Shah, 2014). In recent years, the traditional PMT approach has been enriched with innovative techniques leveraging machine learning algorithms which led to significant improvements in the ability to predict the target outcomes (Follett and Henderson, 2023; McBride and Nichols, 2018; Araar and Tiberti, 2020; Gram-Hansen et al., 2019; Caria et al., 2020; Aiken et al., 2022; Hall et al., 2023). However, it is still unclear whether and how different PMT modelling choices translate into measurable differences in overall program impact, as the existing literature is often limited to assessing their precision by simulating their implementation on a population for which the “true values” of the targeted outcomes are known. The studies that have experimentally investigated the consequences of using alternative targeting approaches on the wellbeing of the target population have typically compared PMT against other approaches. The most established evidence comparing PMT to categorical and community-based targeting suggests that PMT exhibits slightly greater precision in identifying vulnerable households, but this does not translate into significant differences in overall programme impact (Alatas et al., 2012; Beaman et al., 2021;

³In a related study from the same setting, the authors use the same variation to investigate the consequences of delays in the disbursement of assistance, finding again negative effects on various dimensions of households’ wellbeing (Bruni and Sterck, 2025).

⁴Additional evidence on program cuts comes from high-income countries, where researchers have used rich administrative data to assess the impact of social welfare cuts. Findings from these settings suggest that reductions or discontinuation of transfers influence labour force participation only in the short term, but have negative long-term effects on children’s well-being and human capital accumulation, and may also lead to an increase in crime (e.g. Dustmann et al., 2024a,b).

Premand and Schnitzer, 2021; Stoeffler et al., 2016; Hillebrecht et al., 2023). Some recent studies also compared PMT to self-enrolment, targeting through local leaders, geographic targeting, and targeting on expected programme impact rather than on initial vulnerability. These studies show that although some of these alternatives improve precision relative to PMT, this often comes at the expense of higher administrative, social, psychosocial, or political costs, without generally translating into measurable gains in programme effectiveness (Grosh et al., eds, 2022; Alatas et al., 2016; Devereux et al., 2017; Altındağ et al., 2021; Schnitzer and Stoeffler, 2024; Haushofer et al., 2022).⁵ This study contributes to filling this gap by experimentally comparing four different targeting approaches within the context of the prioritization of an existing humanitarian assistance programme. Our design thus asks a different targeting question from most of this literature: when a program cut forces some current recipients out, which rules preserve assistance for households with larger losses from exit? We compare three different PMT modelling choices and one categorical approach and focus on their wellbeing implications for the target population. The study is ideally suited to compare variations in PMT modelling choices, given the availability of a high-quality training dataset, which was collected from a comparable population just a few months before our study.

Finally, the long-notice experiment connects to consumption smoothing, liquidity constraints, and trust. In frictionless models, predictable income losses should be smoothed over time (Friedman, 1957). In practice, spending responds strongly to anticipated income and benefit timing when households face liquidity and credit constraints (Gelman et al., 2014; Goldin et al., 2022). In humanitarian settings, delay and aid-cut evidence points to informal credit and market channels as key adjustment margins (Bruni and Sterck, 2025; Sterck and Bruni, 2025). Our evidence shows that notice affects some behavior and trust, but does not by itself prevent food-security and wellbeing losses.⁶

While conducted in a specific context, the study provides some broader policy lessons. First, we show that cuts to food assistance severely impact the wellbeing of the households. We show that this is the case even in a setting where households are not living in extreme poverty and have some room to compensate for the end of assistance by increasing their own food expenditure, and despite the fact that the original emergency peak had subsided and assistance was scaled back in a way that attempted to preserve the most vulnerable households in the program. This provides a strong warning signal in light of the dramatic cuts to assistance programs that we are currently witnessing across the globe. Our results also highlight a broader systemic problem that is common to most assistance programs, and especially humanitarian ones. By design, these

⁵An important exception is the study by Haushofer et al. (2022), which finds that targeting based on predicted programme impact can lead to better outcomes for households with higher expected returns.

⁶In doing so, this analysis will also speak to the long-standing literature on the permanent income hypothesis (PIH), which proposes that individuals base consumption on expected long-term income rather than current income, implying that temporary changes in available resources should have limited effects on consumption if credit or savings are available (Friedman, 1957). The existing evidence, primarily from high-income settings, suggests that households do not fully smooth expenditures and behave more in line with hyperbolic discounting models, placing disproportionate weight on immediate rewards against future ones (Gelman et al., 2014; Goldin et al., 2022). Moreover, economic theory suggests that the effects of income shocks may vary depending on the level of certainty or uncertainty households face (Khan and Knotek, 2011). Evidence on these dynamics, especially in the context of humanitarian assistance, remains limited. With funding gaps and aid cuts expected to persist, understanding the impact of cuts in assistance as a form of economic shocks, and identifying ways to mitigate their adverse consequences remains a critical policy concern.

programs are meant to fade away over time, once the emergency is over and beneficiaries can sustain their own wellbeing without external support. It is however very rare for these programs to have clear plans for their conclusion, and their survival thus ends up being driven by funding availability and policy priorities. More thoughtful exit strategies might enable households to better sustain their wellbeing, and, importantly, increase the long-term benefits of the programs. Our study also provides a first step in this direction, testing the consequences of providing a longer notice window. Results show that the longer notice might protect institutional trust, but is not enough to smooth the adverse consequences of the end of assistance on food security and psychological wellbeing. Policy makers and organizations should thus test alternative strategies to facilitate their exit in a way that minimizes disruption. These could range from behavioural interventions that help households plan in advance how they will cope with the end of assistance, to heavier-handed interventions that provide psychological assistance and/or additional support to find ways to compensate for the drop in available resources. Finally, our results suggest that rather than picking one targeting method from the menu of available ones, organisations can enhance their targeting performance by running multiple methods and prioritising based on their intersections. Informed by this result, WFP recently adopted this approach in Lebanon for targeting its new programs.

2 Setting and Program

2.1 Lebanon’s crisis and the role of food assistance

In recent years, Lebanon suffered the most devastating, multi-pronged crisis in its modern history ([World Bank, 2022](#)). The economic and financial crisis started in August 2019, when Lebanese banks withheld deposits in US dollar accounts and was compounded in the following months by two other major shocks: the Beirut Port explosion and the COVID-19 pandemic. The port explosion in August 2020 killed nearly 200 people, injured over 6,500, and displaced 300,000 individuals, while destroying critical infrastructure, including the country’s primary grain silos. At the same time, the COVID-19 pandemic and the associated containment measures further deepened the economic downturn, resulting in widespread business closures, job losses, and a severe strain on Lebanon’s already fragile healthcare system. Together, these shocks led to a dramatic collapse in purchasing power, which, in turn, led to a sharp rise in food insecurity across the country, heightening social tensions and exacerbating existing inequalities. Lebanon’s gross domestic product (GDP) per capita declined by more than 50% between 2019 and 2021, resulting in the country’s downgrading in July 2022 to lower-middle-income status after decades as an upper-middle-income country ([World Bank, 2022](#)).

In 2020, as a response to these shocks, WFP, in collaboration with the Lebanese government, started providing emergency assistance in the form of monthly in-kind food parcels to Lebanese households in need. Initial beneficiary identification combined information from government agencies, partner organizations, and a self-enrollment portal, followed by WFP validation.⁷

⁷More specifically, WFP initially relied on lists of vulnerable households provided by government agencies, community partners, as well as a self-registration portal. WFP then validated the inclusion of each household by conducting a short survey. Validation was initially based on data collected through phone surveys, while for later inclusions WFP relied on in-person surveys.

Beneficiaries of the program, which is currently still ongoing, are guaranteed the provision of monthly food assistance for at least the duration of an “assistance cycle” (12 months), which is then renewed based on funding availability. Assistance is provided in the form of one parcel of food for every two household members, and the parcels are designed to provide the equivalent of at least 1,300 kcal/person/day. Each parcel includes 20 kg of commodities⁸ and is worth \$32 (at January 2023 market prices), which corresponds to approximately 30% of the median Lebanese household’s monthly food expenditure and 13% of their total expenditure.⁹ Households are informed each month of the arrival of their parcel through a message on their phone and are then authorised to collect it from one of the many collection points scattered across the country. WFP administrative data from 2023 show that 89% of beneficiaries collect their parcels every month, and virtually everyone (99%) collects them at least half of the time within the cycle.

The program was constantly renewed and expanded to reach almost 60,000 households by 2023; Figure A12 in the Appendix shows the geographic distribution of beneficiaries across the country. Due to limited funding and shifts in priorities, WFP had to drastically reduce the beneficiary caseload to about 25,000 households in 2024, i.e. only 43% of the existing caseload.¹⁰

2.2 Communication, Appeals, and Emergency Cash

Following its standard practice, in December 2023 WFP sent a message to all beneficiaries updating them on their status. Communication was conducted entirely via SMS to ensure timely, direct, and consistent messaging across all beneficiaries. The messages followed standard WFP communications and contained only short, generic information on the inclusion or exclusion from the assistance program for the 2024 cycle. Households selected for continued assistance were informed that assistance would continue for them (at least) throughout 2024. The remaining households were instead informed that their assistance would end as a consequence of a prioritisation exercise motivated by limited funding. Within this latter group, most households were informed that assistance would end in December 2023, while a random subset of households got informed that assistance would continue until June 2024 (more details on the selection of these households below). For households excluded from the program, the message also included information for submitting an appeal either through WFP dedicated phone line or through an online form, within a month from the notification. Whenever an appeal was submitted, WFP reviewed the case to determine whether the household should be reassessed and readmitted. Households whose appeals were successful were informed of any resulting changes to their assistance status and were readmitted into the programme. In practice, this happened very rarely.¹¹

⁸Each parcel contains 10 items: Rice (3Kg), Bulgur (3Kg), Pasta (4Kg), Lentils (4Kg), Beans (1Kg), Chickpeas (1Kg), Tomato Paste (400g), Oil (1.8l), Salt (500g), and Sugar (1Kg).

⁹Author’s estimation based on the nationally representative Lebanon Vulnerability Assessment Panel (LVAP) dataset, which was collected between January and May 2023.

¹⁰The 43% target was based on WFP’s estimation of the number of beneficiaries with consumption levels below the minimum expenditure basket (ME), which was set at LBP 3,238 or USD 71 per capita per month (at January 2023 prices). Such estimation was based on extrapolation from the nationally representative LVAP dataset, collected between January and May 2023 and on a post-distribution monitoring survey that WFP conducted in the first half of 2023 among a representative subset of beneficiaries.

¹¹52% of households excluded from assistance submitted an appeal. By the time of our first round, in June 2024, less than 1% of the appeals had resulted in a change in status. An additional 2% of appeals were accepted,

During the study period, the country experienced yet another dramatic shock with the escalation of tensions with Israel along the southern border. Between September and November of 2024, the conflict caused 3,961 fatalities and over 16,500 injuries. The conflict also led more than a quarter of the population to evacuate: 1 million individuals were internally displaced, while 640,000 crossed the border towards Syria. A ceasefire agreement was signed on November 29th, and an estimated 85% of the internally displaced had returned to their homes by December (UNHCR, 2024). In response to this humanitarian crisis, in October 2024 WFP started providing temporary emergency cash transfers to all households that were: i) registered in the WFP system; ii) located in “priority areas”, as defined based on conflict and displacement intensity (Figure A3 in Appendix)¹²; iii) not already supported through other WFP programs. This means that beneficiaries of the food assistance program were excluded from the emergency cash assistance.¹³ This new assistance continued for at least three months, with some locations still receiving it as late as May 2025.¹⁴ Overall, 58% of the country (1,590 cadasters out of 2,735) fell into the “priority zone” definition, and as a consequence, 74% of the households assigned to stop receiving WFP food assistance in our study had received three months of emergency cash support by the time of our second-round survey in January-February 2025. In the following sections, we will discuss in more detail how the emergency cash program might be expected to affect our second-round analysis and how we address this.

3 Study Design

We structure the study design description into three parts. First, we focus on the random assignment of beneficiaries to the four targeting methods. Second, we discuss how to move from experimental variation in targeting methods to experimental variation in discontinuation of assistance. Third, we discuss the randomization of the subset of discontinued households into the long-notice arm.

Random allocation into different targeting methods

The first and key step in our design is the random allocation of beneficiaries into four equal-sized groups, each assigned to a different targeting method. The four methods can be interpreted as empirical implementations of distinct vulnerability concepts, and are selected among approaches effectively used in the field.

Three of these approaches are based on the Proxy Mean Tests (PMTs) methodology, which is currently the most popular approach for approximating latent household welfare in contexts where direct measurement is not feasible or likely inaccurate. PMTs can be viewed as prediction functions that map observable household characteristics into proxies of underlying vulnerability. The PMT models must first be developed using a training dataset – ideally from a population and households readmitted into the program, after July 2024.

¹²A given area (cadastre) was deemed priority area if it fulfilled any of the following three criteria: (i) it has a value of 3 or higher on the LBCO’s Conflict Impact Index; (ii) it hosts at least 500 internally displaced persons; or (iii) more than 20 percent of the local population is IDPs.

¹³The monthly cash transfer was equal to \$25 per HH plus \$20 per HH member and was thus somewhat more generous than the food support.

¹⁴Assistance was designed to be temporary, with a progressive phase-out expected to begin in January 2025.

as similar as possible to the target population – that contains both the target vulnerability measure of interest and a rich set of socio-economic variables. Modern applications of this methodology rely on machine learning algorithms to select the most relevant variables associated with the outcome of interest and estimate their weights. The model is then applied to the target population to generate a predicted score for each potential beneficiary. Inclusion in the assistance program is then determined by ranking these scores and selecting households until the desired threshold, which can be defined either by the total number of beneficiaries (like in our case) or by a specific predicted score.

In our setting, the three PMT models were trained on the Lebanon Vulnerability Assessment Panel (LVAP) dataset, which was collected between January and May 2023, across a nationally representative sample of 3,597 households. The LVAP dataset records multiple welfare measures and shares a total of 126 socio-economic variables with the WFP census, which was collected just a few months later from the entire WFP beneficiary population. The three PMT models differ in the vulnerability concept they seek to approximate. The first (*PMT-Exp*) targets per capita expenditure, corresponding to a standard measure of poverty and aligning closely with canonical welfare models. The second (*PMT-FCS*) targets food consumption, as measured through the Food Consumption Score (FCS), which captures short-run nutritional adequacy and can be interpreted as a more direct measure of food needs. The third (*PMT-CARI*) targets a multidimensional vulnerability index based on the CARI framework, combining expenditure, food consumption, and coping capacity to capture broader constraints on household welfare. The comparison across these PMT specifications provides a test of how alternative definitions of vulnerability - poverty-based, needs-based, or multidimensional - affect targeting and subsequent outcomes. While these dimensions are correlated, their overlap is far from perfect. When ranking households according to these three outcomes, within the LVAP sample, their average pairwise correlation in the ranking is only 0.39. A few features make our setting ideal for testing the consequences of selecting these alternative methods: we have a training dataset that was collected from a similar population as our target population, almost at the same time as the target population census (more details below); moreover, the training dataset includes multiple vulnerability measures, plus a large set of covariates (considering interactions and square terms, the PMT models are built from a starting pool for about 300 variables).

The fourth targeting method (*Categorical*) is instead a score-based categorical approach based on expert opinions. Rather than estimating a statistical mapping from observables to welfare, categorical targeting relies on a set of observable characteristics - such as household composition, disability status, or gender of the household head - to define eligibility. This approach can be viewed as approximating targeting under informational constraints, where decision-makers rely on salient and easily verifiable proxies rather than a fully specified prediction model. This method is thus cheaper, easier to implement and more transparent, but might also be coarser and less precise in identifying the most vulnerable households. The central design choice in categorical targeting lies in selecting the variables that should define eligibility. In our case, the categorical targeting arm was implemented using a score-based approach informed by expert judgement. A survey was administered by WFP in November 2023 to a pool of 28 experts with operational or research experience in food security and humanitarian assistance

in Lebanon. Their answers were then pooled to define the dimensions to be considered for selecting beneficiaries, with simple weights based on the frequency with which each dimension was mentioned by the experts.¹⁵

More details on the definition of the four methods and on their overall performances are reported in Appendix A.

In practice, in November 2023, the 58,501 assistance recipients were individually randomly assigned to the four targeting groups of equal size.¹⁶ Within each group, the 43% most vulnerable households, as determined by the specific targeting approach assigned to that group, were selected to keep receiving assistance (*continued* assistance households), while the remaining 57% were excluded from the program (*discontinued* assistance households). Table A1 in Appendix, based on the entire census, shows that the initial randomisation created four observationally equivalent groups.¹⁷ Importantly, the balance that we observe in the composition of the four groups does not translate into balance in the profiles of households selected for continued assistance (Table A2 in Appendix), reflecting the fact that different targeting approaches select different profiles. When considering the full set of baseline characteristics, statistical tests reject the null hypothesis of equality of means across the selected in the four groups for 85.6 percent of the characteristics at the 5-percent significance level. Even when excluding the categorical approach and restricting the analysis to only the three PMT approaches, 75 percent of the variables show statistically significant differences at the 5-percent level. These differences can be further confirmed by simulating the application of each method to the entire population, and comparing the selection outcomes (i.e. who gets included in the 43% most vulnerable). Figure A4 in Appendix shows that, although there is a positive correlation in the selection across the four methods, it is quite low (on average, just .32). As one might expect, the three PMT-based methods are relatively more similar, but even in their case, correlation in their final selection is only 0.55, on average.

Experimental variation in the discontinuation of assistance

The differences in the selection across targeting methods allow transforming the experimental variation in targeting methods into experimental variation in the discontinuation of assistance. By simulating the application of the four targeting methods across the full population, we can indeed map each of the 58,501 households into one of the 16 “eligibility spaces” depicted in Figure A1. Each eligibility space groups together households that are selected for continued assistance under a specific set of methods, ranging from E0, which indicates households that are never selected under any of the four methods, to E1E2E3E4, which indicates households that are selected by all four methods. All other spaces include households that are selected by at least some methods, but not all; e.g., E1E2 indicates households that are selected under methods 1 and 2, but not 3 and 4. Table A3 in Appendix shows how the full population is

¹⁵The average pairwise correlation in households vulnerability ranking between categorical score and the other three outcomes targeted by PMT models within LVAP data is 0.24.

¹⁶In practice, the four targeting methods were first simulated across the full population and the randomization was then stratified by eligibility under different combinations of the four targeting methods, to ensure balance along this dimension. In other words, the randomization was stratified by the 16 “eligibility spaces” that will be better discussed in the following section.

¹⁷This table, like all other balance check tables, only report a subset of 20 variables available from the baseline census. When considering the entire set of variables included in the baseline survey, there is a total of 139 possible comparisons. The notes below each balance table report the statics when considering all these comparisons.

distributed across these spaces. The table shows that the population is spread across the full eligibility spectrum, with at least 458 households in each eligibility space. The table also shows that for 39% of the sample, the four methods provide consistent selection: 26% of the households are excluded from assistance by all four methods (Eligibility Space E0), while another 13% are always selected for continued assistance (Eligibility Space E1E2E3E4). This means that for the remaining 61% of the households, the four methods disagree in their inclusion decision. This implies that the random allocation of households into the four targeting arms, described in the first step, translates into the random allocation of the discontinuation of assistance *within a specific eligibility space*. Table A4 in Appendix confirms that, conditional on the inclusion of eligibility space fixed effects, the characteristics of households assigned to receive assistance and households assigned to stop receiving it are well balanced. This means that, once we hold the eligibility space fixed, we can learn about the causal effect of ending assistance (for the “marginal” households) by comparing outcomes between continued and discontinued households.

Random allocation into the long notice window

In order to study whether and how households prepare for the end of assistance, we randomly selected a subset of discontinued households (stratified by eligibility space) for which WFP agreed to extend the provision of assistance for six additional months. In practice, this long-notice sample was identified by randomly selecting 30 households from within each of the 32 combinations of eligibility space and targeting method that led to the discontinuation of assistance (indicated in red in Table A3 in Appendix), for a total of 960 households. When discontinued households received the message informing them of the end of assistance, these 960 households were informed that their assistance would continue for an additional 6 months. They also received a reminder in April 2024, two months before the actual end of assistance. This step thus defines three sets of households in our dataset: *continued assistance* households, which received food assistance throughout the study period; *discontinued assistance* households, which stopped receiving food assistance in January 2024; and discontinued *long-notice* households, which received assistance until June 2024. Importantly, our first data collection took place while the long-notice households were still receiving assistance. By comparing continued assistance and long-notice households in the first round, we can thus learn whether and how households were preparing for the upcoming end of assistance. At that time, indeed, both groups were still receiving assistance, and the only difference was that long-notice households had been informed their assistance would end in June. By the second round, instead, long-notice households had stopped receiving their food assistance. The comparison with the discontinued assistance group in the second round can thus reveal the combined effect of the longer notice window and the six extra months of assistance. Table A5 in Appendix confirms that, conditional on the inclusion of eligibility fixed effects, there is perfect balance in the characteristics of households assigned to the long-notice window and households in the other two groups.

Ethical considerations

Although part of the analysis of this study builds on experimental variation in the discontinuation of food assistance, it is important to stress that inclusion in the discontinued assistance group

was always based on the assessment of a household’s vulnerability, as defined by the different targeting methods. Households ranked in the 43% most-vulnerable group by each targeting method kept receiving assistance, while the remaining 57%, deemed relatively less vulnerable, stopped receiving it. The four targeting methods were developed in collaboration with WFP, to reflect distinct vulnerability concepts, based on actual methodologies that WFP already applies for regular targeting activities across its programs. The random variation in the discontinuation of assistance only emerges for “marginal households” as a consequence of the differences in targeting decisions across the four methods. Following WFP regular practices, households excluded from assistance always had the possibility to appeal the decision through the dedicated hotline or an online form. Appeals were handled directly by WFP in accordance with its standard procedures, without any involvement from the research team. Finally, the definition of a random long-notice sample, implied that a subset of 960 households whose assistance should have been stopped in January 2024 continued to receive it for six additional months. The study design thus never accelerated assistance discontinuation. The study was implemented in partnership with WFP and received ethical clearance from the ethics committees of Humboldt University of Berlin (#2023-08-W) and from HML International Review Board (#2766).

4 Sample, Timeline, and Data

4.1 Sampling

The study sample is a stratified random sample from the overall beneficiary population. We randomly selected 100 households from each of the 64 cells defined by the combination of the 16 eligibility spaces and the four targeting methods (Table A3 in Appendix). By construction, of the total sample of 6,400 households, half were selected for continued assistance, and the other half was instead assigned to stop receiving assistance. This latter group includes 2,240 “short-notice” discontinued assistance households, which stopped receiving assistance in December 2023, and 960 (randomly selected) long-notice households (30 households from each of the 32 discontinuation cells), which stopped receiving assistance in June 2024. By giving equal representation to each cell, this sample maximises power for the comparison between continued and discontinued assistance, though it does not provide a representative distribution of the population across eligibility spaces. By applying the corresponding population weights, it is however straightforward to retrieve population-wide estimates (see below for details).

4.2 Timeline

The detailed timeline of the study is summarised in Figure A11 in the Appendix. In preparation for the prioritisation exercise, between August and October 2023, WFP collected a census among all 58,501 beneficiaries. In December 2023, following the completion of the targeting exercises, all households received a message from WFP to update them on their status. The message either confirmed that assistance would continue (at least) throughout 2024 or that assistance would end. Within the latter group, most households were informed that assistance would end on that same month, while the 960 long-notice households got informed that their assistance would end in June 2024. From January 2024, discontinued households stopped receiving WFP

assistance. The long-notice households were then reminded of the approaching end of assistance through a second SMS in April 2024.

The first round of data collection from the study sample (first-round) took place in May-June 2024, just before the end of assistance for the long-notice households. The second round of data collection (second-round) took place in January-February 2025. All continued assistance households kept receiving food assistance beyond the study period, at least until June 2025, when WFP implemented a new program adjustment.

Between the two rounds of data collection, the conflict between Lebanon and Israel along the southern Lebanese border escalated, leading to a ground invasion in September 2024, intense bombing across the country, and population displacement. In October 2024, WFP started providing emergency cash transfers to households located in affected areas (as defined by conflict and population displacement intensity) that were not already supported through their food assistance program. The second round of data collection thus took place while this WFP cash emergency support was ongoing.

4.3 Data Sources

We make use of four main data sources in our study.

LVAP Database – This database (recently renamed LHS) is the result of a collaborative initiative between the World Bank, UNHCR, and WFP, with support from the Joint Data Center on Forced Displacement, that aims to monitor and analyse the socioeconomic conditions of the Lebanese population. It is structured as a nationally representative household survey of approximately 3,500 households. The dataset is very rich and records detailed information on poverty levels, food security, employment status, access to essential services, and coping mechanisms among households. An LVAP round was collected between January and May 2023, just a few months before the start of our study. We rely on this dataset to train the three PMT targeting models.

Household Census – The baseline census was collected by WFP between August and September 2023, to inform its forthcoming prioritisation exercise. The census covered the full set of 58,501 beneficiaries and contained close to 150 variables, capturing household characteristics, demographics, and assets. It did not contain any direct vulnerability measures. This decision was driven by both the need to preserve the survey’s length and the high risk of strategic misreporting on direct vulnerability-related questions. The census serves two key purposes for our study. First, it provides updated information on the entire population of beneficiaries, which is essential to implement the prioritisation. Second, it provides the baseline information for the study sample.

Administrative Data – WFP stores detailed administrative records of all its operations. Administrative data was collected independently from this study and followed standard WFP procedures. We make use of two sets of administrative data in our analysis: (1) monthly records of households that collected their food parcels, to assess adherence to the study protocol; and (2) records of appeals submitted through phone calls or online form, to assess beneficiaries’ acceptance of the exclusion decision. In addition, to support the second-round analysis, WFP shared with us information on priority areas affected by the conflict with Israel and records of households that received emergency cash transfers.

Survey Data – Our main analysis draws on the two rounds of survey data collected from the study sample of 6,400 households. The questionnaires were developed by the research team to cover a broad range of household characteristics and wellbeing measures, detailed in the following section. Data collection was implemented by the Central Arab Statistical Network (CASN), an independent survey firm with extensive experience in large-scale data collection across Lebanon. Overall, over 60 enumerators were deployed across the country for both rounds of data collection. Survey teams were familiar with local customs and were masked to the assignment status of the households they were interviewing. Surveys were always conducted in-person, using SurveyCTO, a secure data collection tool built on Open Data Kit (ODK).

4.4 Outcome Measures

In this section we provide a short description of the key variables used in our analysis, divided by family of outcomes. A detailed table reporting the components, the exact survey items, the response scales, and the construction of each outcome is provided in Appendix B (Table A3).

Expenditure

Monthly household food expenditure, is the first outcome we consider to understand the consequences of the drop in food assistance on households’ decisions. Households were asked about expenditure across 9 different food groups using a standard 30-day recall period. Importantly, the survey collected information not only on actual disbursements but also on any food purchased on credit, as well as on the estimated value (at current prices) of own food production.

Monthly household non-food expenditure. We also recorded a (non-exhaustive) list of non-food expenses, by considering the most common sources of expenditure among the target population. We did so considering two different recall periods: 30 days for 10 regular expenses such as electricity and transport, and six months for 7 less frequent expenditures such as health care or house repairs. Expenses in the latter category were divided by six to obtain a monthly estimate. To limit the influence of outliers and potential data-entry errors, all expenditure items are winsorized at the 95th percentile before aggregation.

Food Security

We consider two proxies of food security: *Food Consumption Score (FCS)* and *(reduced) Coping Strategy Index*.

The *Food Consumption Score (FCS)* is the food security proxy most commonly used by WFP and its partners. It is based on nine survey questions that capture the frequency with which nine different food categories have been consumed in the household over the previous week. The index ranges from 0 to 112 with a higher value indicating higher consumption ([World Food Programme, 2024](#)).

The *(reduced) Coping Strategy Index (rCSI)* consists of a set of five questions measuring the frequency and severity of harmful behaviours that households might adopt to cope with food shortages, such as reducing the number of meals or decreasing portion sizes. The index ranges

from 0 to 56, where a higher index indicates greater reliance on harmful coping behaviours (Maxwell et al., 2008; Maxwell and Caldwell, 2008).

Psychological Wellbeing

We consider three common measures of psychological well-being: life satisfaction, worriedness, and WHO-5 mental health.

The *life satisfaction index* is an index that combines five questions on satisfaction with household nutrition, health, finances, community relations, and overall life (OECD, 2013). Responses range from 1 (Very satisfied) to 4 (Very unsatisfied). The answers are summed together and the index is then standardized within each survey round, to have mean zero and standard deviation one.

The *worriedness index* is constructed in a similar way, from a set of ten questions designed to capture concerns about food, health care, education, debt, employment, family, extremism, conflict, natural disasters, and displacement (Hidrobo et al., 2023). Responses range from 1 (Not worried at all) to 4 (Very worried). The answers are also in this case summed together and the index is then standardized within each survey round, to have mean zero and standard deviation one.

Finally, the *WHO-5 mental health index* consists of five items measuring mental well-being over the previous two weeks. These cover feeling cheerful, calm, well-rested, active, and interested in daily life. Each item is rated on a 6-point scale from 0 (“At no time”) to 5 (“All of the time”), giving a raw total between 0 and 25, which we rescale by a factor of four to the conventional 0–100 WHO-5 scale (World Health Organization, 2024). The index is then standardized within each survey round, to have mean zero and standard deviation one.

Trust Index

We collect information on generalised trust, following the standard World Values Survey formulation (World Values Survey Association, 2022), as well as seven Likert-scale items on trust in specific institutions and groups such as family, neighbours, religious organisations, NGOs, local and national government, and migrants. The index is then standardized within each survey round, to have mean zero and standard deviation one.

Additional Measures

The survey collected additional information on a range of secondary outcomes.

The end of assistance might force households to modify their financial and employment situation, to the extent that households have some flexibility in those dimensions. To capture the financial situation, the survey initially recorded information on *savings*, *debt*, and *remittances*. The questions were designed to capture both formal and informal transactions, including resources held abroad, in Lebanese pounds or in US dollars. Amounts from different sources were combined and converted into US dollars, winsorized at the 95th percentile, and transformed using the inverse hyperbolic sine. Given that at baseline only 0.3% of households reported any positive

saving, the question was dropped in the second round and we therefore only report results on debt and remittances.

Employment information comes from the household roster. The survey recorded for each household member whether they engaged in any paid work in the previous seven days. For each job reported, respondents provided information on the type of job, start date, and number of hours worked. This enables us to create two variables: the *share of household members working* and the *total number of hours worked* in the household.

Finally, we recorded information on social cohesion, by recording membership in community groups, volunteering, and perceived tension in the community.

5 Empirical Strategy

5.1 Discontinuation and Long Notice

The main discontinuation analysis compares households assigned to stop receiving food assistance with households assigned to continue receiving it, within the same eligibility space. Given the conflict with Israel potentially affecting second-round results, we will present the estimates separately for the first and second rounds, by estimating the following basic equation:

$$Y_{h,e,t} = \beta \text{EndOfAssistance}_{h,e} + \gamma \text{LongNotice}_{h,e} + \Theta C_{h,e,t} + \eta_e + \mu_{h,e,t} \quad (1)$$

where $Y_{h,e}$ is the outcome of interest (e.g., food security) for household h , belonging to eligibility space e , at time t with $t \in \{\text{First}, \text{Second}\}$. $\text{EndOfAssistance}_{h,e}$ is an indicator variable for households assigned to stop receiving WFP food assistance, while $\text{LongNotice}_{h,e}$ is an indicator variable for discontinued households randomly assigned to the long-notice arm. $C_{h,e}$ indicates a vector of baseline control variables selected through a post-double-selection lasso procedure from the household characteristics recorded in the beneficiary census. The eligibility-space fixed effects and the long-notice indicator are always partialled out rather than subjected to selection. In the second round, the specification additionally always includes controls for the randomised order of the survey modules and for enumeration issues flagged during the interview. Importantly, the inclusion of eligibility space fixed effects η_e ensures that the estimation relies on variation in assistance *within* eligibility spaces. This also implies that, as discussed in the design section, the estimated effects holds for households that belong to eligibility spaces where there is variation in assistance status, i.e. all spaces except for E0 and E1E2E3E4. The coefficient β captures the average impact of terminating food assistance on the outcome Y by comparing discontinued (short notice) households to continued assistance households.

The same specification will be used to study the impact of providing long notice. Given that long-notice households are always a subset of the discontinued households, the coefficient γ captures the *differential* impact of the provision of long-notice on the outcome Y , compared to standard short-notice discontinued households. To learn about the comparison between long-notice and continued assistance households, we can sum β and γ . The long-notice interpretation differs by wave. In the first round, long-notice households still receive assistance, so $\beta + \gamma$ tell us whether knowing assistance will end, while still receiving it, changes outcomes relative to

continued households. In the second round, long-notice households have exited, so $\beta + \gamma$ tell us the (short-term) consequences of the end of assistance for households that had six months to prepare for it. The comparison with discontinued (“short-notice”) households (γ) will instead reveal the combined (potential) benefit of providing additional six months of food assistance and a longer notice window. We will always report robust standard errors to account for heteroskedasticity.

As discussed above, one concern in the discontinuation analysis is that second-round results might be affected by the conflict with Israel and the emergency cash provided by WFP to discontinued households. The assigned-discontinuation estimates remain the clearest results because they preserve the original randomization, and reveal the consequences of the original food-assistance scale-down in the realized environment. But to isolate the consequences of actually ending WFP assistance in the second round, we provide two additional analyses. First, we restrict the analysis to non-priority areas. Second, we instrument the actual discontinuation of *any* WFP assistance with the original experimental variation in the end of *food* assistance. In practice, we will estimate the following two-stage model

$$Y_{h,e} = \beta^{IV} \widehat{\text{ActualEndOfAssistance}}_{h,e} + \gamma^{IV} \widehat{\text{ActualLongNotice}}_{h,e} + \Theta^{IV} C_{h,e} + \eta_e + \epsilon_{h,e} \quad (2)$$

Where `ActualEndOfAssistance` is an indicator variable for discontinued households that did not receive any WFP assistance, while `ActualLongNotice` is an indicator variable for the subset of those households that had been randomly assigned to long notice. The two variables are instrumented using the random assignment to discontinuation and long notice, respectively. What we test in this case is the impact of being fully excluded from any WFP support over the study period.

5.2 Different Targeting Strategies

To compare targeting methods, we estimate the following basic equation:

$$Y_{h,e} = \alpha_1 \text{PMT_FCS}_{h,e} + \alpha_2 \text{PMT_CARI}_{h,e} + \alpha_3 \text{Categorical}_{h,e} + \Omega X_{h,e} + \theta_e + \varepsilon_{h,e} \quad (3)$$

where `PMT_FCSh,e` and `PMT_CARIh,e` are indicator variables for households assigned to the PMT arms that target FCS or CARI, respectively, while `Categoricalh,e` is an indicator variable for households assigned to the categorical targeting arm. The PMT model that targets per-capita expenditure is the excluded reference category. $X_{h,e}$ indicates a vector of baseline control variables selected through a post-double-selection lasso procedure from the household characteristics recorded in the beneficiary census, alongside a control for households randomly assigned to the long-notice window that is always partialled out rather than subjected to selection. As in equation (1), in the second round the specification additionally always includes controls for the randomised order of the survey modules and for enumeration issues flagged during the interview.. θ_e indicate eligibility space fixed effects, which in this case is not relevant for identification, but is included as the random sampling was stratified by the 16 eligibility spaces.

The coefficients α_i capture the average difference in Y between households assigned to a given targeting method (PMT targeting FCS, PMT targeting CARI, or categorical) and those assigned to the reference targeting method (PMT targeting expenditure). By comparing α_i coefficients, it is also possible to assess the relative performance across all targeting methods.

When considering the number of appeals, equation (3) can be run on the entire population of 58,501 beneficiaries. For all other outcomes, instead, information is only available for the study sample of 6,400 households.

5.3 Weights and Multiple Hypotheses

For simplicity and transparency, our main specification will report unweighted estimates. There are however two different weightings that one might want to apply to translate our estimated effects into average treatment effects across the population. First, population weights can be applied to account for the fact that the study sample is a stratified randomized sample.¹⁸ Second, weights can be applied in the discontinuation analysis to account for the fact that households that belong to different eligibility spaces have different likelihood of being discontinued. All our results and conclusions remain unaffected by these weighting, that we are reported in in Appendix.

To address concerns about multiple hypothesis testing, we follow [Anderson \(2008\)](#) and construct the inverse-covariance-weighted indices for each family of related outcomes. These indices aggregate closely related measures into a single primary outcome, reducing the number of tests while avoiding over-weighting highly correlated components; component-level estimates are reported as supporting evidence.

6 Descriptive Statistics and Internal Validity

Table A6 reports the baseline characteristics of households in our sample, by study arms. The table indicates that households have fewer than four members and less than one school-age child, on average. About one-third of households are female-headed. Socioeconomic conditions point to substantial vulnerability: 60% of household heads have at most primary education, 74% of households have a member with a chronic illness, and about one third of them have no household member regularly employed. Asset ownership and access to basic services is limited, with only 30% of households owning a car and 24% having piped water, although smartphone ownerships and access to gas are more widespread (81% and 74%, respectively). Overall, the table provides the picture of a vulnerable population within a low-middle-income country. Table A6 in Appendix compares these characteristics to the nationally representative LVAP dataset. The table shows that, as expected, program beneficiaries are more vulnerable than the average Lebanese households (e.g. they have larger households, are more likely to be headed by a woman,

¹⁸For population-weighted estimates, we construct wave-specific weights starting from the inverse respondent share within each targeting-arm-by-eligibility sampling cell: $w_{h,t} = N_h/n_{h,t}$ where N_h is the full beneficiary-population count in cell h and $n_{h,t}$ is the number of surveyed households in that cell in the first round or second-round. We then calibrate these design weights by raking so that the weighted respondent sample matches known full-population margins from the administrative frame, including the 64 sampling cells, governorate, household-size bins, wealth quintiles, appeal status by targeting arm, and selected baseline characteristics ([Deville and Särndal, 1992](#)). None of our main results is affected by the use of population weights.

more likely to suffer from health issues, and have lower wealth). The bottom part of Table A6 summarises the scores associated with the four targeting methods. The three PMT-related scores are predictions from the respective PMT models, while the categorical score is the result of applying the weights derived from the expert-based surveys.

Importantly, Table A6 shows that household characteristics are very similar across the four targeting groups. Table A7 in the Appendix provides a similar comparison across long-notice, discontinued, and continued assistance households (while controlling for eligibility space). Taken together, these balance checks support our study design and confirm that the randomizations were successful in creating observationally equivalent groups across the different comparisons we are interested in.¹⁹

In terms of attrition, when considering the sample of continued assistance and discontinued (short-notice) households, we successfully surveyed around 99% of the target sample in the first round and 94% in the second round, with no differential attrition across targeting arms or discontinuation status (Table A7, Panel A). Whenever a target household could not be surveyed, we replaced it with another household from the same cell. In our main analysis we will focus on the full sample, but all our results are robust to excluding replacements. When it comes to the long-notice households, however, in the first round we only surveyed 31% of them, due to an error in the list of households shared with the survey firm.²⁰ Although substantially smaller, this subsample was still randomly selected and remains well balanced across arms (Table A8 in Appendix). In the second round, the mistake was corrected and we surveyed 94% of the long-notice sample.

We verify program implementation and study design compliance through administrative data. WFP systematically tracks each food parcel and records who collects it. We matched this data with the study data and confirm that 97% of households expected to keep receiving assistance received it, while 99% of households expected to stop receiving assistance did not receive it. We also collected information from the households themselves during our first- and second-round surveys. While such data are noisier, potentially due to recollection error or strategic misreporting, they mirror the expected pattern, with 83% (80%) of the households expected to keep receiving food assistance reporting it in the first round (second-round), and 89% (76%) of households expected to stop receiving food assistance reporting indeed no assistance by the first round (second-round).

Overall, balance, attrition, and implementation checks reassure us on the integrity of the original experimental design, with two important caveats. First, the error in the first-round data collection means that our first-round analysis for long-warning households is underpowered compared to the original design. In discussing those results we will therefore place more emphasis on the point estimates than on statistical significance. Second, the escalation of tensions with

¹⁹The tables only include a subset of 20 variables available from the baseline census. When considering the entire set of variables included in the baseline survey, there are a total of 139 possible comparisons. The notes below each table report the statistics when considering all these comparisons. For instance, when comparing the four targeting arms, there are 14 instances (10.1%) in which differences are statistically significant at the 10% level (p-value < 0.1), 10 instances (7.2%) at the 5% level (p-value < 0.05), and 4 instances (2.9%) at the 1% level (p-value < 0.01).

²⁰When selecting the random sample of households to be surveyed, by accident the code also re-randomized the long-notice households, without accounting for the fact that those households had been already identified by an earlier randomization.

Israel along the southern border and the emergency cash provided by WFP are expected to influence our second-round estimates. To address this challenge, we will also discuss and report in Appendix alternative specifications to shed light on the second-round impact of exclusion from *any* WFP assistance program.

7 Results

7.1 The Consequences of the End of Assistance

We start our analysis by studying the consequences of the end of assistance on food expenditure decisions. Figure A2 summarizes the results. The figure reports the estimated coefficient of interest, along with the associated 95% confidence intervals, and distinguishes between the first round (blue) and the second round (red) results. Estimates are based on equation (1) and the full regression table is reported in Table A9 in Appendix. As explained above, food expenditure takes into account all food purchases (cash or credit) as well as own food production.²¹ The aggregated variable on top of the chart indicates that in the short run, households react to the end of food assistance by increasing their food expenditure by 12.5 USD, on average. This corresponds to about 39% of the market value of a food parcel. Given that the average household received two parcels per month, this figure suggests that, on average, households compensate only for about 19% of the loss in assistance through higher food expenditure. The figure also opens up aggregate expenditure to look into the different food categories, distinguishing between categories included and not included in the WFP parcels. As expected, households excluded from assistance increase their expenditure specifically on food items that were previously covered by the assistance. In the short run, there is some evidence of substitution, with households (slightly) decreasing their purchases of other types of food. As time passes, however, households recover their level of expenditure on excluded food items, while somewhat adjusting downwards their expenses on WFP food categories.

One obvious question is whether this increase in food expenditure is enough to preserve the food security level of the household. We can look at both food consumption and the adoption of food-related risky coping strategies. Our food consumption data are coarser than food expenditure, as they only record whether the households consumed a certain category of food – and if so, how often – over the previous 7 days, without recording the amount. Panel A of Figure A3 shows that in the short run, there is a clear drop in consumption in food items that were previously provided through assistance (the corresponding regression estimates are reported in Table A10). Over time, the gap shrinks, but remains large and significant for pulses (-.13SD). Panel B of Figure A3 considers instead food-related risky coping strategies and shows that discontinued households experienced an increase in the use of all coping strategies (the corresponding regression estimates are reported in Table A11). In the short run, the effect is generally small and imprecisely estimated, but as time goes by, households increase their reliance on borrowing food (+.11 SD), reducing the frequency of meals (+.07 SD), and reducing adult

²¹Households were explicitly asked to estimate the value of each type of home-grown food they consumed over the previous week. These assessments are expected to be noisier than market purchases, but are unlikely to play a major role in our analysis, as the population mostly lives in urban centers and only 3.2% reports consuming home-grown food.

food consumption (+.06SD), leading to an overall increase in the rCSI index of .11SD by the time of our second round.

Overall, our results indicate that the end of assistance negatively impacted households' food security. This is confirmed when we combine food consumption and adoption of risky coping strategies in a single food security Anderson-style index, as reported in Figure A4. This figure shows the standardized effect of discontinuation on all our main outcomes (the corresponding regression estimates are reported in Table A11). The first item confirms the substantial increase in food expenditure documented earlier (+.16SD in the short run). In the short run, households excluded from assistance experience a .1 SD drop in food security and a worsening in their psychological well-being of the same magnitude (-.1 SD). Table A13 in the Appendix disaggregates the psychological well-being index and shows that this decline reflects worsening across all three components: life satisfaction, mental well-being, and worriedness.

The figure also shows that, by the second round roughly seven months later, the negative effects persisted. This persistence is notable given that the second-round estimates are likely attenuated by the emergency cash assistance received by 74% of discontinued households. By the second round, households whose assistance had been discontinued continued to experience lower food security (-0.08 SD) and an even larger decline in psychological well-being (-0.12 SD), driven primarily by lower life satisfaction (Table A13).

Figure A4 also shows that the end of assistance caused a significant drop in trust (-.14SD). To better understand this result, Figure A5 opens up the trust index, showing that the drop is driven by a large and sustained drop in trust towards aid organizations (International Organizations and NGOs)²². In the short run, there is also a smaller drop in trust towards local government – possibly due to some confusion in attributing the responsibility for the end of assistance – and migrants – possibly due to the fact that assistance towards migrants is provided through other programs that were not affected by this cut. However, in the longer run, it is really trust towards aid agencies that remains low (-.13SD), and again, this despite the fact that by the second round, the majority of the discontinued households in our sample had received emergency cash support from WFP. This suggests that once assistance is interrupted and trust is lost, it is hard to recover it.

Table A13 in the Appendix also reports results for additional secondary outcomes. The discontinuation of assistance has no detectable effect on employment or other measured financial outcomes. This raises the question of how households financed the observed increase in food expenditure. The data on non-food expenditure are noisy and incomplete, and do not allow us to precisely identify the source of the additional USD 10–13 per month spent on food. However, results in Table A15 show that households discontinued from assistance increased purchases on credit for both food and non-food items. This is consistent with the rise in harmful coping strategies documented above and points to a potential longer-run risk for household food security and financial stability.

²²International Organizations (IOs) and NGOs were grouped together in the survey question, as piloting revealed that households often struggle to distinguish the two. This is not surprising, given that IOs often rely on local NGOs to implement their programs.

Heterogeneity

Average effects may mask important heterogeneity across population subgroups. The analysis tests for heterogeneity along three pre-specified dimensions that capture distinct forms of vulnerability: baseline poverty (measured by the asset index); household size; and the gender of the household head. The results, reported in Tables A17–A19, suggest that initially wealthier households and larger households increase food expenditure more following the withdrawal of assistance, likely reflecting greater spending capacity in the former case and greater household needs in the latter. However, effects on food security and other dimensions of well-being remain broadly similar across groups. The point estimates also suggest that female-headed households adjust food expenditure less in response to the loss of support and experience larger declines in well-being than male-headed households, consistent with greater vulnerability. These differences, however, are small and imprecisely estimated. We also relied on a data-driven approach, following Chernozhukov et al. (2025), to explore broader patterns in the data. The analysis suggests that the clearest ML heterogeneity pattern is for second-round food expenditure, where households with the highest predicted treatment effects increase food spending substantially more than those with the lowest (Figure A15). Other subgroup signals are weaker and not consistent across learners.²³

Additional Checks

Results are confirmed when using population weights to account for the fact that the study sample is a stratified random sample (tables A20–A21), as well as when accounting for the fact that the likelihood of discontinuation vary across different eligibility spaces, following the fully interacted regression-adjustment logic of Lin (2013), as suggested by Humphreys (2025) (tables A22–A23).²⁴

Scaling down assistance programs necessarily entails discontinuing support for some households while maintaining it for others. The adverse effects of discontinuation may therefore operate through both direct and indirect channels. The direct channel reflects the loss of resources available to the household. The indirect channel may arise from relative comparisons with nearby households that continue to receive assistance. Administrative data from WFP make it possible to explore these mechanisms. The administrative records contain indeed the exact location of 97% of households included in the program. For each household, we can therefore estimate

²³The data-driven heterogeneity analysis follows the Generic ML approach of Chernozhukov et al. (2025). For each primary outcome and survey round, we study heterogeneity in the effect of assigned discontinuation relative to continued assistance, using only pre-treatment baseline and census covariates. Predicted treatment-effect scores are estimated on an auxiliary half-sample using three learners—a lasso, a random forest, and a support vector machine—with 1,000 random sample splits stratified by eligibility space. We then validate whether these scores capture systematic treatment-effect heterogeneity using BLP slopes and GATES contrasts between households in the highest and lowest predicted-effect quartiles. Finally, we use CLAN profiles to describe which baseline characteristics distinguish the high- and low-score groups. Across the 24 learner by outcome-wave validation cells, significant BLP or GATES evidence appears only in a few cases, of which the clearest is second-round food expenditure.

²⁴The tables show estimates of the impact of discontinuation using a centered-strata interaction correction. Instead of only absorbing eligibility-strata fixed effects, they demean each eligibility-stratum indicator within the outcome-wave estimation sample and interact those centered stratum indicators with assigned discontinuation; this lets the discontinuation effect vary flexibly across eligibility strata, while the reported coefficient is the sample-share weighted average effect across included strata.

how many other households nearby were included in the assistance program at baseline and how many kept receiving assistance vs were discontinued. Because the study design generates experimental variation in discontinuation status, it also induces variation in the number of nearby households that remained in or exited the program (conditional on the number of nearby households included in the program). We therefore can test whether such proximity affects our primary outcomes. Results are reported in Table A23 in Appendix. The first column restricts the sample to discontinued households and examines whether the probability of appealing the discontinuation decision varies with the number of households within a 200-meter radius that continue to receive assistance.²⁵ The specification controls for the total number of program households in the same radius, thereby accounting for differences in local beneficiary density. The estimates indicate that the likelihood of appealing increases with the number of nearby households that continue to receive assistance. The effect is modest in magnitude but precisely estimated. The remaining columns consider the full study sample and test whether the effects of discontinuation vary with proximity to continued beneficiaries. This is done by estimating a specification similar to equation (1), augmented with an interaction between the discontinuation indicator and the number of continued-assistance households within 200 meters. The results in Table A23 do not provide evidence that the negative effects of discontinuation are systematically larger for households that are closer to continued beneficiaries. However, our data allow us to go a step further and refine our proximity measure to only focus on “similar” households, i.e. households that belong to the same eligibility space. Table A24 shows the results with this more precise measure. This analysis again shows that the probability of appealing increases with the number of similar nearby households that continue to receive assistance. In addition, the negative effects of discontinuation on food security and psychological well-being are now significantly larger for households located near similar households that remained in the program. The coefficient on the main discontinuation effect remains large and statistically significant in most specifications, with the exception of food security in the second round. Overall, this analysis suggests that both channels are at play: relative comparisons with nearby continued beneficiaries appear to amplify the direct welfare losses associated with the withdrawal of assistance.

Finally, as discussed above, a major concern is that second-round estimates may be affected by the emergency cash assistance provided by WFP in response to the conflict with Israel. A first naive way to recover a cleaner estimate of the second-round discontinuation effect is to exclude WFP-defined “priority areas,” where emergency assistance was provided. Table A24 in the Appendix reports this analysis. First round effects are broadly comparable to those estimated in the full sample, although in some cases less precisely estimated because of the smaller sample size. By the second round, however, the estimated effects are substantially larger: psychological well-being and trust decline by more than twice the magnitude of the average effects reported above. The experimental variation in food assistance discontinuation, however, was not stratified by geographic area. An alternative approach is therefore to use the full sample while adopting an instrumental variables strategy, instrumenting actual discontinuation of all WFP assistance with the original experimental assignment to food assistance discontinuation. This specification estimates the effect of being excluded from WFP food support while not receiving any other

²⁵Results remain very similar when considering alternative radius, such as 100mt or 500mt.

WFP assistance. Results, reported in Tables A25 and A26 for main and secondary outcomes, respectively, show substantially larger effects than the previous estimates. These estimates allow for two non-mutually exclusive interpretations, which we cannot disentangle in this setting. First, they may indicate that the adverse effects of assistance discontinuation grow over time. Second, they may reflect the fact that, during an additional shock such as the conflict with Israel, WFP assistance served as an important buffer, so that exclusion from support had especially severe consequences for household well-being. More broadly, the results suggest that withdrawing assistance in fragile settings exposed to recurrent shocks can generate substantial and persistent welfare losses

7.2 Long Notice Window

The previous section documented the adverse consequences of assistance discontinuation. This section examines whether households are able to prepare for the loss of support once they are informed that it is approaching. The analysis focuses on households assigned to the long-notice arm, which received an additional six months of assistance before exit.

Figure A6 summarizes the effects on food expenditure and the main outcome indices. Panel A compares long-notice households to households that continued receiving assistance, while Panel B compares them to households whose assistance was discontinued under the standard short-notice protocol. By the first round, long-notice households (who were still receiving assistance but had been informed that it would end) had increased food expenditure by 0.10 SD relative to continued-assistance households. The estimate is economically meaningful, although imprecisely estimated ($p = 0.116$), likely reflecting the limited statistical power generated by the sampling error discussed above. Disaggregating food purchases, Table A10 shows that the clearest increase is in protein-rich foods, a category not included in the WFP food basket. This increase in food expenditure, however, is the only notable behavioral adjustment observed in the first round. Effects on the other outcome domains are close to zero, indicating that long-notice households otherwise resembled continued-assistance households. Consistent with this interpretation, Panel B shows that in the first round, long-notice households had higher food security, psychological well-being, and trust than households whose assistance had already been discontinued.

The second-round results show that the extended notice period had done little to mitigate the adverse consequences of assistance withdrawal. Long-notice households in the second round were spending significantly more on food, and the composition of their expenditures resembled that of discontinued households, with higher spending on items previously included in the assistance basket. This pattern suggests that households did not stockpile assistance before exit. Once support ended, food security and psychological well-being declined relative to continued-assistance households. Because the second round was conducted roughly seven months after assistance ended for long-notice households, these estimates could be broadly compared to the first-round effects for short-notice households, measured approximately six months after their own exit.²⁶ The magnitudes are strikingly similar. The increase in food expenditure is identical (+0.10 SD), the decline in food security is slightly smaller (-0.07 SD), and the

²⁶A major caveat of this comparison relates to the conflict with Israel and new emergency assistance provided at the time of the second round. Results should therefore be interpreted as merely suggestive.

decline in psychological well-being is somewhat larger (-0.14 SD). The main exception is trust: despite losing assistance, long-notice households exhibit only a small decline in trust relative to continued-assistance households, which contrasts sharply with the large decline in trust observed among short-notice households in the first round.

Overall, the results suggest that advance notice may improve the perceived legitimacy of program exit and allow limited adjustment in the timing and composition of food purchases. However, information alone does not enable households to cushion the socioeconomic consequences of losing assistance. This finding appears consistent with standard models of consumption smoothing under liquidity constraints (Zeldes, 1989; Deaton, 1991): even when income losses are anticipated, households may be unable to smooth consumption if they lack savings, access to credit, or sufficient margins of adjustment. This is also consistent with households’ self-reported information, which tells us that 90% of households fully consume their assistance within the month of receiving it.

7.3 Targeting Methods

Given the adverse impact of the discontinuation of assistance, we now switch our focus to which rule should determine who remains in the program. As a first step, we are interested in understanding whether it is possible to identify a targeting method (out of the four we considered) that leads to overall higher wellbeing among the target population. One important consideration is that while the discontinuation analysis focused on “marginal” households that were eligible under some, but not all methods, the comparison across targeting arms can take into account the entire population, to assess the aggregated impact of adopting one targeting method over another. We will nevertheless also zoom in on the “marginal” households to better investigate disagreements across methods.

Households Selection

We earlier discussed that the four targeting methods selected different households for continued assistance (Table A.2). A central question is whether any one method dominates the others in identifying households most in need of food assistance. Assessing targeting precision, however, requires a benchmark against which selection can be evaluated. This is challenging both conceptually and empirically. Conceptually, vulnerability is multidimensional, and each method in the study is exactly designed to prioritize a different aspect of it. It is therefore not obvious which welfare metric should serve as the “ground truth” for evaluating targeting accuracy. Empirically, the baseline census contains limited information on household vulnerability, allowing only the construction of a standard asset-based wealth index, based on 33 asset-ownership questions. With this caveat in mind, Table A28 and Figure A6 in the Appendix show that households with higher baseline asset wealth were more likely to be discontinued, and this pattern holds across all four targeting methods. The relationship between asset wealth and discontinuation is somewhat weaker under the categorical approach. To the extent that lower asset wealth proxies for greater vulnerability, this provides suggestive evidence that categorical targeting may be less precise than PMT-based methods, although the difference is modest.

A second, indirect measure of targeting precision is provided by appeal behavior. Although the cost of appealing was low, households still had to take the time to submit a formal appeal to WFP, and indeed only about half of discontinued households did so. More vulnerable households may be relatively more likely to incur this cost if they expect a greater chance of reversal. Table A34 provides suggestive evidence consistent with this interpretation: column 1 shows that households with higher baseline asset wealth were less likely to appeal, while column 2 shows that households farther from the eligibility cutoff (i.e., those classified by the assigned targeting method as less vulnerable) were also less likely to appeal. At the same time, appeal behavior is also shaped by access costs: column 3 shows that owning a phone or smartphone increased the likelihood of appealing, presumably by lowering the cost of contacting WFP support. Subject to this caveat, Figure A7 compares appeal rates across targeting arms. The share of discontinued households submitting an appeal is higher under categorical targeting, at 56%, than under the PMT-based methods, where appeal rates range from 50% to 52%. Taken together, the evidence from asset-based selection and appeal behavior suggests that categorical targeting may underperform relative to PMT-based approaches in terms of precision. By relying on coarser demographic and socioeconomic criteria, categorical targeting may be more prone to exclusion errors, leaving out some asset-poor households and generating higher rates of appeal. This appears consistent with evidence suggesting that richer data-driven approaches, such as PMTs, can improve the identification of household-level vulnerability relative to simpler categorical rules (Alatas et al., 2012; Brown et al., 2018). By contrast, there is little evidence of meaningful differences in precision across the three PMT-based methods, either in identifying asset-poor households or in reducing appeals, despite the fact that they target different outcomes and select partly distinct beneficiary profiles. The next question is whether these differences in beneficiary selection translate into differences in program effectiveness.

Wellbeing

Table A4 reports the estimates from regression (3), comparing the primary outcomes across the four targeting arms and distinguishing between first-round (Panel A) and second-round (Panel B). The analysis includes both selected and excluded households and thus compare targeting methods in terms of average impact across the sample.

Overall, the table shows that no method stands out from the others in terms of average impact. There are no systematic differences in terms of food expenditure (column 1), food security (column 2), psychological wellbeing (columns 3), or trust (column 4). Table A30 in Appendix shows that similar results hold when considering the specific components of the food security and psychological wellbeing index, as well as the additional secondary outcomes, related to social cohesion, employment status, and financial situation of the household. This means that even when compared against the specific outcomes that they were designed to target (e.g. FCS for PMT-FCS or food expenditure for PMT-Exp), no method appears to outperform the others. Results are confirmed when applying population weights to generate population-wide estimates (Table A31 and A32 in Appendix).

Heterogeneity

Although the four targeting methods generate similar average outcomes, they may still differ in their effects across population subgroups. Also in this case, we focus on the pre-specified heterogeneity dimensions: baseline wealth, household size, and female headship. The results, reported in Tables A33–A35 in the Appendix broadly mirror the way these dimensions are mapped into the different selection methods: subgroups tend to fare relatively better under the methods that were more likely to select them. For example, the categorical approach was more likely than the PMT-based methods to select female-headed households, and this is reflected in slightly higher psychological well-being and trust among these households in the second round, under that method. Conversely, because the categorical approach selected smaller households on average, larger households tend to exhibit slightly worse food security and psychological well-being under this method in the second round. These differences, however, are generally small and imprecisely estimated.

Eligibility-space diagnostic

The analysis so far indicates that, although the four targeting methods select different households and generate some differences in acceptance, as measured by appeals, these differences do not translate into substantial differences in average outcomes for the targeted population. This finding is consistent with the broader targeting literature, which shows that even meaningful differences in targeting precision do not always generate detectable differences in program impacts (Ravallion, 2009).

The analysis so far, however, has focused on average outcomes for the full target population, including households that all methods agree should be included or excluded. While this is arguably the primary parameter of policy interest, it may obscure differences in performance at the margin. Appendix D therefore provides a complementary analysis focusing on marginal households, defined as those in the 14 eligibility spaces where some methods classify households as eligible for continued assistance while others do not. For each eligibility space, we estimate discontinuation effects and then we compare those effects between the spaces selected vs not selected by each method. This provides a summary measure of whether a given targeting rule systematically includes households that would suffer more from discontinuation, or excludes households that would be less affected.

The results broadly confirm the main findings: no targeting method systematically outperforms the others. PMT-Expenditure tends to select households that, in the medium run, would otherwise increase food expenditure more in response to the loss of assistance, while the opposite pattern is observed for categorical targeting. However, these differences do not translate into systematic differences in food security, psychological well-being, or other welfare outcomes.

Methods Combination

One possible explanation for the preceding results is that targeting is inherently noisy, with performance depending on the precision of both the underlying model (especially for PMT-based approaches) and the data used to implement it. Such noise may generate targeting errors

and reduce the contrast across methods, leading to similar average welfare outcomes despite differences in selection. Consistent with this interpretation, Figure A6 showed that each method selected some households with relatively high baseline wealth for continued assistance. Although high wealth does not automatically imply the absence of vulnerability, those outliers might reflect the imprecision of the targeting methods. Our design makes it possible to test whether combining information across methods can improve precision and thus reduce the adverse consequences of discontinuation. The simple underlying idea behind this test is that agreement across independent but noisy signals may be informative, even when no single signal clearly dominates.

To test this, we compare the effects of assistance discontinuation across households deemed eligible by relatively more versus fewer targeting methods. We start by considering appeal behavior. Figure A8 shows a clear positive relationship between the number of methods classifying a household as eligible for assistance and the likelihood of appealing the discontinuation decision. To the extent that appeals reflect underlying need, this suggests that the intersection of multiple targeting methods may indeed help identify more vulnerable households. This interpretation is further supported by baseline asset data: Figure A7 shows that households eligible under more methods had significantly lower asset ownership at baseline. In other words, by combining the selection, it is possible to get rid of some of the high-wealth households that were included by the individual methods.

Figure A9 shows how this translates in terms of effects on the main outcomes, with corresponding regression estimates in Table A36. Discontinued households that were eligible under three methods were less able to increase food expenditure after losing assistance than households eligible under fewer methods. In the short run, the difference is substantial: households eligible under three of the four methods did not increase at all food spending, while those eligible under only one method increased food expenditure by 0.26 SD, equivalent to approximately USD 20. The gap narrows over time, but the point estimates still indicate that the increase in food expenditure remains 64% larger among households eligible under one method than among those eligible under three. A symmetric pattern emerges for food security: discontinued households eligible under three methods experience larger declines, although the point estimates cannot be distinguished at conventional levels. Results in Table A37 suggests that this pattern is driven by greater reliance on emergency coping strategies, with the gap widening over time. By contrast, effects on psychological well-being and trust are less clear, with no clear ranking across eligibility groups and mixed patterns over time.

Overall, these results suggest that, although no single targeting method clearly dominates the others, agreement across methods is informative. Households identified as eligible by multiple approaches appear more asset-poor at baseline, are more likely to appeal discontinuation, and are less able to compensate for the loss of food assistance. Combining targeting methods may therefore help identify households that face larger needs when assistance is withdrawn.

8 Discussion and Conclusion

This study provides the first experimental evidence of the consequences of cutting a large-scale humanitarian assistance program. While conducted in a specific context, the study results offer four main insights with broader relevance for the literature on social protection and targeting, as well as for humanitarian policy and practice.

First, our analysis shows that discontinuing assistance, even in a context where emergency needs had partially subsided and efforts were made to retain the most vulnerable recipients, has significant negative effects on households' welfare. Despite households' attempt to compensate for end of assistance with increased food expenditure, discontinued households report reduced food security and lower psychological well-being that persist over time. These results come from cutting a program that provided the equivalent of 30% of the median household's monthly food expenditure and 13% of its total expenditure. It appears likely that the recent cuts to more generous support programs—such as those following the reduction in the UK aid budget and the termination of the large majority of USAID contracts in early 2025, discussed in Section 1—might lead to even more dramatic consequences.

Second, assistance discontinuation also led to a large decline in trust in aid agencies. Although this outcome may be less directly tied to household welfare in the short run, it is central to the effectiveness of humanitarian organizations, as trust can shape beneficiaries' willingness to engage with future programs. Consistent with this interpretation, Table A38 in the Appendix uses Arab Barometer data to show that trust in the United Nations and civil society organizations is positively associated with willingness to accept aid, as well as with support for international organizations' involvement in addressing socioeconomic and governance challenges.²⁷ Importantly, the decline in trust appears persistent: trust remained significantly lower in the second round, despite the introduction of a new assistance program in response to the escalation of conflict along Lebanon's southern border.

Third, the results show that the provision of a longer warning might have some benefits especially in terms of sustaining trust towards the aid organization, but other strategies should be tested to reduce the consequences of the discontinuation of assistance on food security and psychological wellbeing. Within our context, households do not seem to take preventive action that enable them to smooth their transition out of assistance, and ultimately experience welfare losses similar in magnitude to those who received only one month of notice. These findings suggest that information alone is not enough: households facing chronic vulnerability may lack the resources or flexibility to meaningfully prepare for shocks, even when forewarned. Additional forms of support and assistance should therefore be developed and tested.

Fourth, while we found no single targeting method clearly outperforms others in terms of program impact, each method identifies a distinct set of beneficiaries, with only partial overlaps.

²⁷The Arab Barometer includes two measures that proxy trust in international organizations and NGOs: whether the respondent holds a favorable view of the United Nations, and whether the respondent trusts civil society organizations. These measures are correlated with responses to questions on whether the respondent would accept foreign aid and whether they would support the United Nations working on socioeconomic or governance-related issues. All specifications control for age, gender, education, employment status, settlement type, household size, income hardship, and location fixed effects. The analysis is reported both for the full set of available countries and for Lebanon separately. Further details are provided in the table note.

By combining methods and prioritizing households identified as vulnerable across multiple approaches, we find that it is possible to reach the relatively more vulnerable households, in greatest need of support. This insight has already informed WFP’s ongoing programming in Lebanon, where intersection-based targeting strategies are being adopted across multiple programs.

Finally, our results and analysis highlight a broader structural issue in the design of humanitarian and development programs: while substantial attention is devoted to initial targeting and design, relatively little guidance or evidence exists on how to exit programs responsibly. Assistance scale-downs are often dictated by donor priorities and budget availability, rather than by rigorous assessments of when support is no longer essential. Without clear exit strategies, even well-targeted programs risk losing the gains achieved during the implementation phase. More deliberate and evidence-based exit planning, including behavioral or bridging interventions to support transitions, could help preserve program impacts and mitigate harm.

Our study has a number of limitations. First, as with any empirical evaluation, the findings are tied to the specific population and context studied; the national spread of the sample strengthen external validity, but more research on these topics shall be conducted across multiple contexts to enrich the evidence provided here. Second, the discontinuation analysis focuses on marginal households, namely those eligible under at least one but not all targeting methods, so the estimates cannot be directly generalized to households consistently classified as vulnerable or non-vulnerable by all methods. The observed gradient in appeals and negative outcomes by number of eligibility methods suggests that discontinuation effects would likely be strongest (weakest) among households identified as vulnerable (non-vulnerable) by all methods. Third, emergency cash assistance introduced in conflict-affected areas during the study period likely reduced the contrast between continued and discontinued households. The discontinuation estimates should therefore be interpreted as conservative, lower-bound estimates of the effects of ending assistance.

By generating the first experimental evidence on the consequences of cutting aid program, our study contributes to a more complete understanding of the program lifecycle and informs both academic debates and operational decisions in humanitarian settings. Humanitarian organizations increasingly face the need to scale down or reprioritize programming in the face of volatile funding and overlapping emergencies. Our findings suggest that relatively simple modifications to existing program design, such as intersecting targeting methods can help improve both the perceived fairness and effectiveness of scale-down decisions. At the same time, they point to the need for greater attention to the end phases of assistance programs, which remain understudied despite their high practical relevance. Future research should test additional models of transition support, explore scalable alternatives to precise targeting, and continue to build the evidence base on how to end assistance in fragile settings.

References

- Aiken, Emily, Suzanne Bellue, Dean Karlan, Christopher Udry, and Joshua E. Blumenstock, “Machine Learning and Phone Data Can Improve Targeting of Humanitarian Aid,” *Nature*, 2022, 603 (7903), 864–870.
- , Tim Ohlenburg, and Joshua Blumenstock, “Moving Targets: When Does a Poverty Prediction Model Need to Be Updated?,” in “Proceedings of the 6th ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies” 2023, pp. 117–117.
- Alatas, Vivi, Abhijit Banerjee, Rema Hanna, Benjamin A. Olken, and Julia Tobias, “Targeting the Poor: Evidence from a Field Experiment in Indonesia,” *American Economic Review*, 2012, 102 (4), 1206–1240.
- , – , – , – , Ririn Purnamasari, and Matthew Wai-Poi, “Self-Targeting: Evidence from a Field Experiment in Indonesia,” *Journal of Political Economy*, 2016, 124 (2), 371–427.
- Altındağ, Onur, Stephen D. O’Connell, Aytuğ Şaşmaz, Zeynep Balcioglu, Paola Cadoni, Matthew Jerneck, and Aled K. Foong, “Targeting Humanitarian Aid Using Administrative Data: Model Design and Validation,” *Journal of Development Economics*, 2021, 148, 102564.
- Anderson, Michael L., “Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects,” *Journal of the American Statistical Association*, 2008, 103 (484), 1481–1495.
- Araar, Abdelkrim and Luca Tiberti, “Optimal Targeting and Poverty Reduction with Fixed Budget and Under Imperfect Information,” Technical Report, PEP Working Paper 2020.
- Beaman, Lori, Ariel BenYishay, Jeremy Magruder, and Ahmed Mushfiq Mobarak, “Can Network Theory-Based Targeting Increase Technology Adoption?,” *American Economic Review*, 2021, 111 (6), 1918–1943.
- Brown, Caitlin, Martin Ravallion, and Dominique van de Walle, “A Poor Means Test? Econometric Targeting in Africa,” *Journal of Development Economics*, 2018, 134, 109–124.
- Bruni, Valentina and Olivier Sterck, “The Welfare and Market Effects of Delays in Humanitarian Assistance,” *SSRN Scholarly Paper*, 2025, (5190878).
- Cameron, Lisa and Manisha Shah, “Can Mistargeting Destroy Social Capital and Stimulate Crime? Evidence from a Cash Transfer Program in Indonesia,” *Economic Development and Cultural Change*, 2014, 62 (2), 381–415.
- Caria, Stefano, Grant Gordon, Maximilian Kasy, Simon Quinn, Soha Shami, and Alexander Teytelboym, “An Adaptive Targeted Field Experiment: Job Search Assistance for Refugees in Jordan,” *SSRN Electronic Journal*, 2020.

- Chernozhukov, Victor, Mert Demirer, Esther Duflo, and Iván Fernández-Val**, “Generic Machine Learning Inference on Heterogeneous Treatment Effects in Randomized Experiments,” *Working Paper*, 2025.
- Crosta, Tommaso, Dean Karlan, Finn Ong, Julius Rüschepöhler, and Christopher R. Udry**, “Unconditional Cash Transfers: A Bayesian Meta-Analysis of Randomized Evaluations in Low and Middle Income Countries,” Working Paper 32779, National Bureau of Economic Research 2024.
- Deaton, Angus**, “Saving and Liquidity Constraints,” *Econometrica*, 1991, *59* (5), 1221–1248.
- Devereux, Stephen, Edoardo Masset, Rachel Sabates-Wheeler, Michael Samson, Althea-Maria Rivas, and Dolf Te Lintelo**, “The Targeting Effectiveness of Social Transfers,” *Journal of Development Effectiveness*, 2017, *9* (2), 162–211.
- Deville, Jean-Claude and Carl-Erik Särndal**, “Calibration Estimators in Survey Sampling,” *Journal of the American Statistical Association*, 1992, *87* (418), 376–382.
- Dustmann, Christian, Rasmus Landersø, and Lars H. Andersen**, “Refugee Benefit Cuts,” *American Economic Journal: Economic Policy*, 2024, *16* (2), 406–441.
- , —, —, and —, “Unintended Consequences of Welfare Cuts on Children and Adolescents,” *American Economic Journal: Applied Economics*, 2024, *16* (4), 161–185.
- Filmer, Deon and Lant H. Pritchett**, “Estimating Wealth Effects without Expenditure Data—or Tears: An Application to Educational Enrollments in States of India,” *Demography*, 2001, *38* (1), 115–132.
- Follett, Lendie and Heath Henderson**, “A Hybrid Approach to Targeting Social Assistance,” *Journal of Development Economics*, 2023, *160*, 103002.
- Friedman, Milton**, “The Permanent Income Hypothesis,” in “A Theory of the Consumption Function,” NBER / Princeton University Press, 1957, pp. 20–37.
- Gelman, Michael, Shachar Kariv, Matthew D. Shapiro, Dan Silverman, and Steven Tadelis**, “Harnessing Naturally Occurring Data to Measure the Response of Spending to Income,” *Science*, 2014, *345* (6193), 212–215.
- Goldin, Jacob, Tatiana Homonoff, and Katherine Meckel**, “Issuance and Incidence: SNAP Benefit Cycles and Grocery Prices,” *American Economic Journal: Economic Policy*, 2022, *14* (1), 152–178.
- Gram-Hansen, Bradley J., Patrick Helber, Indhu Varatharajan, Faiza Azam, Alejandro Coca-Castro, Veronika Kopackova, and Piotr Bilinski**, “Mapping Informal Settlements in Developing Countries Using Machine Learning and Low Resolution Multi-spectral Data,” in “Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society” 2019, pp. 361–368.

- Grosh, Margaret, Phillippe Leite, Matthew Wai-Poi, and Emil Tesliuc, eds**, *Revisiting Targeting in Social Assistance: A New Look at Old Dilemmas*, The World Bank, 2022.
- Hall, Ola, Francis Dompae, Ibrahim Wahab, and Fred M. Dzanku**, “A Review of Machine Learning and Satellite Imagery for Poverty Prediction: Implications for Development Research and Applications,” *Journal of International Development*, 2023, 35 (7), 1753–1768.
- Haushofer, Johannes, Paul Niehaus, Carlos Paramo, Edward Miguel, and Michael Walker**, “Targeting Impact versus Deprivation,” Working Paper 30138, National Bureau of Economic Research 2022.
- Hidrobo, Melissa, John Hoddinott, Amber Peterman, Amy Margolies, and Vanessa Moreira**, “Cash, Food, or Vouchers? Evidence from a Randomized Experiment in Northern Ecuador,” *Journal of Development Economics*, 2014, 107, 144–156.
- , **Naureen Karachiwalla, and Shalini Roy**, “The Impacts of Cash Transfers on Mental Health and Investments: Experimental Evidence from Mali,” *Journal of Economic Behavior & Organization*, 2023, 216, 608–630.
- Hillebrecht, Michael, Stefan Klonner, and Noraogo A. Pacere**, “The Dynamics of Poverty Targeting,” *Journal of Development Economics*, 2023, 161, 103033.
- Hoddinott, John, Susanna Sandström, and Joanna Upton**, “The Impact of Cash and Food Transfers: Evidence from a Randomized Intervention in Niger,” *American Journal of Agricultural Economics*, 2018, 100 (4), 1032–1049.
- Humphreys, Macartan**, “Notes on Regression Adjustment with Stratified Randomization,” 2025. Working paper; reference details to be verified.
- Iqbal, Muhammad et al.**, “Unanticipated Program Exit and Household Welfare: Regression Discontinuity Evidence from Pakistan,” 2026. Working paper; reference details to be verified.
- Jeong, Dahyeon and Iva Trako**, “Cash and In-Kind Transfers in Humanitarian Settings: A Review of Evidence and Knowledge Gaps,” Policy Research Working Paper 10026, World Bank Group 2022.
- Khan, Aubhik and Edward S. Knotek**, “How Do Households Respond to Uncertainty Shocks?,” *Federal Reserve Bank of Kansas City Economic Review*, 2011, 96 (2).
- Lin, Winston**, “Agnostic Notes on Regression Adjustments to Experimental Data: Reexamining Freedman’s Critique,” *The Annals of Applied Statistics*, 2013, 7 (1), 295–318.
- Maxwell, Daniel and Richard Caldwell**, “The Coping Strategies Index: Field Methods Manual,” Technical Report, Cooperative for Assistance and Relief Everywhere (CARE) and World Food Programme 2008.
- , —, and **Mark Langworthy**, “Measuring Food Insecurity: Can an Indicator Based on Localized Coping Behaviors Be Used to Compare across Contexts?,” *Food Policy*, 2008, 33 (6), 533–540.

- McBride, Linden and Austin Nichols**, “Retooling Poverty Targeting Using Out-of-Sample Validation and Machine Learning,” *The World Bank Economic Review*, 2018, 32 (3), 531–550.
- McKenzie, David J.**, “Measuring Inequality with Asset Indicators,” *Journal of Population Economics*, 2005, 18 (2), 229–260.
- OCHA**, “Global Humanitarian Overview 2024: October Update,” <https://www.unocha.org/attachments/b9488007-b44e-4a35-9827-6a9f8488194e/10%20GH0%202024%20October%20Update.pdf> 2024.
- OECD**, *OECD Guidelines on Measuring Subjective Well-being*, Paris: OECD Publishing, 2013.
- Premand, Patrick and Pascale Schnitzer**, “Efficiency, Legitimacy, and Impacts of Targeting Methods: Evidence from an Experiment in Niger,” *The World Bank Economic Review*, 2021, 35 (4), 892–920.
- Ravallion, Martin**, “How Relevant Is Targeting to the Success of an Antipoverty Program?,” *World Bank Research Observer*, 2009, 24 (2), 205–231.
- Schnitzer, Pascale and Quentin Stoeffler**, “Targeting Social Safety Nets: Evidence from Nine Programs in the Sahel,” *The Journal of Development Studies*, 2024, 60 (4), 574–595.
- Sterck, Olivier and Valentina Bruni**, “The Welfare and General-Equilibrium Impacts of Aid Cuts,” *SSRN Scholarly Paper*, 2025, (5270846).
- Stoeffler, Quentin, Bradford Mills, and Carlo Del Ninno**, “Reaching the Poor: Cash Transfer Program Targeting in Cameroon,” *World Development*, 2016, 83, 244–263.
- UNHCR**, “Lebanon Emergency Flash Update,” https://www.unhcr.org/sites/default/files/2024-12/UNHCR%20Lebanon%20-%20External%20Flash%20update%20%2829%2011%202024%29_.pdf November 2024. 29 November 2024.
- Vyas, Seema and Lilani Kumaranayake**, “Constructing Socio-economic Status Indices: How to Use Principal Components Analysis,” *Health Policy and Planning*, 2006, 21 (6), 459–468.
- World Bank**, “Lebanon Economic Monitor, Fall 2022: Time for an Equitable Banking Resolution,” Lebanon Economic Monitor, World Bank, Washington, DC 2022. License: CC BY 3.0 IGO.
- World Food Programme**, “Food Consumption Score,” <https://resources.vam.wfp.org/data-analysis/quantitative/food-security/food-consumption-score> 2024.
- World Health Organization**, “The World Health Organization-Five Well-Being Index (WHO-5),” <https://www.who.int/publications/m/item/WHO-UCN-MSD-MHE-2024.01> 2024.
- World Values Survey Association**, “World Values Survey: Generalized Trust Item,” <https://www.worldvaluessurvey.org> 2022. Standard generalized-trust item: “Generally speaking, would you say that most people can be trusted or that you need to be very careful in dealing with people?”

Zeldes, Stephen P., “Consumption and Liquidity Constraints: An Empirical Investigation,”
Journal of Political Economy, 1989, 97 (2), 305–346.

A Targeting Models

We compare four targeting methods. All are applied to the same beneficiary census and retain the 43% most vulnerable households; what differs is how vulnerability is defined and measured. Three are proxy means tests (PMTs) that predict a welfare outcome from observable household characteristics: PMT-Exp targets per-capita expenditure (chronic poverty), PMT-FCS targets the Food Consumption Score (food deprivation), and PMT-CARI targets a composite measure of food insecurity that also incorporates households’ coping capacity. The fourth method, the Expert (Categorical) score, does not predict an outcome but instead ranks households directly using vulnerability criteria elicited from local food security experts. The construction of each targeting method is described below.

A proxy means test (PMT) predicts household welfare from observable, easily verified characteristics such as demographics, assets, housing conditions, and access to services. The prediction model is estimated using a training survey in which both the welfare outcome and household characteristics are observed, and is then applied to a beneficiary registry (WFP beneficiary census) containing the same predictors to generate a welfare score for each household. Households are ranked by their predicted score, and the neediest share—here the bottom 43%—is selected for assistance.

The PMTs are trained using the Lebanese Household Survey (LHS), a nationally representative survey of 3,597 households collected between January and May 2023. The survey contains detailed household characteristics alongside three welfare measures: log per-capita household expenditure, the Food Consumption Score (FCS), and the Consolidated Approach to Reporting Indicators of Food Security (CARI). PMT-Exp predicts expenditure, PMT-FCS predicts the FCS, and PMT-CARI predicts the probability of food insecurity derived from the CARI classification. Because the programme targets poorer households, and to reduce the influence of high-income outliers, the estimation sample is restricted to the poorest 75% of the expenditure distribution. This yields a poverty rate of 43%, matching both the programme’s coverage and the latest WFP post-distribution monitoring.

The Expert (Categorical) method does not estimate a statistical model. Instead, it ranks households using a weighted vulnerability index constructed from the judgments of 28 experts of the Lebanese food security contexts. Each expert selected and ranked the ten household characteristics they considered most indicative of need from the same set of variables available to the PMTs. These rankings were aggregated using a Borda count to assign a weight to each criterion.

The criteria were then classified as either inclusion criteria, which increase a household’s vulnerability score, or exclusion criteria, which reduce it. A household’s final score is obtained by adding the weights of all inclusion criteria it satisfies and subtracting the weights of any exclusion criteria. The most important inclusion criteria include having no income, a widowed, divorced, or single household head, a household head who is a minor or has a disability, and the absence of working-age members. The strongest exclusion criteria capture indicators of relatively higher living standards, such as owning a vehicle or durable appliances (e.g., a dryer, washing machine, air conditioner, or microwave) or deriving the main household income from a large business. Households are ranked by their final score, and the top 43% are selected for assistance.

The prediction models for the PMTs use only variables observed in both the LHS and the WFP beneficiary census. After harmonizing the two datasets, standardizing continuous variables, expanding categorical variables into indicator variables, and adding squared and interaction terms, the estimation includes 192 candidate predictors. For each outcome, predictors are selected using the Least Absolute Shrinkage and Selection Operator (LASSO), which outperforms or matches alternative methods while producing a parsimonious and interpretable model.

The estimated coefficients are then applied to the WFP beneficiary census (CP3), covering 58,501 households surveyed between August and September 2023. Because the census contains the same household characteristics but not the welfare outcomes, the models generate a predicted welfare score for every household. Households are ranked by their predicted score, and the bottom 43% are selected for assistance, matching the programme’s coverage.

A.1 Targeting Accuracy

We assess each method’s predictive performance in the training database by comparing its predicted score with the welfare outcome it is designed to predict. Table A1 reports both classification accuracy at the 43% eligibility threshold and ranking performance across the full outcome distribution. The expenditure-based PMT performs best across all measures, achieving the highest precision (69%) and the most accurate ranking of households (AUC = 0.83). PMT-CARI (precision = 67%, AUC = 0.81) and PMT-FCS (precision = 64%, AUC = 0.77) also perform well, while the categorical approach performs substantially worse (precision = 55%, AUC = 0.67). Nevertheless, all methods outperform random targeting (precision = 41%, AUC = 0.50) by a considerable margin.

Table A2 incorporates the categorical arm and benchmarks the targeting methods against an asset index, the only pre-intervention welfare measure available for the target population. Households with fewer assets are more likely to be selected under every method, although the degree of progressivity varies. A one-standard-deviation increase in assets reduces the odds of selection by 87% under PMT-FCS, 83% under PMT-CARI, 77% under the expenditure-based PMT, and 66% under the categorical approach, indicating that the latter is least effective at prioritizing asset-poor households.

Table A1: Targeting accuracy by method and ground-truth outcome (full matrix)

Method	Spearman ρ	AUC	Precision	Recall
<i>Panel A. Ground truth: log per-capita consumption</i>				
PMT-Exp	0.57 [0.51, 0.62]	0.83 [0.80, 0.86]	0.69 [0.64, 0.75]	0.72 [0.67, 0.77]
PMT-FCS	0.46 [0.40, 0.52]	0.77 [0.73, 0.81]	0.64 [0.58, 0.69]	0.69 [0.63, 0.74]
PMT-CARI	0.52 [0.46, 0.58]	0.81 [0.77, 0.84]	0.67 [0.62, 0.73]	0.71 [0.66, 0.77]
Expert	0.29 [0.22, 0.36]	0.67 [0.63, 0.71]	0.55 [0.48, 0.60]	0.56 [0.50, 0.61]
Random	-0.00 [-0.08, 0.07]	0.50 [0.45, 0.54]	0.41 [0.35, 0.47]	0.43 [0.38, 0.47]
<i>Panel B. Ground truth: Food Consumption Score</i>				
PMT-Exp	0.25 [0.18, 0.32]	0.65 [0.61, 0.69]	0.54 [0.49, 0.60]	0.57 [0.50, 0.63]
PMT-FCS	0.35 [0.28, 0.41]	0.71 [0.66, 0.74]	0.58 [0.52, 0.63]	0.62 [0.56, 0.68]
PMT-CARI	0.29 [0.22, 0.36]	0.67 [0.63, 0.71]	0.57 [0.51, 0.62]	0.60 [0.54, 0.66]
Expert	0.17 [0.09, 0.24]	0.60 [0.55, 0.64]	0.49 [0.43, 0.54]	0.50 [0.44, 0.56]
Random	0.00 [-0.07, 0.08]	0.50 [0.46, 0.54]	0.42 [0.36, 0.48]	0.43 [0.38, 0.47]
<i>Panel C. Ground truth: CARI food insecurity</i>				
PMT-Exp	0.45 [0.38, 0.51]	0.77 [0.72, 0.80]	0.62 [0.57, 0.67]	0.68 [0.62, 0.74]
PMT-FCS	0.40 [0.32, 0.46]	0.74 [0.69, 0.77]	0.59 [0.53, 0.64]	0.66 [0.60, 0.71]
PMT-CARI	0.45 [0.39, 0.51]	0.77 [0.73, 0.80]	0.62 [0.56, 0.67]	0.69 [0.64, 0.75]
Expert	0.21 [0.13, 0.28]	0.62 [0.58, 0.67]	0.50 [0.44, 0.55]	0.54 [0.47, 0.59]
Random	-0.00 [-0.08, 0.07]	0.50 [0.46, 0.54]	0.40 [0.34, 0.45]	0.43 [0.38, 0.47]

Notes: Targeting accuracy on the held-out LHS fold. Spearman's ρ and the area under the ROC curve (AUC) are threshold-agnostic; precision and recall are evaluated at the 43% eligibility cut-off. Each method's continuous score is compared to the indicated ground-truth outcome. *Random* targets households at random and anchors the no-information benchmark. 95% bootstrap confidence intervals (1,000 replications) in brackets.

Table A2: Selection on baseline assets, by targeting method

	Asset index
PMT-Exp	-1.450*** (0.061)
PMT-FCS	-2.015*** (0.078)
PMT-CARI	-1.756*** (0.069)
Expert	-1.080*** (0.053)
Joint test of equal slopes (p)	0.000

Notes: Each cell is the coefficient on the welfare index from a separate logit of an indicator for selection by that arm on the index (robust SE in parentheses); negative values mean poorer households are more likely selected. The joint test reshapes to household-by-arm and tests equality of the index slope across the four arms (logit with arm interactions, SE clustered by household). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Outcome Construction

This appendix documents the construction of each outcome used in the analysis. Table A3 lists, for every indicator, its constituent components, the individual survey items, the response scales, the weights applied, and the resulting score range. Item wording and response categories are transcribed from the second-round household questionnaire. Following Anderson (2008), closely related items are aggregated into inverse-covariance-weighted indices for each outcome family (food security, psychological well-being, and trust); the individual components are reported as supporting evidence. The Food Consumption Score and reduced Coping Strategy Index use the standard WFP weights (World Food Programme, 2024; Maxwell et al., 2008; Maxwell and Caldwell, 2008); the worriedness items follow Hidrobo et al. (2023), the WHO-5 items follow World Health Organization (2024), and the generalized-trust item follows the standard World Values Survey formulation (World Values Survey Association, 2022); the life-satisfaction items follow the domain-satisfaction approach set out in OECD (2013). The asset index is the first principal component of the durable-good ownership indicators, standardized so that higher values denote greater wealth (Filmer and Pritchett, 2001; Vyas and Kumaranayake, 2006; McKenzie, 2005).

Table A3: Detailed description of the outcomes

Indicator	Components	Sub-components / items	Possible answers	Weight	Score range
Monthly household food expenditure	Value consumed over past 30 days, summed across 9 food groups. For each group three amounts are asked: (a) cash or credit purchases, (b) in-kind gifts / assistance, (c) own production. The measure sums (a)+(c); in-kind	Cereals and tubers (basket) Pulses, legumes, and nuts (basket)	Monetary value in LBP or USD, 30-day recall. “(basket)” marks the food groups included in the WFP food parcel.		

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Table A3 continued from previous page

Indicator	Components	Sub-components / items	Possible answers	Weight	Score range
		Sugar and sweets (basket)			
		Oil, fat, and butter (basket)			
		Condiments and spices (basket)			
		Vegetables			
		Fruits			
		Meat, fish, and eggs			
		Milk and dairy			
Monthly household non-food expenditure	Recurrent expenses — 10 items (30-day recall)	Rent; electricity; drinking water; heating; soap & cleaning items; communications (phone); transportation & fuel; alcohol & tobacco; cooking gas; household utilities	Monetary value (LBP or USD), 30-day recall	sum	USD/hh/mo
	Infrequent expenses — 7 items (6-month recall, divided by 6)	Healthcare / medical; clothing & shoes; education; celebrations / social events; agricultural inputs; construction / house repairs; other	Monetary value (LBP or USD), 6-month recall		
Purchases on credit	Food purchases on credit	Value of food purchased on credit rather than paid in cash	Monetary value (LBP or USD); indicator for any credit purchase	—	USD/hh/mo
	Non-food purchases on credit	Value of non-food purchased on credit rather than paid in cash	Monetary value (LBP or USD); indicator for any credit purchase		
Food security index (Anderson 2008)	Food Consumption Score (FCS): weighted frequency of food groups consumed over past 7 days	Cereals, roots, and tubers		2	
		Pulses		3	
		Vegetables		1	
		Fruits		1	
		Meat and fish	0 - 7 days	4	0-112
		Milk		4	
		Sugar		0.5	
		Oil		0.5	
		Condiments and spices		0	
	Reduced Coping Strategy Index (rCSI, reversed): frequency of 5 food-related coping strategies used in past 7 days	Rely on less preferred or less expensive foods		1	
		Borrow food or rely on help from a friend or relative	0 - 7 days	2	0-56
		Limit portion size at mealtimes		1	
		Restrict consumption by adults so children can eat		3	
		Reduce number of meals eaten in a day		1	

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Indicator	Components	Sub-components / items	Possible answers	Weight	Score range
Livelihood Coping Strategy Index (LCSI)	Whether the household adopted each strategy over a round-specific reference period (since January 2024 at midline, since July 2024 at endline); classified by the most severe category used	<i>Stress</i> : reduce or cease payments on essential utilities <i>Stress</i> : sell household assets (furniture, fridge, television) <i>Stress</i> : reduce expenditures on education <i>Stress</i> : reduce expenditures on non-essential health <i>Crisis</i> : reduce expenses on essential health (postpone or stop treatment) <i>Crisis</i> : sell productive, income-generating assets <i>Crisis</i> : withdraw children from school <i>Emergency</i> : sell an owned house or land <i>Emergency</i> : engage in begging <i>Emergency</i> : engage in socially degrading, high-risk, or life-threatening work	0 = No, did not need to; 1 = No, already exhausted this strategy; 2 = Yes; 3 = Not applicable; -99 = Don't know; -999 = Prefer not to disclose	n.a.	category
Life satisfaction index	Satisfaction across 5 household domains (summed, then standardized)	... the household's nutrition / food security ... the household's overall health ... the household's financial situation ... interaction with the community and neighbors ... your life in general	1 = Very satisfied; 2 = Rather satisfied; 3 = Rather unsatisfied; 4 = Very unsatisfied (reversed); -999 = Prefer not to disclose	n.a.	z
Worriedness index	Degree of worry across 10 domains (summed, then standardized)	Money for basic food needs Money for health care Money for education Indebtedness Employment Family Extremism Conflict / sectarian tensions Natural disasters Migration / displacement	1 = Not worried at all; 2 = Not too worried; 3 = Moderately/somewhat worried; 4 = Very worried; -999 = Prefer not to disclose	n.a.	z
WHO-5 Mental Health Index	Sum of 5 items on well-being over the past two weeks (then standardized)	I have felt cheerful and in good spirits	0 = At no time; 1 = Some of the time; 2 = Less than half the time; 3 = More than half the time; 4 = Most of the time; 5 = All of the time; -999 = Prefer not to disclose	1	0-25

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Indicator	Components	Sub-components / items	Possible answers	Weight	Score range
		I have felt calm and relaxed I have felt active and vigorous I woke up feeling fresh and rested My daily life has been filled with things that interest me			
Psychological well-being index	Inverse-covariance-weighted (Anderson 2008) index of the three well-being measures	Life satisfaction index; worriedness index (reversed); WHO-5 mental health index	Aggregated from standardized components	ICW	z
Trust index	Generalised trust	In general, do you think people can be trusted or that they cannot be trusted?	1 = Most people can be trusted; 2 = Need to be very careful (reversed); -99 = Don't know; -999 = No answer	n.a.	z
	Trust in 7 specific groups / institutions (summed, then standardized)	Your family Your neighbors Religious organizations International organizations (UN) / NGOs Your local government Your national government Migrants (from outside Lebanon)	1 = Trust completely; 2 = Trust somewhat; 3 = Do not trust very much; 4 = Do not trust at all (reversed); -999 = Prefer not to disclose		
Financial situation	Savings	Any monetary savings, and estimated total amount (formal + informal, LBP/USD, incl. abroad)	Yes / No; amount in LBP or USD	—	USD
	Debt	Any current debt, and estimated total amount; amount acquired since July 2024	Yes / No; amount in LBP or USD		
	Remittances	Any money received from persons abroad since July 2024, and total amount	Yes / No; count; amount in LBP or USD		
Employment (household roster)	Share of members working	Per member: any work for wage, salary, commission, own-account business, in past 7 days	Yes / No (per member)	—	share (0-1)

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Indicator	Components	Sub-components / items	Possible answers	Weight	Score range
	Total hours worked	Per job (wage, own-account, unpaid): hours worked in past 7 days, summed over members	Hours (continuous)		hours
Social cohesion	Group membership	Number of groups or associations (youth, religious, tontines) a member belongs to	Count (integer)	—	—
	Volunteering	Any household member volunteered in community activities since July 2024	Yes / No		
	Community tension	Perceived tension between community members since July 2024	1 = Not at all; 2 = Very little; 3 = Somewhat; 4 = A lot; -99 = Don't know		
Asset index (baseline)	Ownership of durable goods (first principal component, PCA)	Refrigerator; freezer; cooking burner; oven; microwave; washing machine; dishwasher; dryer; vacuum; flat-screen TV; other TV; DVD player; satellite dish; air conditioner; water heater; computer/laptop; smartphone; other mobile; tablet; internet; water purifier; water pump; medical appliance; motorcycle; ATV; bus; pickup; bicycle; boat; tuk-tuk; car	Owned (1) / not owned (0)	PCA loadings	PCA index

Notes: All monetary amounts are reported in Lebanese pounds or US dollars and converted to US dollars; expenditure and financial amounts are winsorized at the 95th percentile at the item level, and expenditure variables are expressed as monthly household totals (not per capita). Savings, debt, and remittance amounts are additionally transformed using the inverse hyperbolic sine; the savings question was dropped at endline. The family indices (food security, psychological well-being, and trust) are inverse-covariance-weighted (Anderson 2008) z -score indices, whose components are standardized to mean zero and standard deviation one within each survey round before aggregation; components marked “reversed” are reversed so that higher index values denote better outcomes. Codes -999 (“Prefer not to disclose”) and -99 (“Don’t know”) are treated as missing.

Notes: Construction of the outcome variables. Items marked “(reversed)” are recoded so that higher index values denote better outcomes (more trust, greater satisfaction). Codes -999 (“Prefer not to disclose”) and -99 (“Don’t know”) are treated as missing. “n.a.” indicates an unweighted (equal-weight) sum of the listed items, standardized within survey round to mean zero and standard deviation one (z). The FCS and rCSI use the standard WFP weights. Expenditure and financial variables are winsorized at the 95th percentile at the item level before aggregation, converted to US dollars, and expressed as a monthly household total (not per capita); six-month expenses are divided by six to obtain a monthly figure. The asset index reports PCA loadings as “weights” and its score is standardized rather than a bounded sum.

C Additional Tables and Figures

C.1 Main Tables and Figures

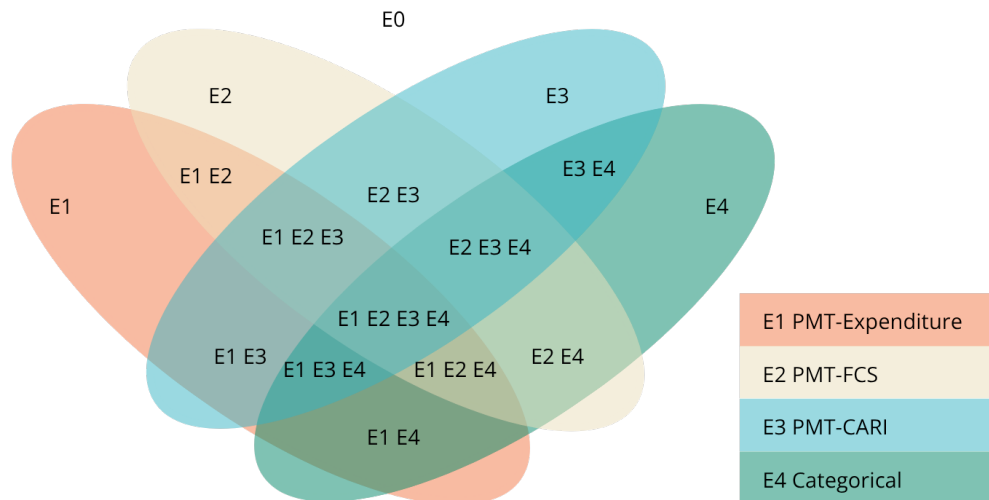


Figure A1: Eligibility spaces defined by the four targeting methods

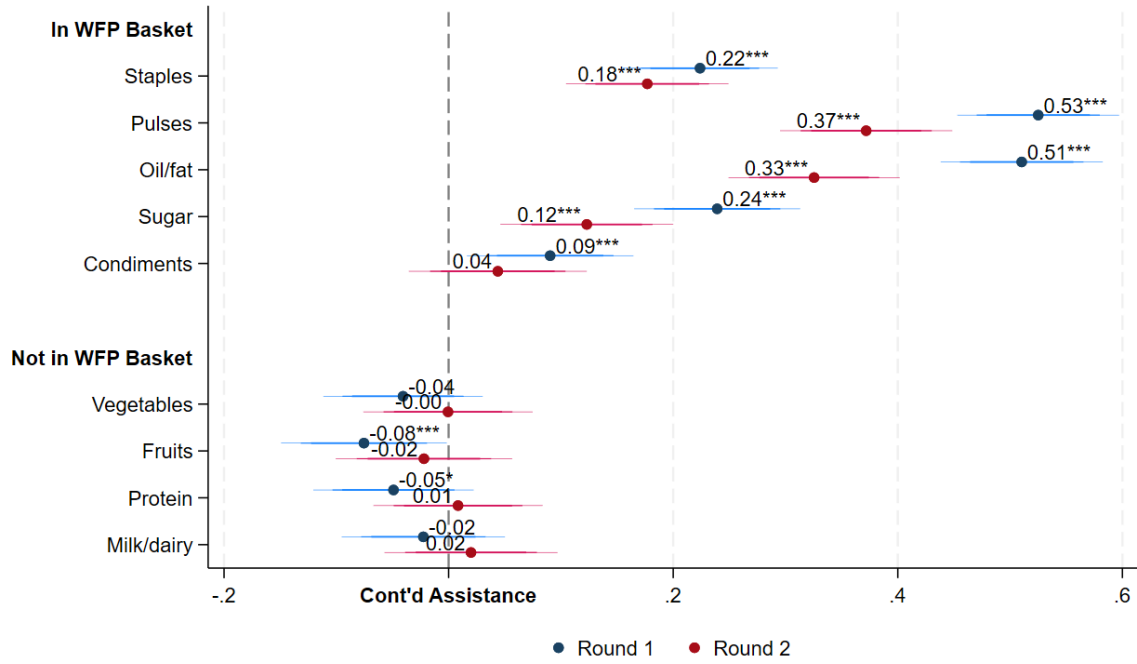
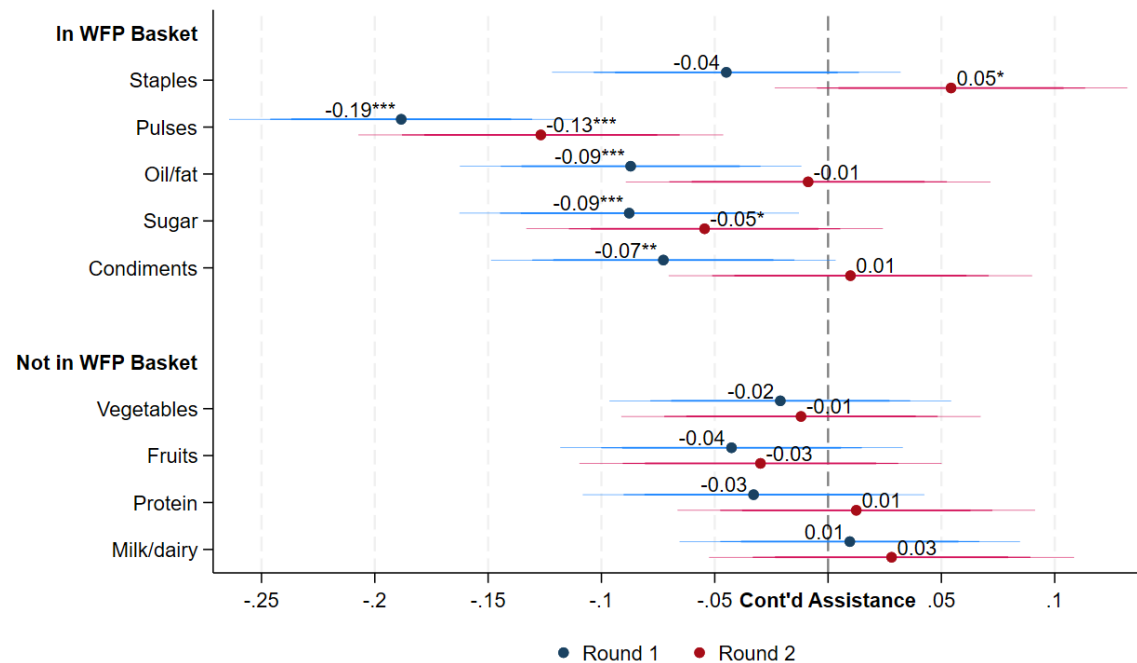


Figure A2: The effect of the end of assistance on food expenditure

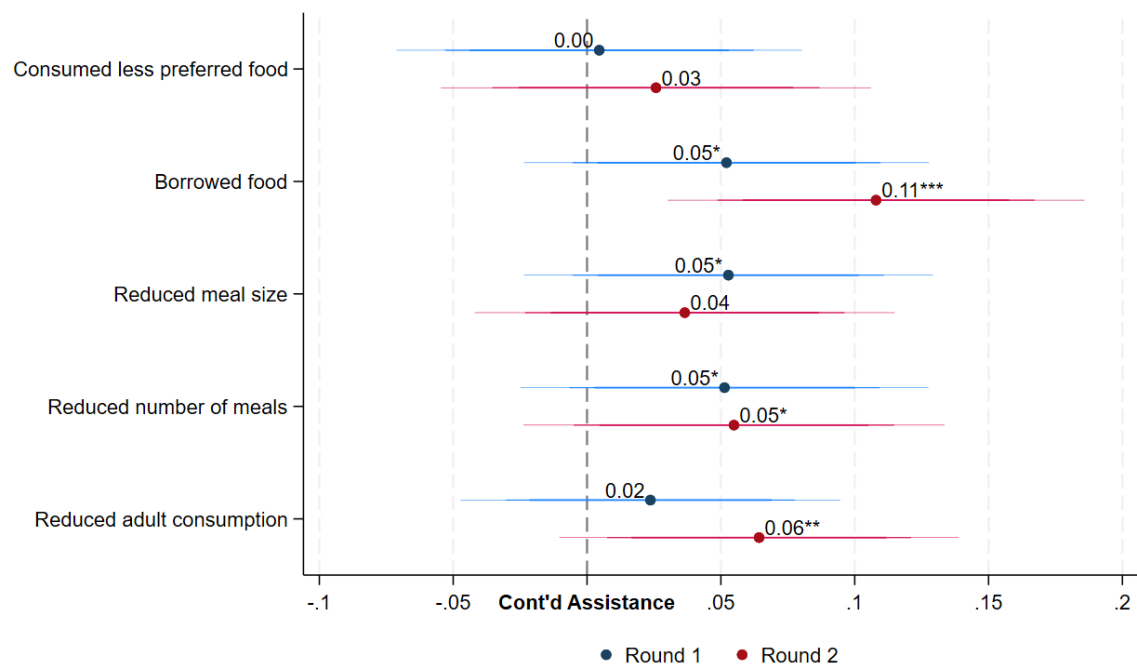
Table A4: Impacts of Targeting on Downstream Outcomes (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	Life Satsf. Index (SD)	Worriedness Index (SD)	WHO-5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A: Round 1							
PMT-FCS	0.15 (0.58)	-0.54 (0.44)	6.79** (2.94)	16.11** (7.88)	-0.02 (0.04)	-0.02 (0.04)	0.37 (0.68)
PMT-CARI	1.07* (0.59)	0.34 (0.44)	2.07 (2.96)	6.12 (7.85)	0.03 (0.04)	-0.05 (0.04)	0.12 (0.68)
Expert	-0.06 (0.59)	0.04 (0.44)	4.40 (2.94)	13.30* (7.90)	0.01 (0.04)	-0.00 (0.04)	-0.06 (0.67)
Mean PMT-Exp.	54.47	16.62	136.64	392.21	-0.00	0.02	23.00
Observations	6,358	6,358	6,358	6,358	6,358	6,357	6,300
p-val(PMT-FCS=PMT-CARI)	0.119	0.049	0.110	0.201	0.238	0.388	0.721
p-val(PMT-FCS=Expert)	0.720	0.193	0.414	0.721	0.531	0.574	0.521
p-val(PMT-CARI=Expert)	0.057	0.505	0.431	0.360	0.574	0.149	0.783
Panel B: Round 2							
PMT-FCS	0.24 (0.58)	0.06 (0.39)	3.01 (2.88)	-2.51 (8.53)	0.00 (0.04)	0.04 (0.04)	-0.37 (0.60)
PMT-CARI	0.23 (0.58)	-0.19 (0.39)	-0.18 (2.84)	-10.75 (8.49)	0.08** (0.04)	0.01 (0.03)	0.71 (0.61)
Expert	0.28 (0.58)	0.60 (0.39)	4.72 (2.89)	-0.06 (8.40)	0.04 (0.04)	0.03 (0.03)	0.06 (0.61)
Mean PMT-Exp.	53.30	14.96	142.28	470.22	-0.03	-0.02	20.54
Observations	6,465	6,465	6,465	6,465	6,465	6,465	6,453
p-val(PMT-FCS=PMT-CARI)	0.989	0.531	0.265	0.335	0.021	0.346	0.071
p-val(PMT-FCS=Expert)	0.940	0.174	0.555	0.772	0.285	0.642	0.477
p-val(PMT-CARI=Expert)	0.930	0.049	0.087	0.204	0.212	0.632	0.288

Notes: PMT-Expenditure targeting serves as the reference group. The table shows the regression results on pre-specified primary outcomes at the household level using survey data. Each column is its own dependent variable. The independent variable of interest is the ITT treatment indicator and includes beneficiaries and non-beneficiaries households. Food Consumption Score (FCS, 0–112) measures dietary diversity and frequency. The reduced Coping Strategy Index (rCSI, 0–56) measures frequency and severity of negative coping strategies. Expenditure includes credit purchases and own production but excludes program transfers; food and total expenditure are winsorized at the 95th percentile. The life satisfaction index is a standardized composite (nutrition, health, finances, community



(a) Food consumption, by food group



(b) Food-related risky coping strategies

Figure A3: The effect of the end of assistance on food consumption and coping strategies

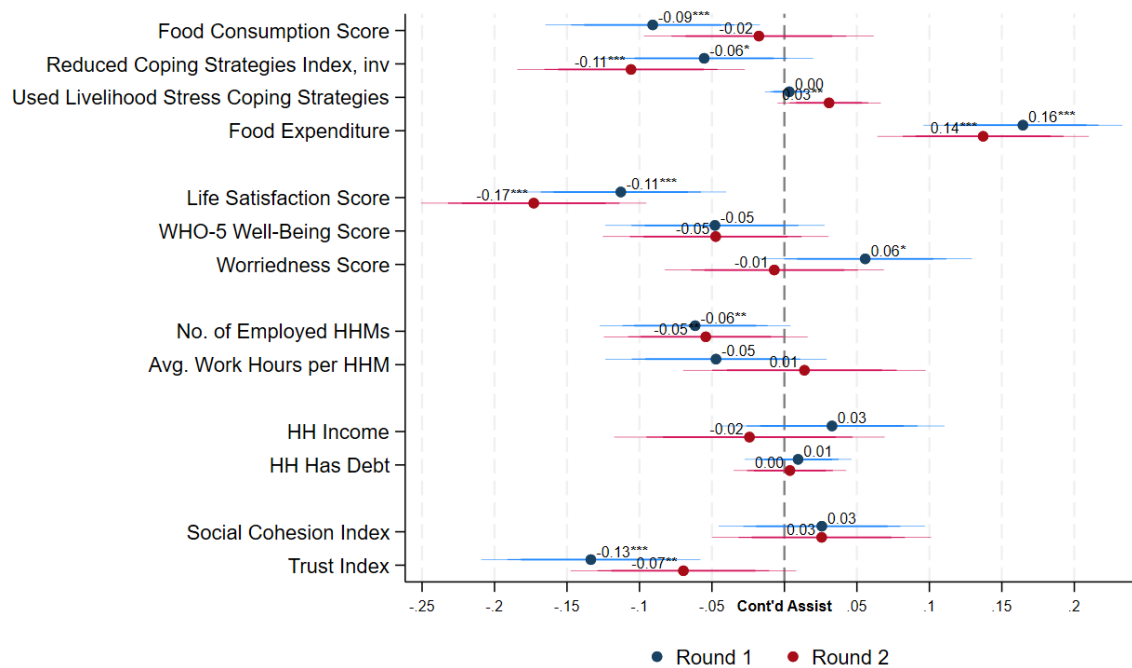


Figure A4: The effect of the end of assistance on main outcomes

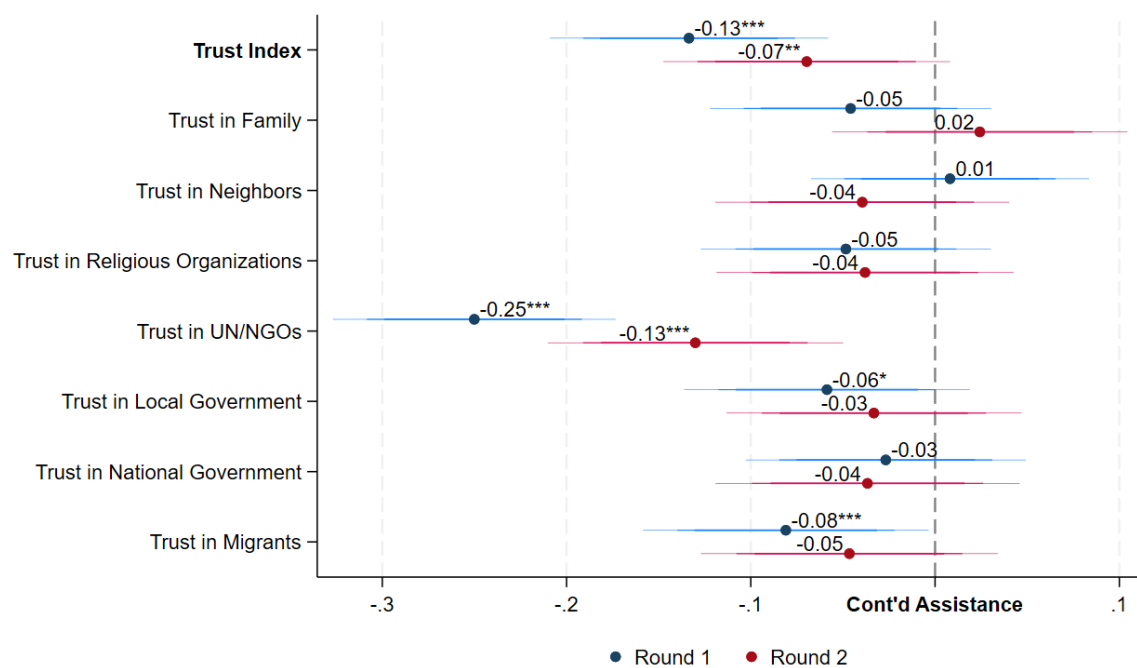


Figure A5: The effect of the end of assistance on trust

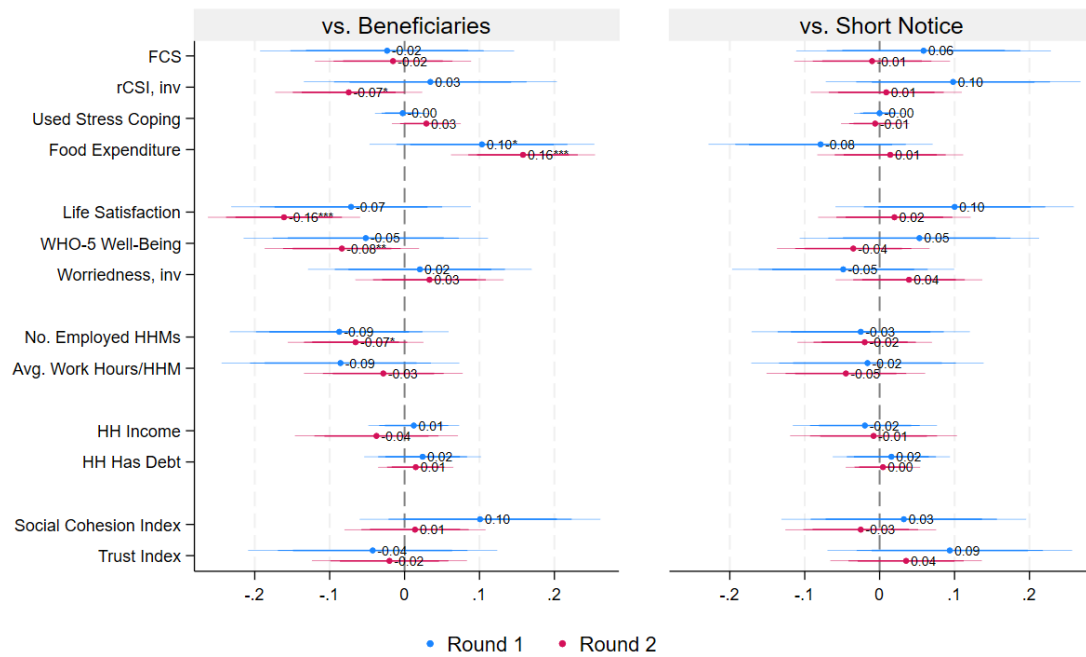


Figure A6: The effect of the long notice window on main outcomes

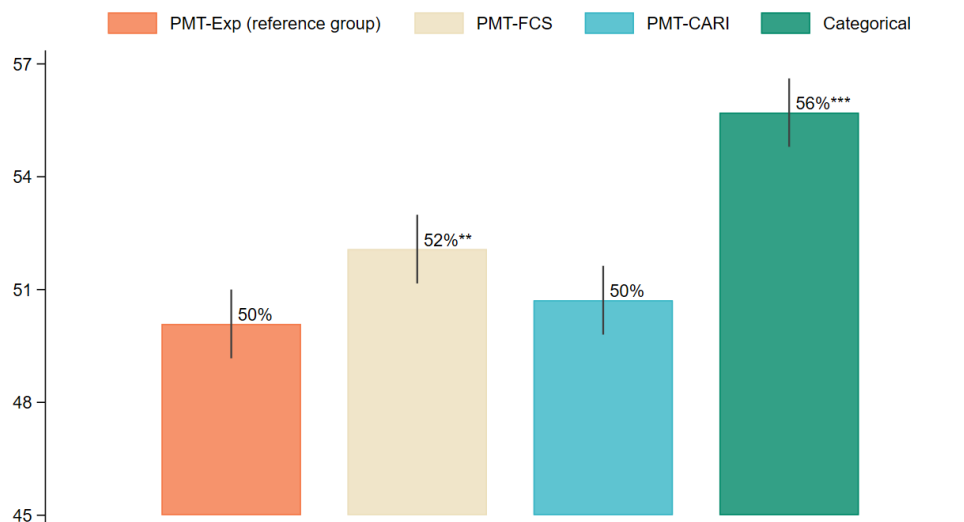


Figure A7: Appeal rates by targeting arm

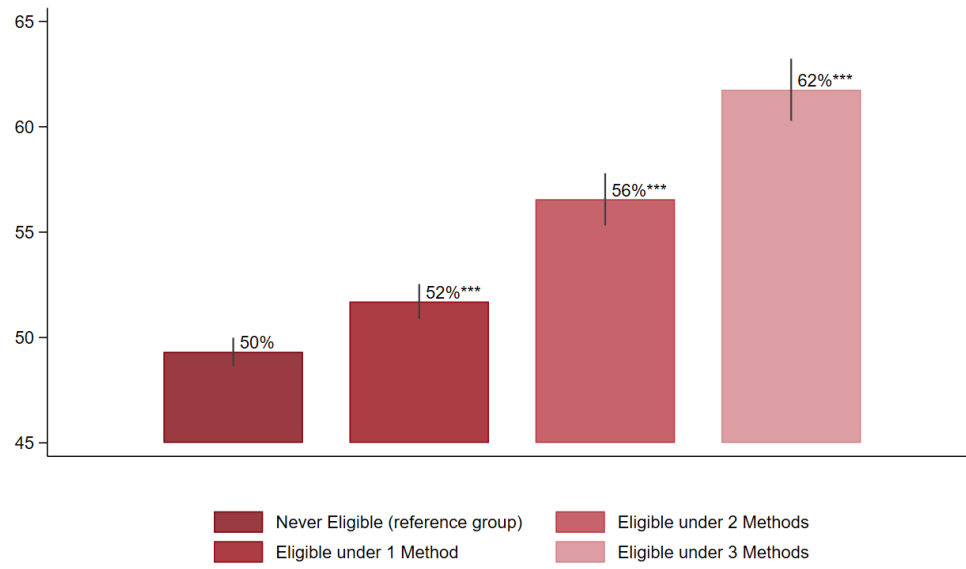


Figure A8: Appeals by number of eligible methods

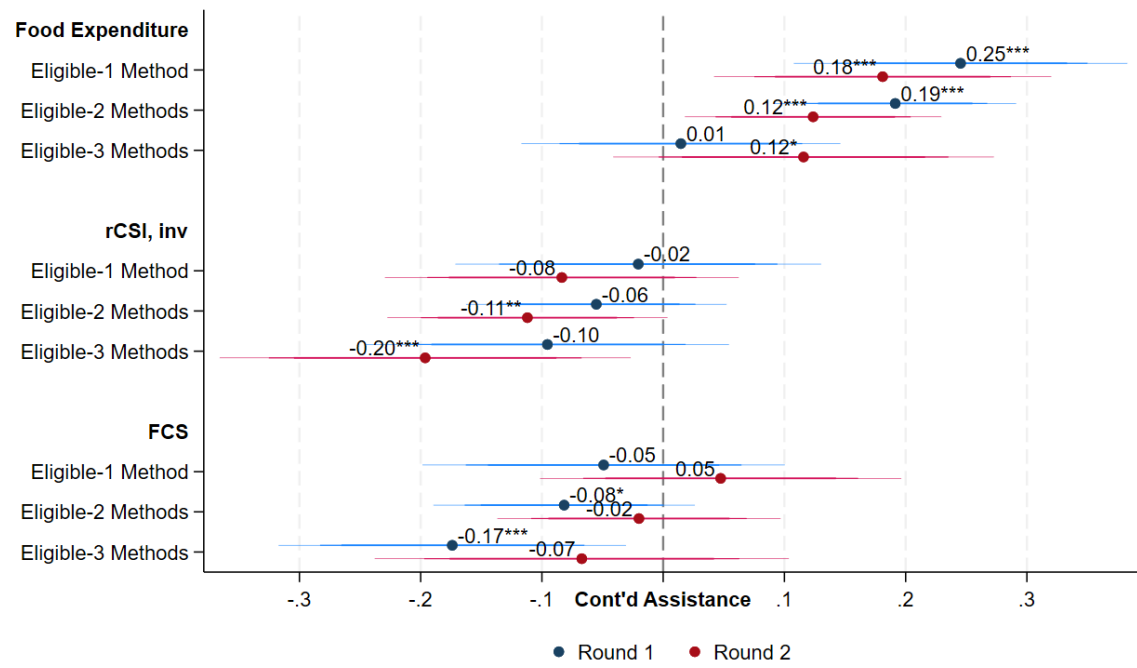


Figure A9: Effects of discontinuation by number of eligible methods

C.2 Setting and Design

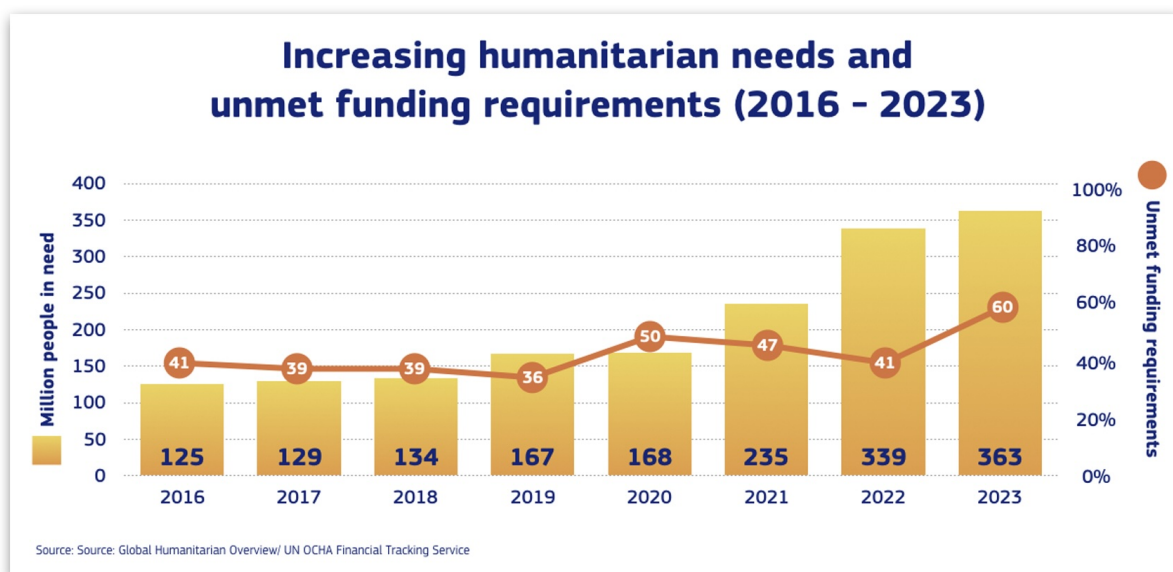


Figure A10: Global humanitarian needs and funding

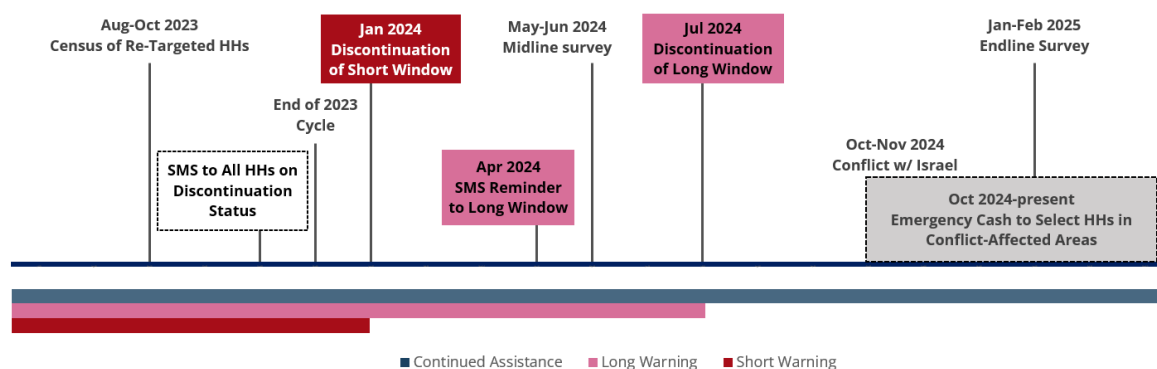


Figure A11: Study timeline

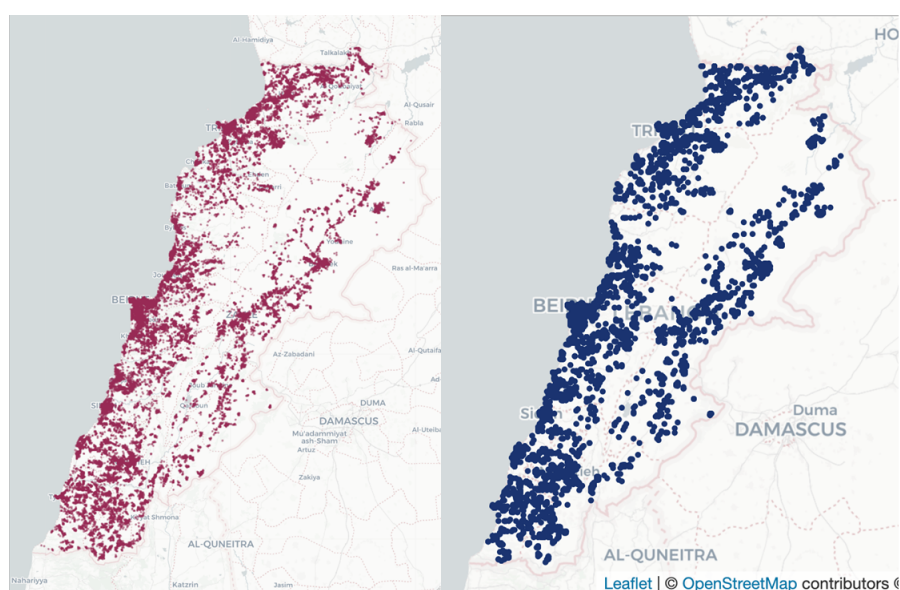


Figure A12: Geographic distribution of programme beneficiaries

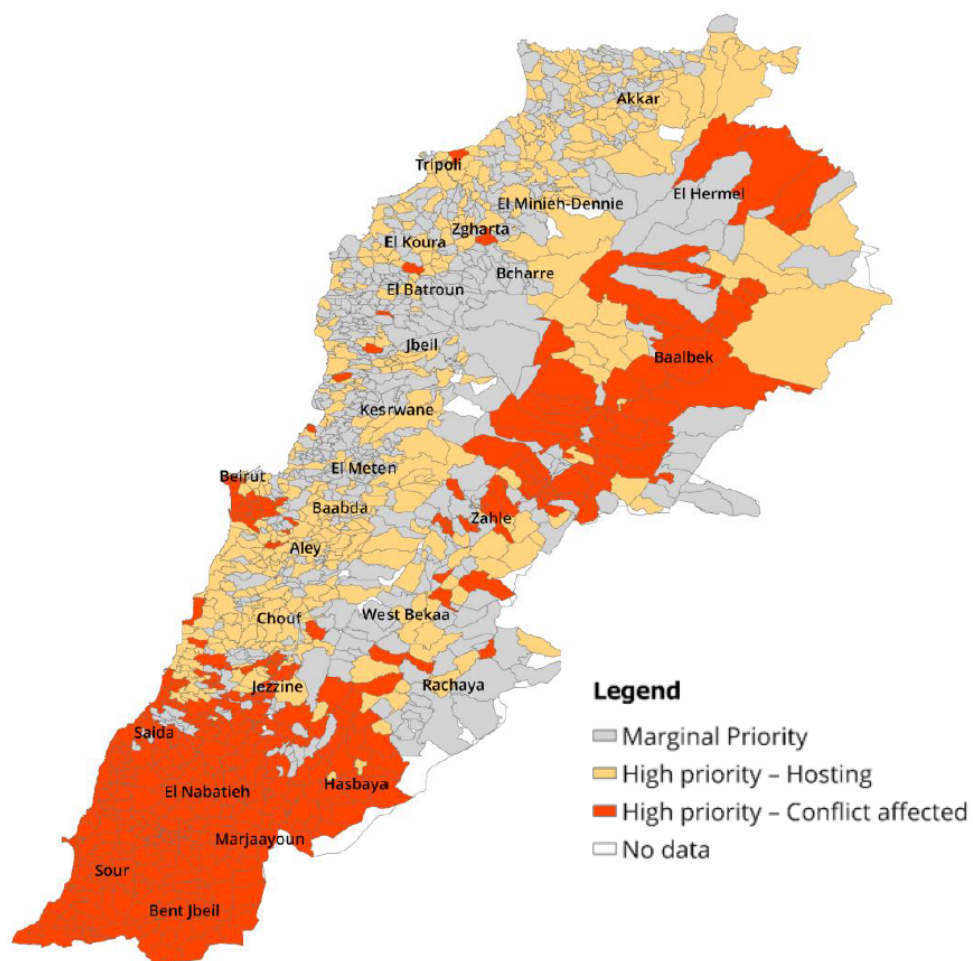


Figure A13: Conflict-based prioritisation of cadastres

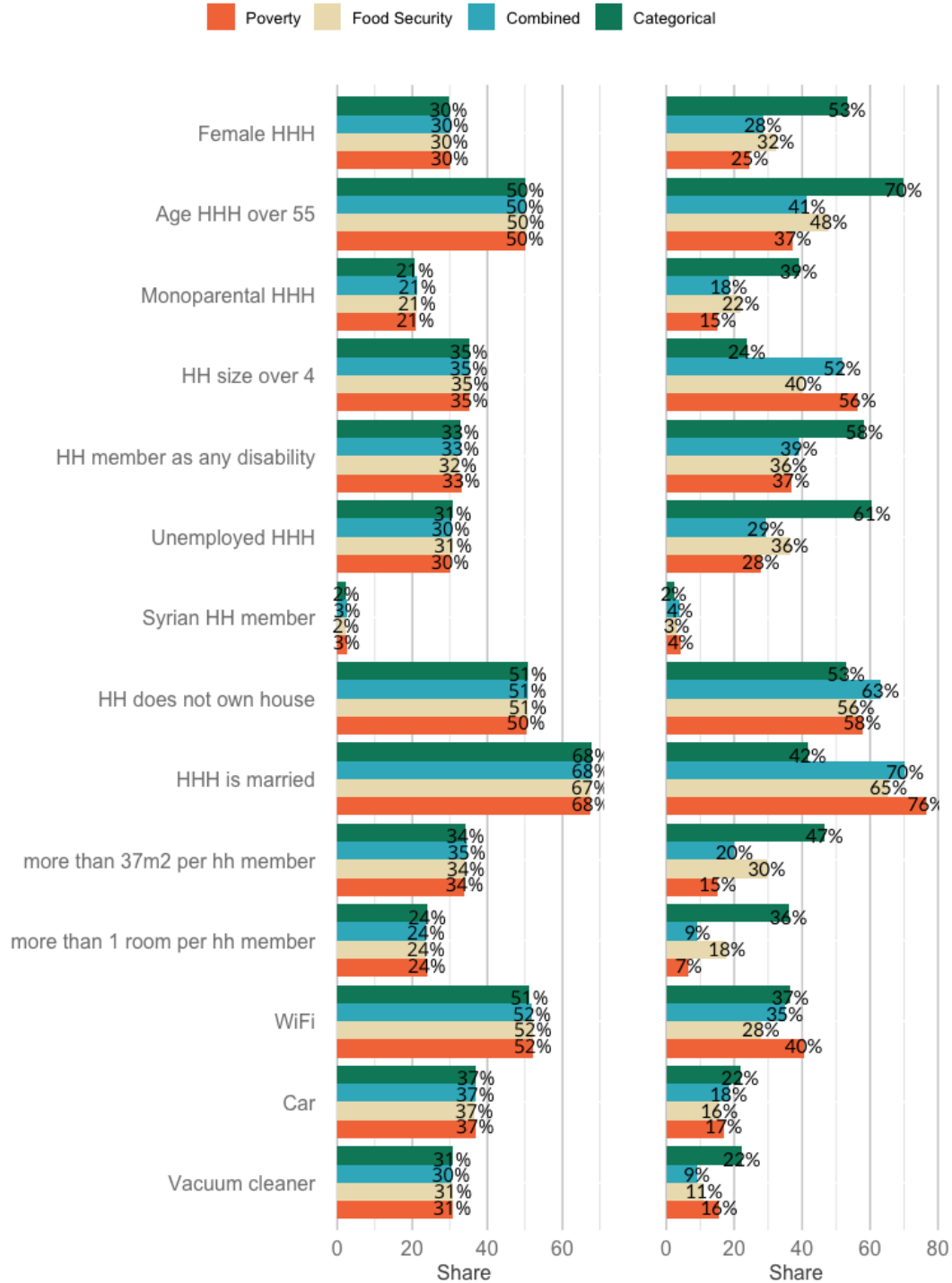


Figure A14: Overlap in household selection across the four targeting methods

Table A5: Allocation of HHs Across Eligibility Spaces and Targeting Methods

Eligibility Space	Total (%)	Random allocation to 4 targeting methods			
		Applied Method T1	Applied Method T2	Applied Method T3	Applied Method T4
Total	58501	14624	14626	14625	14626
Selected for assistance	25160 (43%)	6289	6289	6290	6292
Not selected	33341 (57%)	8335	8337	8335	8334
E0	15295 (26.14%)	3824	3824	3824	3823
E1E2E3E4	7727 (13.21%)	1931	1932	1932	1932
E1	2724 (4.66%)	681	681	681	681
E1E2	706 (1.21%)	176	177	176	177
E1E2E3	7753 (13.25%)	1939	1938	1938	1938
E1E2E4	458 (0.78%)	114	115	114	115
E1E3	2825 (4.83%)	707	706	707	705
E1E3E4	1810 (3.09%)	453	452	452	453
E1E4	1153 (1.97%)	288	289	287	289
E2	1945 (3.32%)	485	486	487	487
E2E3	1390 (2.38%)	347	347	348	348
E2E3E4	2332 (3.99%)	583	583	583	583
E2E4	2845 (4.86%)	712	711	711	711
E3	703 (1.20%)	176	176	176	175
E3E4	616 (1.05%)	154	154	154	154
E4	8219 (14.05%)	2054	2055	2055	2055

Notes: Counts and percentages by eligibility space. Green numbers indicate selected; red numbers indicate not selected.

C.3 Balance

Table A6: Baseline characteristics by study arm

Variable	PMT-Exp Mean/(SD) (1)	PMT-FCS Mean/(SD) (2)	PMT-CARI Mean/(SD) (3)	Expert Mean/(SD) (4)	Across all groups F-stat/P-value (5)
Female headed household	0.34 (0.47)	0.35 (0.48)	0.31 (0.46)	0.32 (0.47)	2.06 0.10
Household head divorced/separated or widowed	0.23 (0.42)	0.24 (0.43)	0.22 (0.41)	0.21 (0.41)	1.10 0.35
Senior household head (>64)	0.26 (0.44)	0.27 (0.44)	0.27 (0.44)	0.27 (0.45)	0.12 0.95
HH size	3.81 (1.79)	3.83 (1.81)	3.77 (1.80)	3.81 (1.82)	0.28 0.84
HH in Beirut	0.06 (0.24)	0.07 (0.25)	0.05 (0.22)	0.05 (0.22)	1.11 0.34
Dependency ratio	0.35 (0.30)	0.34 (0.30)	0.35 (0.31)	0.35 (0.30)	0.40 0.75
School-age children	0.88 (1.18)	0.87 (1.21)	0.86 (1.16)	0.88 (1.20)	0.21 0.89
Household head with primary education or lower	0.60 (0.49)	0.61 (0.49)	0.60 (0.49)	0.59 (0.49)	0.65 0.58
Household head suffers from any disability	0.21 (0.40)	0.19 (0.39)	0.21 (0.41)	0.20 (0.40)	1.00 0.39
No members employed	0.33 (0.47)	0.34 (0.47)	0.34 (0.47)	0.34 (0.47)	0.36 0.78
HH owns smartphone	0.80 (0.40)	0.81 (0.40)	0.80 (0.40)	0.82 (0.39)	1.02 0.38
HH owns car	0.28 (0.45)	0.29 (0.46)	0.30 (0.46)	0.31 (0.46)	1.32 0.27
Piped water	0.23 (0.42)	0.25 (0.43)	0.25 (0.43)	0.24 (0.43)	1.05 0.37
Gas for cooking	0.75 (0.43)	0.73 (0.45)	0.72 (0.45)	0.72 (0.45)	1.89 0.13
Wealth index - PCA	-0.20 (0.89)	-0.17 (0.87)	-0.16 (0.87)	-0.17 (0.86)	0.53 0.66
Score: PMT-CARI (predicted pr.)	0.40 (0.13)	0.40 (0.14)	0.40 (0.14)	0.40 (0.13)	0.16 0.92
Score: PMT-Exp (predicted in ln)	15.19 (0.38)	15.19 (0.37)	15.19 (0.39)	15.19 (0.38)	0.07 0.98
Score: PMT-FCS (predicted)	54.26 (7.87)	54.52 (7.65)	54.42 (8.05)	54.13 (7.81)	0.82 0.48
Score: Expert	39.54 (43.74)	39.52 (43.54)	39.21 (44.55)	39.19 (43.84)	0.03 0.99
Number of observations	1600	1600	1600	1600	6400

Notes: The table reports baseline means, with standard deviations in parentheses. Column (5) reports the F-statistic and p-value of a test of joint equality across the four targeting arms. Robust standard errors; tests are two sided. The table reports a subset of the variables available from the baseline census. When considering the full set of baseline variables, there are 139 possible comparisons. Among these, for the comparison across the four targeting arms, there are 14 instances (10.1%) in which differences are statistically significant at the 10% level (p-value < 0.1), 10 instances (7.2%) at the 5% level (p-value < 0.05), and 4 instances (2.9%) at the 1% level (p-value < 0.01). Statistical significance is denoted by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A7: Attrition and survey completion rates

		Non-response rate at	
	N	Round 1	Round 2
		(1)	(2)
Panel A: Targeting arms			
PMT-Exp	1600	0.11 (0.31)	0.07 (0.25)
PMT-FCS	1600	0.11 (0.32)	0.06 (0.24)
PMT-CARI	1600	0.10 (0.30)	0.07 (0.25)
Expert	1600	0.11 (0.31)	0.06 (0.24)
F-test		0.41	0.30
Panel B: Continued vs. discontinued (short notice)			
Discontinued (short notice)	2254	0.01 (0.07)	0.06 (0.24)
Continued	3200	0.01 (0.08)	0.06 (0.24)
F-test		0.76	0.06
Panel C: Long notice vs. all others			
Long notice	946	0.69 (0.46)	0.07 (0.25)
Others	5454	0.01 (0.08)	0.06 (0.24)
F-test		2058.01***	1.00

Notes: The table reports non-response (attrition) rates, with standard deviations in parentheses, for the first (Round 1, May–June 2024) and second (Round 2, January–February 2025) rounds of data collection. Panel A compares the four targeting arms; Panel B compares continued beneficiaries with households discontinued under the standard short-notice protocol, excluding long-notice households; Panel C compares long-notice households with all others. F-tests assess balance in attrition across groups within each panel. The high Round 1 non-response among long-notice households in Panel C reflects the sampling error described in the text, which was corrected before Round 2. Statistical significance is denoted by: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Balance: targeting arms, full population

Variable	PMT-Exp Mean/(SD) (1)	PMT-FCS Mean/(SD) (2)	PMT-CARI Mean/(SD) (3)	Expert Mean/(SD) (4)	Across all groups F-stat/P-value (5)
Female headed household	0.30 (0.46)	0.30 (0.46)	0.30 (0.46)	0.30 (0.46)	0.16 0.92
Household head divorced/separated or widowed	0.21 (0.41)	0.21 (0.41)	0.21 (0.41)	0.21 (0.41)	0.88 0.45
Senior household head (>64)	0.25 (0.43)	0.25 (0.43)	0.25 (0.43)	0.25 (0.43)	0.09 0.97
HH size	3.86 (1.83)	3.85 (1.81)	3.86 (1.82)	3.85 (1.83)	0.36 0.78
HH in Beirut	0.05 (0.22)	0.05 (0.22)	0.05 (0.22)	0.05 (0.22)	0.34 0.80
Dependency ratio	0.35 (0.30)	0.35 (0.30)	0.35 (0.30)	0.35 (0.30)	0.20 0.90
School-age children	0.93 (1.23)	0.93 (1.23)	0.94 (1.23)	0.92 (1.23)	0.42 0.74
Household head with primary education or lower	0.56 (0.50)	0.56 (0.50)	0.57 (0.50)	0.56 (0.50)	1.20 0.31
Household head suffers from any disability	0.19 (0.39)	0.18 (0.38)	0.19 (0.39)	0.18 (0.38)	1.09 0.35
No members employed	0.30 (0.46)	0.31 (0.46)	0.30 (0.46)	0.31 (0.46)	0.82 0.48
HH owns smartphone	0.82 (0.38)	0.83 (0.38)	0.83 (0.38)	0.82 (0.38)	0.97 0.41
HH owns car	0.37 (0.48)	0.37 (0.48)	0.37 (0.48)	0.37 (0.48)	0.22 0.88
Piped water	0.23 (0.42)	0.22 (0.42)	0.23 (0.42)	0.22 (0.42)	0.41 0.74
Gas for cooking	0.74 (0.44)	0.73 (0.44)	0.73 (0.44)	0.74 (0.44)	0.20 0.90
Wealth index - PCA	-0.00 (1.01)	0.01 (1.00)	0.00 (1.00)	-0.01 (1.00)	0.90 0.44
Score: PMT-CARI (predicted pr.)	0.37 (0.19)	0.37 (0.19)	0.37 (0.19)	0.37 (0.19)	0.32 0.81
Score: PMT-Exp (predicted in ln)	15.25 (0.49)	15.25 (0.49)	15.25 (0.49)	15.25 (0.49)	0.21 0.89
Score: PMT-FCS (predicted)	55.42 (10.03)	55.46 (10.00)	55.37 (9.96)	55.35 (9.87)	0.35 0.79
Score: Expert	33.80 (44.56)	33.40 (44.41)	33.77 (44.75)	34.04 (44.52)	0.51 0.68
Number of observations	14624	14626	14625	14626	58501

Notes: Baseline means with standard deviations in parentheses, across the four targeting arms in the full beneficiary census. Column (5) reports the F-statistic and p-value of a test of joint equality. Robust standard errors; tests are two sided.

Table A9: Balance: households selected for continued assistance, population

Variable	PMT-Exp Mean/(SD) (1)	PMT-FCS Mean/(SD) (2)	PMT-CARI Mean/(SD) (3)	Expert Mean/(SD) (4)	Across all groups F-stat/P-value (5)
Female headed household	0.25 (0.43)	0.32 (0.47)	0.28 (0.45)	0.53 (0.50)	444.12*** 0.00
Household head divorced/separated or widowed	0.15 (0.36)	0.22 (0.41)	0.18 (0.39)	0.39 (0.49)	347.53*** 0.00
Senior household head (>64)	0.17 (0.38)	0.25 (0.43)	0.19 (0.40)	0.47 (0.50)	523.55*** 0.00
HH size	4.74 (1.81)	4.02 (2.03)	4.52 (1.97)	3.19 (1.98)	799.02*** 0.00
HH in Beirut	0.02 (0.15)	0.07 (0.26)	0.03 (0.18)	0.04 (0.21)	57.36*** 0.00
Dependency ratio	0.36 (0.26)	0.38 (0.30)	0.37 (0.28)	0.43 (0.36)	59.13*** 0.00
School-age children	1.44 (1.41)	1.11 (1.40)	1.37 (1.44)	0.52 (1.05)	800.94*** 0.00
Household head with primary education or lower	0.62 (0.48)	0.72 (0.45)	0.67 (0.47)	0.67 (0.47)	47.40*** 0.00
Household head suffers from any disability	0.19 (0.40)	0.21 (0.41)	0.22 (0.41)	0.38 (0.49)	230.37*** 0.00
No members employed	0.28 (0.45)	0.36 (0.48)	0.29 (0.46)	0.61 (0.49)	631.36*** 0.00
HH owns smartphone	0.76 (0.42)	0.75 (0.43)	0.77 (0.42)	0.74 (0.44)	7.72*** 0.00
HH owns car	0.17 (0.37)	0.16 (0.36)	0.18 (0.39)	0.22 (0.41)	28.80*** 0.00
Piped water	0.25 (0.43)	0.22 (0.41)	0.24 (0.43)	0.26 (0.44)	8.88*** 0.00
Gas for cooking	0.66 (0.47)	0.67 (0.47)	0.65 (0.48)	0.71 (0.45)	22.55*** 0.00
Wealth index - PCA	-0.51 (0.89)	-0.65 (0.85)	-0.65 (0.85)	-0.41 (0.98)	99.19*** 0.00
Score: PMT-CARI (predicted pr.)	0.52 (0.15)	0.51 (0.15)	0.55 (0.12)	0.41 (0.19)	861.16*** 0.00
Score: PMT-Exp (predicted in ln)	14.80 (0.29)	14.99 (0.43)	14.88 (0.36)	15.22 (0.47)	1,283.32*** 0.00
Score: PMT-FCS (predicted)	49.91 (9.73)	46.65 (7.56)	48.19 (9.01)	53.28 (9.18)	690.93*** 0.00
Score: Expert	36.48 (41.26)	45.96 (45.38)	42.90 (44.00)	75.87 (32.57)	1,522.59*** 0.00
Number of observations	6289	6289	6290	6292	25160

Notes: As in Table A8, restricted to households selected for continued assistance under each targeting method.

Table A10: Balance: households selected for continued assistance, survey sample

Variable	PMT-Exp Mean/(SD) (1)	PMT-FCS Mean/(SD) (2)	PMT-CARI Mean/(SD) (3)	Expert Mean/(SD) (4)	Across all groups F-stat/P-value (5)
Female headed household	0.25 (0.43)	0.32 (0.47)	0.28 (0.45)	0.53 (0.50)	444.12*** 0.00
Household head divorced/separated or widowed	0.15 (0.36)	0.22 (0.41)	0.18 (0.39)	0.39 (0.49)	347.53*** 0.00
Senior household head (>64)	0.17 (0.38)	0.25 (0.43)	0.19 (0.40)	0.47 (0.50)	523.55*** 0.00
HH size	4.74 (1.81)	4.02 (2.03)	4.52 (1.97)	3.19 (1.98)	799.02*** 0.00
HH in Beirut	0.02 (0.15)	0.07 (0.26)	0.03 (0.18)	0.04 (0.21)	57.36*** 0.00
Dependency ratio	0.36 (0.26)	0.38 (0.30)	0.37 (0.28)	0.43 (0.36)	59.13*** 0.00
School-age children	1.44 (1.41)	1.11 (1.40)	1.37 (1.44)	0.52 (1.05)	800.94*** 0.00
Household head with primary education or lower	0.62 (0.48)	0.72 (0.45)	0.67 (0.47)	0.67 (0.47)	47.40*** 0.00
Household head suffers from any disability	0.19 (0.40)	0.21 (0.41)	0.22 (0.41)	0.38 (0.49)	230.37*** 0.00
No members employed	0.28 (0.45)	0.36 (0.48)	0.29 (0.46)	0.61 (0.49)	631.36*** 0.00
HH owns smartphone	0.76 (0.42)	0.75 (0.43)	0.77 (0.42)	0.74 (0.44)	7.72*** 0.00
HH owns car	0.17 (0.37)	0.16 (0.36)	0.18 (0.39)	0.22 (0.41)	28.80*** 0.00
Piped water	0.25 (0.43)	0.22 (0.41)	0.24 (0.43)	0.26 (0.44)	8.88*** 0.00
Gas for cooking	0.66 (0.47)	0.67 (0.47)	0.65 (0.48)	0.71 (0.45)	22.55*** 0.00
Wealth index - PCA	-0.51 (0.89)	-0.65 (0.85)	-0.65 (0.85)	-0.41 (0.98)	99.19*** 0.00
Score: PMT-CARI (predicted pr.)	0.52 (0.15)	0.51 (0.15)	0.55 (0.12)	0.41 (0.19)	861.16*** 0.00
Score: PMT-Exp (predicted in ln)	14.80 (0.29)	14.99 (0.43)	14.88 (0.36)	15.22 (0.47)	1,283.32*** 0.00
Score: PMT-FCS (predicted)	49.91 (9.73)	46.65 (7.56)	48.19 (9.01)	53.28 (9.18)	690.93*** 0.00
Score: Expert	36.48 (41.26)	45.96 (45.38)	42.90 (44.00)	75.87 (32.57)	1,522.59*** 0.00
Number of observations	6289	6289	6290	6292	25160

Notes: As in Table A9, for the stratified survey sample.

Table A11: Programme beneficiaries versus the nationally representative LHS sample

	LHS	WFP Census	Across all groups
Female-headed HH	0.17 (0.38)	0.30 (0.46)	349.66*** 0.00
Monoparental HH	0.15 (0.36)	0.21 (0.41)	93.31*** 0.00
Senior HHH (>64)	0.28 (0.45)	0.25 (0.43)	16.24*** 0.00
HH size	3.77 (1.80)	3.86 (1.82)	7.70*** 0.01
HH in Beirut	0.03 (0.18)	0.05 (0.22)	26.45*** 0.00
Dependency ratio	0.40 (0.31)	0.35 (0.30)	108.60*** 0.00
School-age children	0.96 (1.30)	0.93 (1.23)	1.64 0.20
HHH education primary or lower	0.42 (0.49)	0.56 (0.50)	253.01*** 0.00
HHH with disability	0.08 (0.28)	0.18 (0.39)	389.92*** 0.00
No members employed	0.29 (0.45)	0.31 (0.46)	2.85* 0.09
HH owns smartphone	0.94 (0.24)	0.83 (0.38)	627.29*** 0.00
Piped water	0.57 (0.49)	0.22 (0.42)	1,673.07*** 0.00
Gas for cooking	0.96 (0.21)	0.73 (0.44)	3,156.88*** 0.00
Number of observations	3511	58501	62012

Notes: Comparison of the WFP beneficiary census (CP3) with the Lebanese Household Survey, the nationally representative survey used to train the proxy means tests.

Table A12: Balance: continued versus discontinued households, population

Variable	Discontinued Mean/(SD) (1)	Continued Mean/(SD) (2)	Across all groups F-stat/P-value (3)
Female headed household	0.26 (0.44)	0.35 (0.48)	0.31 0.58
Household head divorced/separated or widowed	0.19 (0.39)	0.24 (0.42)	0.91 0.34
Senior household head (>64)	0.23 (0.42)	0.27 (0.44)	0.75 0.39
HH size	3.66 (1.62)	4.12 (2.04)	0.00 0.97
HH in Beirut	0.06 (0.23)	0.04 (0.20)	0.46 0.50
Dependency ratio	0.32 (0.30)	0.38 (0.30)	0.00 0.94
School-age children	0.79 (1.09)	1.11 (1.38)	0.01 0.93
Household head with primary education or lower	0.48 (0.50)	0.67 (0.47)	0.21 0.65
Household head suffers from any disability	0.13 (0.34)	0.25 (0.43)	3.74* 0.05
No members employed	0.24 (0.43)	0.39 (0.49)	1.26 0.26
HH owns smartphone	0.88 (0.33)	0.75 (0.43)	0.02 0.89
HH owns car	0.51 (0.50)	0.18 (0.39)	1.30 0.25
Piped water	0.21 (0.41)	0.24 (0.43)	0.09 0.76
Gas for cooking	0.78 (0.41)	0.67 (0.47)	0.03 0.87
Wealth index - PCA	0.42 (0.85)	-0.56 (0.90)	0.05 0.83
Score: PMT-CARI (predicted pr.)	0.28 (0.16)	0.50 (0.16)	0.00 0.95
Score: PMT-Exp (predicted in ln)	15.47 (0.42)	14.97 (0.43)	0.02 0.89
Score: PMT-FCS (predicted)	59.84 (7.99)	49.51 (9.24)	1.04 0.31
Score: Expert	21.26 (40.90)	50.31 (43.80)	0.23 0.63
Number of observations	33341	25160	58501

Notes: Baseline means by discontinuation status, conditional on eligibility space, in the full population.

Table A13: Balance: long-notice households, population

Variable	Long Notice Mean/(SD) (1)	Discontinued Mean/(SD) (2)	Continued Mean/(SD) (3)	Across all groups F-stat/P-value (4)
Female headed household	0.30 (0.46)	0.26 (0.44)	0.35 (0.48)	0.17 0.85
Household head divorced/separated or widowed	0.22 (0.41)	0.19 (0.39)	0.24 (0.42)	0.49 0.61
Senior household head (>64)	0.25 (0.44)	0.23 (0.42)	0.27 (0.44)	0.38 0.68
HH size	3.81 (1.76)	3.66 (1.61)	4.12 (2.04)	0.17 0.84
HH in Beirut	0.06 (0.25)	0.06 (0.23)	0.04 (0.20)	0.44 0.64
Dependency ratio	0.34 (0.31)	0.32 (0.30)	0.38 (0.30)	1.42 0.24
School-age children	0.83 (1.12)	0.79 (1.09)	1.11 (1.38)	0.79 0.46
Household head with primary education or lower	0.59 (0.49)	0.47 (0.50)	0.67 (0.47)	0.67 0.51
Household head suffers from any disability	0.19 (0.40)	0.13 (0.34)	0.25 (0.43)	2.14 0.12
No members employed	0.30 (0.46)	0.24 (0.43)	0.39 (0.49)	0.64 0.53
HH owns smartphone	0.82 (0.38)	0.88 (0.32)	0.75 (0.43)	0.11 0.89
HH owns car	0.32 (0.47)	0.51 (0.50)	0.18 (0.39)	1.30 0.27
Piped water	0.23 (0.42)	0.21 (0.41)	0.24 (0.43)	0.06 0.95
Gas for cooking	0.74 (0.44)	0.78 (0.41)	0.67 (0.47)	0.05 0.95
Wealth index - PCA	-0.06 (0.77)	0.43 (0.85)	-0.56 (0.90)	0.39 0.68
Score: PMT-CARI (predicted pr.)	0.38 (0.11)	0.28 (0.16)	0.50 (0.16)	0.05 0.95
Score: PMT-Exp (predicted in ln)	15.23 (0.34)	15.47 (0.42)	14.97 (0.43)	0.67 0.51
Score: PMT-FCS (predicted)	55.43 (6.74)	59.98 (7.99)	49.51 (9.24)	0.76 0.47
Score: Expert	34.47 (43.12)	20.86 (40.77)	50.31 (43.80)	0.12 0.89
Number of observations	980	32361	25160	58501

Notes: Baseline means for households assigned to the long-notice arm versus other discontinued households.

Table A14: Balance: continued versus discontinued households, survey sample

Variable	Discontinued Mean/(SD) (1)	Continued Mean/(SD) (2)	Across all groups F-stat/P-value (3)
Female headed household	0.29 (0.45)	0.36 (0.48)	1.62 0.20
Household head divorced/separated or widowed	0.20 (0.40)	0.24 (0.43)	4.62** 0.03
Senior household head (>64)	0.23 (0.42)	0.30 (0.46)	1.75 0.19
HH size	3.76 (1.67)	3.85 (1.93)	0.01 0.92
HH in Beirut	0.06 (0.24)	0.06 (0.23)	2.81* 0.09
Dependency ratio	0.33 (0.30)	0.37 (0.31)	0.12 0.73
School-age children	0.84 (1.11)	0.90 (1.25)	0.59 0.44
Household head with primary education or lower	0.55 (0.50)	0.65 (0.48)	0.03 0.87
Household head suffers from any disability	0.16 (0.37)	0.24 (0.43)	0.12 0.73
No members employed	0.28 (0.45)	0.40 (0.49)	0.12 0.73
HH owns smartphone	0.84 (0.36)	0.77 (0.42)	0.01 0.94
HH owns car	0.37 (0.48)	0.22 (0.41)	0.79 0.38
Piped water	0.23 (0.42)	0.26 (0.44)	0.26 0.61
Gas for cooking	0.76 (0.43)	0.70 (0.46)	0.12 0.73
Wealth index - PCA	0.06 (0.82)	-0.41 (0.85)	0.06 0.81
Score: PMT-CARI (predicted pr.)	0.36 (0.13)	0.45 (0.13)	0.05 0.83
Score: PMT-Exp (predicted in ln)	15.29 (0.36)	15.08 (0.37)	0.09 0.76
Score: PMT-FCS (predicted)	56.40 (7.22)	52.21 (7.85)	0.02 0.89
Score: Expert	30.11 (41.83)	48.89 (43.55)	0.57 0.45
Number of observations	3875	3524	7399

Notes: As in Table A12, for the stratified survey sample. The table reports a subset of the variables available from the baseline census. When considering the full set of baseline variables, there are 77 possible comparisons. For the comparison between continued and discontinued assistance, there are 9 instances (11.7%) that are statistically significant at the 10% level (p-value < 0.1), 6 instances (7.8%) at the 5% level (p-value < 0.05), and 1 instances (1.3%) at the 1% level (p-value < 0.01).

Table A15: Balance: long-notice households, survey sample

Variable	Long Notice Mean/(SD) (1)	Discontinued Mean/(SD) (2)	Continued Mean/(SD) (3)	Across all groups F-stat/P-value (4)
Female headed household	0.30 (0.46)	0.29 (0.45)	0.36 (0.48)	0.91 0.40
Household head divorced/separated or widowed	0.21 (0.41)	0.20 (0.40)	0.24 (0.43)	2.33* 0.10
Senior household head (>64)	0.26 (0.44)	0.23 (0.42)	0.30 (0.46)	1.21 0.30
HH size	3.82 (1.76)	3.74 (1.64)	3.85 (1.93)	0.77 0.46
HH in Beirut	0.07 (0.25)	0.06 (0.23)	0.06 (0.23)	2.34* 0.10
Dependency ratio	0.33 (0.31)	0.33 (0.30)	0.37 (0.31)	0.98 0.37
School-age children	0.83 (1.12)	0.84 (1.10)	0.90 (1.25)	0.52 0.59
Household head with primary education or lower	0.59 (0.49)	0.54 (0.50)	0.65 (0.48)	0.55 0.58
Household head suffers from any disability	0.19 (0.39)	0.16 (0.36)	0.24 (0.43)	0.38 0.68
No members employed	0.30 (0.46)	0.27 (0.44)	0.40 (0.49)	0.07 0.94
HH owns smartphone	0.83 (0.38)	0.85 (0.36)	0.77 (0.42)	0.03 0.97
HH owns car	0.31 (0.46)	0.39 (0.49)	0.22 (0.41)	0.84 0.43
Piped water	0.23 (0.42)	0.22 (0.42)	0.26 (0.44)	0.19 0.83
Gas for cooking	0.74 (0.44)	0.76 (0.42)	0.70 (0.46)	0.16 0.85
Wealth index - PCA	-0.05 (0.77)	0.09 (0.84)	-0.41 (0.85)	0.03 0.97
Score: PMT-CARI (predicted pr.)	0.38 (0.11)	0.35 (0.13)	0.45 (0.13)	0.07 0.93
Score: PMT-Exp (predicted in ln)	15.23 (0.33)	15.31 (0.37)	15.08 (0.37)	1.91 0.15
Score: PMT-FCS (predicted)	55.45 (6.82)	56.71 (7.32)	52.21 (7.85)	0.02 0.98
Score: Expert	33.96 (42.77)	28.87 (41.46)	48.89 (43.55)	0.32 0.72
Number of observations	946	2929	3524	7399

Notes: As in Table A13, for the stratified survey sample. The table reports a subset of the variables available from the baseline census. When considering the full set of baseline variables, there are 77 possible comparisons. For the comparison across long warning, short warning, and continued assistance, there are 11 instances (14.3%) that are statistically significant at the 10% level (p-value < 0.1), 3 instances (3.9%) at the 5% level (p-value < 0.05), and 1 instances (1.3%) at the 1% level (p-value < 0.01).

Table A16: Balance: long-notice households surveyed in the first round

Variable	Long Notice Mean/(SD) (1)	Discontinued Mean/(SD) (2)	Continued Mean/(SD) (3)	Across all groups F-stat/P-value (4)
Female headed household	0.26 (0.44)	0.29 (0.45)	0.36 (0.48)	1.60 0.20
Household head divorced/separated or widowed	0.18 (0.39)	0.20 (0.40)	0.24 (0.43)	2.66* 0.07
Senior household head (>64)	0.23 (0.42)	0.23 (0.42)	0.30 (0.46)	0.95 0.39
HH size	4.02 (1.71)	3.74 (1.65)	3.84 (1.93)	0.91 0.40
HH in Beirut	0.04 (0.20)	0.06 (0.23)	0.06 (0.23)	2.52* 0.08
Dependency ratio	0.32 (0.29)	0.34 (0.30)	0.37 (0.31)	0.27 0.76
School-age children	0.89 (1.18)	0.84 (1.11)	0.91 (1.26)	0.12 0.89
Household head with primary education or lower	0.65 (0.48)	0.55 (0.50)	0.64 (0.48)	3.08** 0.05
Household head suffers from any disability	0.20 (0.40)	0.16 (0.36)	0.24 (0.43)	0.36 0.70
No members employed	0.28 (0.45)	0.27 (0.45)	0.40 (0.49)	0.07 0.94
HH owns smartphone	0.80 (0.40)	0.85 (0.36)	0.77 (0.42)	0.29 0.75
HH owns car	0.33 (0.47)	0.38 (0.49)	0.22 (0.41)	0.17 0.84
Piped water	0.25 (0.43)	0.23 (0.42)	0.26 (0.44)	0.38 0.68
Gas for cooking	0.74 (0.44)	0.76 (0.43)	0.70 (0.46)	0.17 0.84
Wealth index - PCA	-0.12 (0.76)	0.08 (0.83)	-0.41 (0.85)	0.01 0.99
Score: PMT-CARI (predicted pr.)	0.40 (0.08)	0.35 (0.13)	0.45 (0.13)	0.19 0.82
Score: PMT-Exp (predicted in ln)	15.20 (0.27)	15.31 (0.37)	15.08 (0.37)	1.53 0.22
Score: PMT-FCS (predicted)	55.27 (5.71)	56.58 (7.28)	52.14 (7.86)	0.61 0.54
Score: Expert	33.26 (42.72)	29.30 (41.47)	48.97 (43.54)	0.42 0.66
Number of observations	293	2865	3200	6358

Notes: Restricted to the long-notice households actually surveyed in the first round, i.e. the subsample affected by the sampling error described in Section 6.

C.4 The Consequences of the End of Assistance

Table A17: Impacts of Discontinuation on Food Expenditure (ITT)

	Food Expenditure Groups								
	Staples (1)	Pulses (2)	Oil/fat (3)	Sugar (4)	Condiments (5)	Vegetables (6)	Fruits (7)	Protein (8)	Dairy (9)
Panel A: Round 1									
Discontinued	0.22*** (0.03)	0.52*** (0.03)	0.51*** (0.03)	0.24*** (0.03)	0.09*** (0.03)	-0.04 (0.03)	- 0.08*** (0.03)	-0.05* (0.03)	-0.02 (0.03)
Long Notice	- 0.18*** (0.06)	- 0.50*** (0.06)	- 0.41*** (0.06)	- 0.26*** (0.06)	-0.09 (0.06)	0.10* (0.06)	0.11 (0.06)	0.17*** (0.06)	0.05 (0.06)
Mean Cont. HHs	-0.09	-0.28	-0.26	-0.12	-0.10	-0.01	-0.04	-0.05	-0.04
Observations	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358
Panel B: Round 2									
Discontinued	0.18*** (0.03)	0.37*** (0.03)	0.32*** (0.03)	0.12*** (0.03)	0.04 (0.03)	-0.00 (0.03)	-0.02 (0.03)	0.01 (0.03)	0.02 (0.03)
Long Notice	0.02 (0.04)	-0.00 (0.04)	-0.00 (0.04)	-0.04 (0.04)	-0.01 (0.04)	0.05 (0.04)	0.06 (0.04)	-0.01 (0.04)	-0.02 (0.04)
Mean Cont. HHs	-0.06	-0.21	-0.17	-0.07	-0.04	-0.03	-0.04	-0.07	-0.06
Observations	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465

Notes: Each column reports a separate regression for a food expenditure sub-group. The independent variable of interest is the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Impacts of Discontinuation on FCS Groups (ITT)

	FCS Groups								
	Staples (1)	Pulses (2)	Oil/fat (3)	Sugar (4)	Condiments (5)	Vegetables (6)	Fruits (7)	Protein (8)	Dairy (9)
Panel A: Round 1									
Discontinued	-0.04 (0.03)	- 0.19*** (0.03)	- 0.09*** (0.03)	- 0.09*** (0.03)	-0.07** (0.03)	-0.02 (0.03)	-0.04 (0.03)	-0.03 (0.03)	0.01 (0.03)
Long Notice	0.04 (0.06)	0.09 (0.06)	0.04 (0.06)	0.04 (0.06)	0.00 (0.06)	0.05 (0.06)	0.14** (0.06)	0.03 (0.06)	-0.01 (0.06)
Mean Cont. HHs	0.01	0.07	0.04	0.02	0.00	-0.04	-0.04	-0.04	-0.06
Observations	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358
Panel B: Round 2									
Discontinued	0.05* (0.03)	- 0.13*** (0.03)	-0.01 (0.03)	-0.06* (0.03)	0.01 (0.03)	-0.01 (0.03)	-0.04 (0.03)	0.01 (0.03)	0.03 (0.03)
Long Notice	-0.07* (0.04)	0.06 (0.04)	0.02 (0.04)	-0.00 (0.04)	0.01 (0.04)	0.00 (0.04)	0.02 (0.04)	-0.01 (0.04)	-0.03 (0.04)
Mean Cont. HHs	-0.00	0.05	0.02	0.01	-0.03	-0.04	-0.04	-0.06	-0.05
Observations	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465

Notes: Each column reports a separate regression for an FCS food group. The independent variable of interest is the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Impacts of Discontinuation on rCSI (ITT)

	rCSI Strategies				
	Consumed less preferred food	Borrowed food	Reduced meal size	Reduced number of meals	Reduced adult consumption
	(1)	(2)	(3)	(4)	(5)
Panel A: Round 1					
Discontinued	0.00 (0.03)	0.05* (0.03)	0.05* (0.03)	0.05* (0.03)	0.02 (0.03)
Long Notice	-0.07 (0.06)	-0.03 (0.06)	-0.10* (0.06)	-0.09 (0.06)	-0.03 (0.06)
Mean Cont. HHs	0.00	0.02	0.01	0.01	0.01
Observations	6,358	6,358	6,358	6,358	6,358
Panel B: Round 2					
Discontinued	0.03 (0.03)	0.11*** (0.03)	0.04 (0.03)	0.06** (0.03)	0.06** (0.03)
Long Notice	0.04 (0.04)	-0.02 (0.04)	0.01 (0.04)	0.02 (0.04)	-0.04 (0.04)
Mean Cont. HHs	-0.01	-0.00	-0.01	-0.02	-0.03
Observations	6,465	6,465	6,465	6,465	6,465

Notes: Each column reports a separate regression for a coping strategy component of the rCSI. The independent variable of interest is the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A20: Impacts of Discontinuation on All Outcomes (ITT)

	FCS	rCSI	LCSI Stress	Food Exp.	Life Sats Index	WHO-5	Worry Index	No. Emp HHMs	Avg. Work Hrs.	HH Income	HH Has Debt	SC Index	Trust Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: Round 1													
Discontinued	- 0.09*** (0.03)	-0.06* (0.03)	0.00 (0.01)	0.16*** (0.03)	- 0.11*** (0.03)	-0.05* (0.03)	0.05* (0.03)	- 0.06** (0.03)	-0.05 (0.03)	0.03 (0.03)	0.01 (0.01)	0.03 (0.03)	- 0.13*** (0.03)
Long Notice	0.07 (0.06)	0.09 (0.06)	-0.00 (0.01)	-0.08 (0.06)	0.07 (0.06)	0.03 (0.06)	-0.04 (0.06)	-0.03 (0.06)	-0.04 (0.06)	-0.02 (0.04)	0.01 (0.03)	0.04 (0.06)	0.10 (0.06)
Mean Cont. HHs	-0.03	-0.02	0.94	-0.13	0.04	0.00	-0.04	-0.06	-0.06	-0.02	0.46	-0.03	0.06
Observations	6,358	6,358	6,358	6,358	6,358	6,300	6,357	6,358	5,792	6,358	6,325	6,358	6,356
Panel B: Round 2													
Discontinued	-0.02 (0.03)	- 0.11*** (0.03)	0.03** (0.01)	0.13*** (0.03)	- 0.18*** (0.03)	-0.05 (0.03)	-0.01 (0.03)	- 0.06** (0.03)	0.02 (0.03)	-0.02 (0.04)	0.00 (0.02)	0.03 (0.03)	- 0.07** (0.03)
Long Notice	-0.01 (0.04)	0.01 (0.04)	-0.01 (0.02)	0.01 (0.04)	0.01 (0.04)	-0.04 (0.04)	0.04 (0.04)	-0.02 (0.03)	-0.05 (0.04)	-0.01 (0.04)	0.00 (0.02)	-0.03 (0.04)	0.04 (0.04)
Mean Cont. HHs	-0.04	0.02	0.73	-0.10	0.04	-0.01	-0.02	-0.05	-0.06	-0.04	0.53	-0.06	0.01
Observations	6,465	6,465	6,465	6,465	6,465	6,453	6,465	6,465	5,938	6,465	6,450	6,465	6,465

Notes: Each column reports a separate regression. The independent variable of interest is the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A21: Impacts of Discontinuation on Trust (ITT)

		Institutional Trust						
	Trust Index	Family	Neighbors	Religious Org.	UN/NGOs	Local Gov.	National Gov.	Migrants
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Round 1								
Discontinued	-0.13*** (0.03)	-0.04 (0.03)	0.00 (0.03)	-0.05 (0.03)	-0.25*** (0.03)	-0.06* (0.03)	-0.03 (0.03)	-0.09*** (0.03)
Long Notice	0.10 (0.06)	-0.01 (0.07)	0.08 (0.06)	-0.02 (0.06)	0.21*** (0.06)	-0.01 (0.06)	0.01 (0.06)	0.10 (0.06)
Mean Cont. HHs	0.06	0.00	-0.01	0.02	0.11	0.02	0.06	0.06
Observations	6,356	6,332	6,288	6,092	6,130	6,133	6,117	5,993
Panel B: Round 2								
Discontinued	-0.07** (0.03)	0.02 (0.03)	-0.04 (0.03)	-0.04 (0.03)	-0.13*** (0.03)	-0.03 (0.03)	-0.04 (0.03)	-0.05 (0.03)
Long Notice	0.04 (0.04)	-0.00 (0.04)	0.02 (0.04)	0.04 (0.04)	0.05 (0.04)	0.02 (0.04)	0.02 (0.04)	-0.01 (0.04)
Mean Cont. HHs	0.01	-0.05	-0.00	-0.01	0.05	-0.02	0.02	0.06
Observations	6,465	6,454	6,413	6,241	6,351	6,267	6,160	6,130

Notes: Each column reports a separate regression for a trust outcome. The independent variable of interest is the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.5 Instrumental-Variable Estimates

Table A22: Impacts of Discontinuation on All Outcomes (2SLS)

	FCS	rCSI	LCSI Stress	Food Exp.	Life Sats. Index	WHO-5	Worry Index	No. Emp. HHs	Avg. Work Hrs.	HH Income	HH Debt	SC Index	Trust Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: Round 1													
Discontinued	-	-0.06*	0.00	0.17***	-	-0.05*	0.05*	-	-0.05	0.03	0.01	0.03	-
	0.10*** (0.03)	(0.03)	(0.01)	(0.03)	0.12*** (0.03)	(0.03)	(0.03)	0.06** (0.03)	(0.03)	(0.03)	(0.01)	(0.03)	0.14*** (0.03)
Long Notice	-0.03 (0.07)	0.03 (0.07)	0.00 (0.01)	0.09 (0.06)	-0.04 (0.06)	-0.02 (0.06)	0.02 (0.06)	-0.09* (0.06)	-0.09 (0.06)	0.02 (0.03)	0.02 (0.03)	0.07 (0.06)	-0.04 (0.06)
Mean Cont. HHs	-0.02	-0.01	0.95	-0.12	0.03	0.00	-0.03	-0.05	-0.05	-0.01	0.46	-0.02	0.06
Observations	6,358	6,358	6,358	6,358	6,358	6,300	6,357	6,358	5,792	6,358	6,325	6,358	6,356
Panel B: Round 2													
Discontinued	-0.10	-	0.15**	0.67***	-	-0.24	-0.04	-	0.06	-0.12	0.02	0.13	-
	(0.15)	0.56*** (0.15)	(0.07)	(0.14)	0.87*** (0.15)	(0.15)	(0.14)	0.31** (0.13)	(0.16)	(0.18)	(0.07)	(0.14)	0.34** (0.15)
Long Notice	-0.04 (0.18)	0.08 (0.18)	-0.03 (0.08)	0.03 (0.17)	0.10 (0.18)	-0.16 (0.18)	0.17 (0.17)	-0.08 (0.15)	-0.21 (0.18)	-0.04 (0.19)	-0.00 (0.08)	-0.12 (0.17)	0.20 (0.17)
Mean Cont. HHs	0.01	-0.03	0.73	-0.02	0.01	-0.04	0.02	-0.00	-0.00	-0.00	0.54	-0.02	-0.03
Observations	6,465	6,465	6,465	6,465	6,465	6,453	6,465	6,465	5,938	6,465	6,450	6,465	6,465

Notes: Each column reports a separate 2SLS regression. The independent variable of interest is actual discontinuation, instrumented by the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Impacts of Discontinuation on Food Expenditure (2SLS)

	Food Expenditure Groups								
	Staples (1)	Pulses (2)	Oil/fat (3)	Sugar (4)	Condiments (5)	Vegetables (6)	Fruits (7)	Protein (8)	Dairy (9)
Panel A: Round 1									
Discontinued	0.23*** (0.03)	0.54*** (0.03)	0.53*** (0.03)	0.24*** (0.03)	0.09*** (0.03)	-0.05 (0.03)	- 0.08*** (0.03)	-0.05* (0.03)	-0.03 (0.03)
Long Notice	0.04 (0.06)	0.02 (0.06)	0.10 (0.06)	-0.03 (0.06)	-0.01 (0.06)	0.06 (0.06)	0.04 (0.07)	0.13** (0.06)	0.02 (0.07)
Mean Cont. HHs	-0.10	-0.27	-0.25	-0.12	-0.10	-0.00	-0.02	-0.03	-0.03
Observations	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358
Panel B: Round 2									
Discontinued	0.87*** (0.14)	1.86*** (0.17)	1.62*** (0.16)	0.61*** (0.15)	0.21 (0.15)	-0.01 (0.14)	-0.11 (0.15)	0.04 (0.15)	0.10 (0.15)
Long Notice	0.04 (0.17)	-0.09 (0.22)	-0.08 (0.19)	-0.17 (0.17)	-0.07 (0.17)	0.20 (0.17)	0.25 (0.18)	-0.03 (0.17)	-0.09 (0.17)
Mean Cont. HHs	-0.01	-0.04	-0.04	-0.03	-0.00	-0.02	-0.00	-0.02	-0.01
Observations	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465

Notes: Each column reports a separate 2SLS regression for a food expenditure sub-group. The independent variable of interest is actual discontinuation, instrumented by the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A24: Impacts of Discontinuation on FCS Groups (2SLS)

	FCS Groups								
	Staples (1)	Pulses (2)	Oil/fat (3)	Sugar (4)	Condiments (5)	Vegetables (6)	Fruits (7)	Protein (8)	Dairy (9)
Panel A: Round 1									
Discontinued	-0.05 (0.03)	- 0.19*** (0.03)	- 0.09*** (0.03)	- 0.09*** (0.03)	-0.07** (0.03)	-0.02 (0.03)	-0.04 (0.03)	-0.04 (0.03)	0.01 (0.03)
Long Notice	-0.00 (0.06)	-0.10 (0.06)	-0.05 (0.06)	-0.05 (0.06)	-0.07 (0.06)	0.03 (0.06)	0.11 (0.07)	-0.01 (0.07)	-0.00 (0.06)
Mean Cont. HHs	0.01	0.07	0.03	0.02	0.01	-0.03	-0.03	-0.02	-0.05
Observations	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,358
Panel B: Round 2									
Discontinued	0.25 (0.15)	- 0.63*** (0.16)	-0.03 (0.15)	-0.29* (0.15)	0.04 (0.15)	-0.04 (0.15)	-0.17 (0.15)	0.06 (0.15)	0.16 (0.16)
Long Notice	-0.32* (0.18)	0.28 (0.18)	0.09 (0.18)	0.01 (0.17)	0.03 (0.17)	0.02 (0.18)	0.09 (0.18)	-0.03 (0.18)	-0.14 (0.18)
Mean Cont. HHs	-0.00	0.00	0.02	0.04	-0.00	-0.01	-0.00	0.01	0.01
Observations	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465	6,465

Notes: Each column reports a separate 2SLS regression for an FCS food group. The independent variable of interest is actual discontinuation, instrumented by the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A25: Impacts of Discontinuation on rCSI (2SLS)

	rCSI Strategies				
	Consumed less preferred food	Borrowed food	Reduced meal size	Reduced number of meals	Reduced adult consumption
	(1)	(2)	(3)	(4)	(5)
Panel A: Round 1					
Discontinued	0.00 (0.03)	0.05* (0.03)	0.05* (0.03)	0.05* (0.03)	0.02 (0.03)
Long Notice	-0.07 (0.06)	0.02 (0.06)	-0.05 (0.07)	-0.04 (0.06)	-0.01 (0.06)
Mean Cont. HHs	0.00	0.01	0.01	0.01	0.00
Observations	6,358	6,358	6,358	6,358	6,358
Panel B: Round 2					
Discontinued	0.14 (0.15)	0.56*** (0.15)	0.19 (0.15)	0.31** (0.15)	0.30** (0.14)
Long Notice	0.17 (0.18)	-0.11 (0.18)	0.04 (0.17)	0.10 (0.18)	-0.18 (0.16)
Mean Cont. HHs	0.01	-0.00	0.04	0.02	0.02
Observations	6,465	6,465	6,465	6,465	6,465

Notes: Each column reports a separate 2SLS regression for a coping strategy component of the rCSI. The independent variable of interest is actual discontinuation, instrumented by the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A26: Impacts of Discontinuation on Trust (2SLS)

	Institutional Trust							
	Trust Index	Family	Neighbors	Religious Org.	UN/NGOs	Local Gov.	National Gov.	Migrants
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Round 1								
Discontinued	-0.14*** (0.03)	-0.05 (0.03)	0.00 (0.03)	-0.05 (0.03)	-0.25*** (0.03)	-0.06* (0.03)	-0.03 (0.03)	-0.09*** (0.03)
Long Notice	-0.04 (0.06)	-0.05 (0.07)	0.09 (0.06)	-0.07 (0.06)	-0.03 (0.07)	-0.07 (0.06)	-0.02 (0.06)	0.01 (0.07)
Mean Cont. HHs	0.06	0.00	-0.00	0.02	0.11	0.01	0.04	0.05
Observations	6,356	6,332	6,288	6,092	6,130	6,133	6,117	5,993
Panel B: Round 2								
Discontinued	-0.34** (0.15)	0.13 (0.15)	-0.21 (0.15)	-0.18 (0.15)	-0.64*** (0.15)	-0.16 (0.15)	-0.18 (0.16)	-0.23 (0.15)
Long Notice	0.20 (0.17)	-0.00 (0.17)	0.09 (0.18)	0.22 (0.18)	0.25 (0.18)	0.14 (0.18)	0.10 (0.18)	-0.03 (0.17)
Mean Cont. HHs	-0.03	-0.02	-0.03	-0.02	0.00	-0.02	-0.01	-0.01
Observations	6,465	6,454	6,413	6,241	6,351	6,267	6,160	6,130

Notes: Each column reports a separate 2SLS regression for a trust outcome. The independent variable of interest is actual discontinuation, instrumented by the ITT indicator. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.6 Heterogeneity

Table A27: Heterogeneous ITT effects on downstream outcomes by baseline subgroup

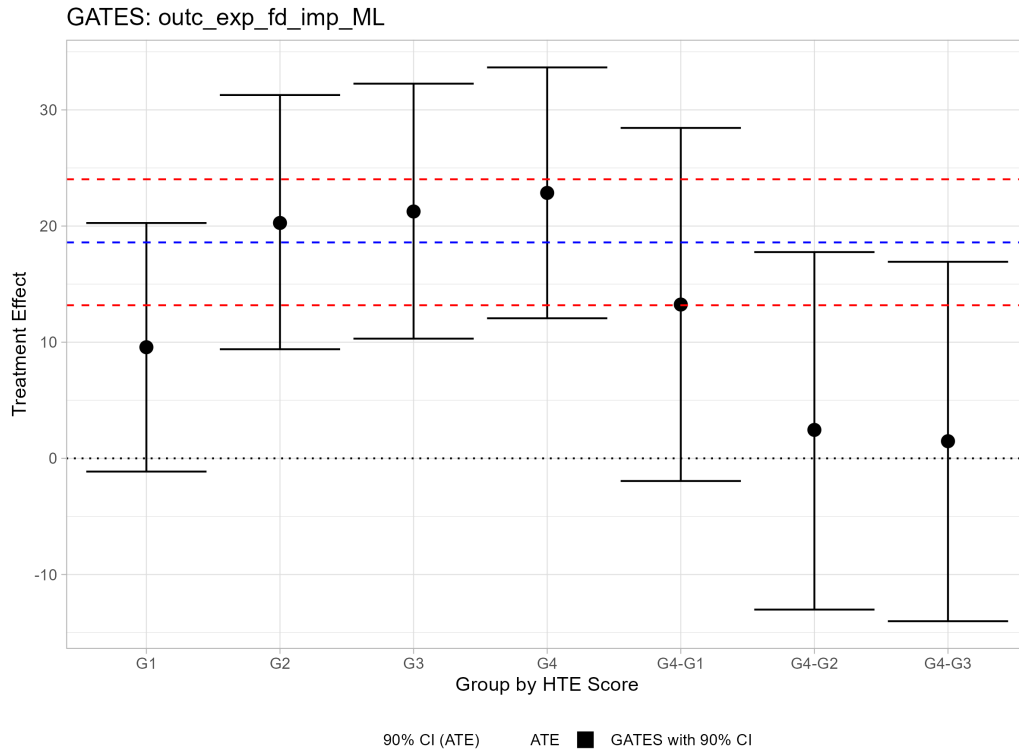
	FCS (1)	rCSI (2)	Food Exp. (USD) (3)	WHO-5 Index (4)
<i>Female-headed household (33% of households)</i>				
PMT-FCS	1.36 (1.01)	-1.23* (0.71)	7.96* (4.83)	1.66 (1.14)
PMT-CARI	2.05** (1.01)	-0.55 (0.72)	1.63 (4.94)	-0.19 (1.16)
Expert	0.72 (1.03)	-0.39 (0.72)	-0.68 (4.79)	0.30 (1.16)
<i>Elderly head (65+) (27% of households)</i>				
PMT-FCS	2.24* (1.19)	-1.27* (0.76)	18.61*** (5.82)	1.31 (1.27)
PMT-CARI	1.69 (1.18)	-0.10 (0.79)	3.51 (5.74)	0.65 (1.26)
Expert	2.57** (1.19)	-1.71** (0.75)	10.62* (5.57)	1.05 (1.24)
<i>Large household (5+ members) (34% of households)</i>				
PMT-FCS	-1.22 (1.01)	1.22 (0.79)	4.04 (4.80)	0.19 (1.14)
PMT-CARI	0.89 (1.01)	1.01 (0.81)	1.70 (5.08)	0.93 (1.17)
Expert	-1.34 (1.00)	1.23 (0.82)	2.58 (5.01)	-0.28 (1.14)
<i>Bottom asset quartile (25% of households)</i>				
PMT-FCS	1.36 (1.09)	-2.44*** (0.82)	15.62*** (5.32)	2.46* (1.28)
PMT-CARI	0.59 (1.09)	-0.18 (0.87)	1.34 (5.29)	0.35 (1.28)
Expert	0.85 (1.12)	-1.09 (0.85)	6.88 (5.13)	0.09 (1.28)
<i>Eligible (continued under assigned method) (50% of households)</i>				
PMT-FCS	-2.20*** (0.80)	-1.08* (0.63)	-4.83 (3.95)	0.39 (0.97)
PMT-CARI	0.05 (0.81)	0.03 (0.65)	0.17 (4.06)	-0.61 (0.97)
Expert	-1.61* (0.83)	-1.64*** (0.62)	-19.41*** (3.90)	-2.21** (0.93)
<i>Eligible under 3 methods (25% of households)</i>				
PMT-FCS	-0.40 (1.13)	-1.47* (0.89)	10.55* (5.44)	2.10 (1.35)
PMT-CARI	0.73 (1.16)	-0.06 (0.88)	3.78 (5.73)	-0.71 (1.34)
Expert	-0.64 (1.19)	-0.92 (0.88)	7.94 (5.53)	-0.06 (1.33)
<i>Eligible under 2 methods (37% of households)</i>				
PMT-FCS	0.58 (0.95)	-0.75 (0.73)	4.88 (4.84)	-0.35 (1.12)
PMT-CARI	1.21 (0.94)	0.38 (0.74)	-1.68 (4.79)	0.10 (1.13)
Expert	0.11 (0.95)	-0.32 (0.73)	3.35 (4.80)	-0.17 (1.10)
<i>Eligible under 1 method (25% of households)</i>				
PMT-FCS	-0.32 (1.17)	0.14 (0.85)	6.66 (5.97)	0.02 (1.34)
PMT-CARI	1.06 (1.21)	0.38 (0.89)	5.42 (5.99)	2.29* (1.33)
Expert	0.06 (1.17)	0.57 (0.85)	2.08 (6.03)	0.86 (1.31)

Notes: Each cell reports the arm×subgroup interaction coefficient from a regression of the downstream outcome on targeting arm, the baseline subgroup indicator, and their interaction (Round 1; PMT-Expenditure is the reference arm; control for the longer warning period). The coefficient is the differential effect of the arm for subgroup members relative to non-members. Subgroups are defined at baseline (pre-randomization). Eligibility (continued versus discontinued) is assigned by each method, so that 3-panel blends targeting with the discontinuation margin. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

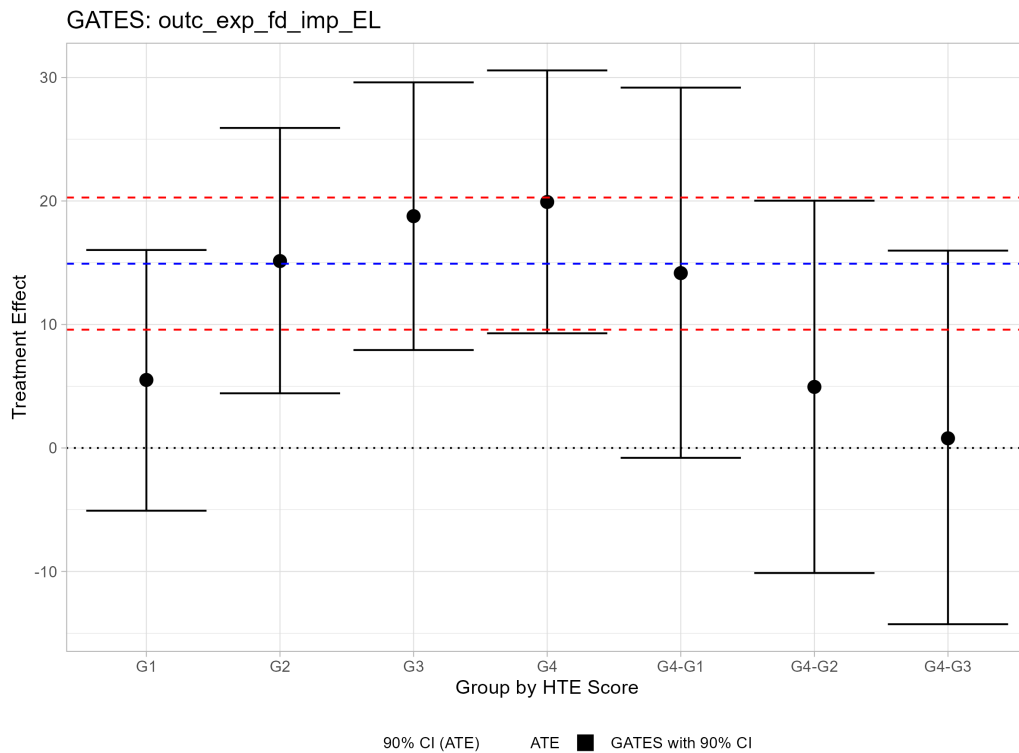
Table A28: Heterogeneous Impacts of Discontinuation by Targeting Arm (ITT)

	FCS	rCSI	LCSI Stress	Food Exp.	Life Sat Index	WHO-5	Worry Index	No. Em HHMs	Alg. Wor Hrs	HH Income	HH Has Debt	SC Index	Trust Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: Round 1													
PMT-FCS	-	0.08	-0.01	-0.02	-0.01	0.05	-0.09	-	-0.05	-0.03	0.02	-0.00	0.04
	0.14**							0.20***					
	(0.06)	(0.07)	(0.02)	(0.06)	(0.06)	(0.07)	(0.06)	(0.06)	(0.06)	(0.02)	(0.03)	(0.08)	(0.07)
PMT-CARI	-0.02	0.01	0.01	0.03	-0.07	-0.02	-0.04	-0.08	0.05	0.04	0.03	-0.01	-0.08
	(0.06)	(0.07)	(0.02)	(0.06)	(0.06)	(0.07)	(0.06)	(0.07)	(0.06)	(0.07)	(0.03)	(0.08)	(0.07)
Expert	-0.04	0.18***	-0.00	-	-0.11	-0.09	-	-	-0.04	0.02	-	0.03	-0.02
				0.29***			0.19***	0.36***			0.09***		
	(0.06)	(0.07)	(0.02)	(0.06)	(0.07)	(0.07)	(0.06)	(0.06)	(0.07)	(0.04)	(0.03)	(0.08)	(0.07)
Discontinued	0.07	0.33***	-0.02	0.04	-0.10	0.06	-0.08	0.00	0.18***	0.13	-	0.13	-
											0.13***		0.15**
	(0.07)	(0.07)	(0.02)	(0.07)	(0.07)	(0.08)	(0.07)	(0.07)	(0.07)	(0.16)	(0.04)	(0.09)	(0.07)
PMT-FCS × Discontinued	0.21**	-0.14	0.04	0.17	-0.04	-0.10	0.06	0.31***	0.09	-0.07	0.02	-0.01	-0.03
	(0.10)	(0.10)	(0.03)	(0.10)	(0.10)	(0.10)	(0.10)	(0.11)	(0.10)	(0.17)	(0.05)	(0.12)	(0.10)
PMT-CARI × Discontinued	0.15	-0.07	0.02	0.06	0.04	-0.08	-0.05	0.11	0.05	-0.20	-0.03	0.04	0.09
	(0.10)	(0.10)	(0.02)	(0.10)	(0.10)	(0.11)	(0.10)	(0.10)	(0.11)	(0.18)	(0.05)	(0.12)	(0.10)
Expert × Discontinued	0.07	-	0.05**	0.63***	0.10	0.02	0.31***	0.49***	0.12	-0.14	0.23***	-	0.10
		0.39***										0.18*	
	(0.10)	(0.10)	(0.02)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.16)	(0.05)	(0.11)	(0.10)
Long Notice	-0.10	-0.11	0.00	-0.07	-0.02	0.04	0.01	-0.12	-0.09	-0.05	0.04	-0.04	0.14
	(0.10)	(0.08)	(0.02)	(0.10)	(0.09)	(0.09)	(0.07)	(0.08)	(0.09)	(0.04)	(0.04)	(0.08)	(0.09)
(PMT-Exp*Continued)	0.03	-0.07	0.94	-0.06	0.08	0.03	0.03	0.08	-0.03	-0.02	0.47	-0.08	0.09
Observations	6,358	6,358	6,358	6,358	6,358	6,300	6,357	6,358	5,792	6,358	6,325	6,358	6,356
Panel B: Round 2													
PMT-FCS	-0.02	0.06	0.02	-0.09	-0.00	0.00	-0.08	-	-0.08	-	-0.00	0.03	-0.05
								0.20***		0.10**			
	(0.06)	(0.06)	(0.03)	(0.06)	(0.07)	(0.07)	(0.07)	(0.06)	(0.06)	(0.04)	(0.03)	(0.07)	(0.07)
PMT-CARI	-	0.01	0.02	-	0.02	0.09	0.02	-0.10	-0.05	-0.05	0.00	-0.01	0.08
	0.17***			0.12*									
	(0.06)	(0.07)	(0.03)	(0.06)	(0.07)	(0.07)	(0.06)	(0.06)	(0.06)	(0.04)	(0.03)	(0.07)	(0.07)
Expert	-0.08	-0.02	-0.02	-	-0.05	-	-	-	-0.04	-	-	0.07	0.14**
				0.32***		0.12*	0.26***	0.35***		0.13***	0.11***		
	(0.07)	(0.06)	(0.03)	(0.06)	(0.07)	(0.07)	(0.07)	(0.06)	(0.07)	(0.05)	(0.03)	(0.07)	(0.07)
Discontinued	0.07	0.10	-0.03	-0.02	0.02	0.11	-	-0.05	0.25***	0.09	-	0.31***	0.16**
							0.22***				0.18***		
	(0.07)	(0.07)	(0.03)	(0.07)	(0.07)	(0.08)	(0.07)	(0.07)	(0.08)	(0.06)	(0.04)	(0.08)	(0.07)
PMT-FCS × Discontinued	0.05	-0.11	-0.02	0.18*	-0.11	-0.03	0.24**	0.32***	0.01	0.20**	0.06	-0.04	0.08
	(0.10)	(0.10)	(0.04)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.05)	(0.14)	(0.10)
PMT-CARI × Discontinued	0.29***	0.02	0.01	0.12	0.03	-	0.04	0.19*	0.07	0.06	0.06	0.07	-0.04
						0.18*							
	(0.10)	(0.10)	(0.04)	(0.10)	(0.10)	(0.11)	(0.10)	(0.10)	(0.11)	(0.08)	(0.05)	(0.11)	(0.10)
Expert × Discontinued	0.08	-0.15	0.06	0.54***	0.07	0.17*	0.58***	0.50***	-0.01	0.15*	0.25***	-0.17	-
												0.25**	
	(0.10)	(0.10)	(0.04)	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)	(0.11)	(0.08)	(0.05)	(0.11)	(0.10)
Long Notice	-0.09	-0.03	-0.01	-0.09	-	-	0.02	-	-	-	-0.02	-	-0.03
					0.12**	0.13**		0.21***	0.22***	0.14***		0.16***	
	(0.06)	(0.05)	(0.02)	(0.06)	(0.05)	(0.06)	(0.05)	(0.06)	(0.05)	(0.05)	(0.03)	(0.06)	(0.05)
(PMT-Exp*Continued)	0.01	0.02	0.73	-0.00	0.06	-0.00	0.06	0.08	-0.01	0.01	0.54	-0.07	0.05
Observations	6,465	6,465	6,465	6,465	6,465	6,453	6,465	6,465	5,938	6,465	6,450	6,465	6,465

Notes: Each column reports a separate weighted regression. Interactions are between the discontinuation indicator and each targeting arm; the reference group is continued households in the PMT-Exp (T1) arm. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.



(a) First round



(b) Second round

Figure A15: Generic ML: sorted group average treatment effects (GATES) on food expenditure

Notes: Group average treatment effects of assigned discontinuation on household monthly food expenditure (USD), by quartile of the predicted treatment-effect score, following Chernozhukov et al. (2025). G1 is the quartile with the lowest predicted effect and G4 the highest; the three rightmost estimates are differences relative to G4. Scores are estimated on an auxiliary half-sample over 1,000 random splits stratified by eligibility space; the plotted learner is the one selected on out-of-sample fit. Dashed blue line is the ATE, red dashed lines its 90% confidence band. Bars are 90% confidence intervals.

C.7 Targeting Methods

Table A29: Impacts of Targeting on Secondary Outcomes (ITT)

	Employment rate	Avg. work hours	Income from work (SD)	Has Debt	Social Cohesion Index (SD)	Trust Index (SD)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Round 1						
PMT-FCS	0.00 (0.01)	0.01 (0.04)	0.00 (0.04)	0.01 (0.02)	0.00 (0.04)	-0.03 (0.04)
PMT-CARI	0.01 (0.01)	0.03 (0.04)	-0.02 (0.03)	0.02 (0.02)	-0.00 (0.04)	-0.03 (0.04)
Expert	-0.00 (0.01)	0.02 (0.04)	0.00 (0.03)	0.02 (0.02)	-0.01 (0.04)	-0.03 (0.04)
Mean PMT-Exp.	0.29	-0.02	0.00	0.44	0.00	0.02
Observations	5,792	5,792	6,358	6,325	6,358	6,356
p-val(PMT-FCS=PMT-CARI)	0.419	0.650	0.536	0.606	0.831	0.963
p-val(PMT-FCS=Expert)	0.801	0.777	0.987	0.394	0.727	0.991
p-val(PMT-CARI=Expert)	0.291	0.860	0.287	0.737	0.899	0.954
Panel B: Round 2						
PMT-FCS	-0.02** (0.01)	-0.08** (0.04)	0.03 (0.04)	0.03* (0.02)	0.00 (0.04)	-0.04 (0.03)
PMT-CARI	-0.00 (0.01)	-0.01 (0.04)	0.00 (0.03)	0.01 (0.02)	0.03 (0.04)	0.06* (0.03)
Expert	0.00 (0.01)	0.02 (0.04)	-0.01 (0.03)	0.03* (0.02)	-0.01 (0.03)	-0.01 (0.04)
Mean PMT-Exp.	0.30	0.02	-0.00	0.50	-0.00	-0.00
Observations	5,938	5,938	6,465	6,450	6,465	6,465
p-val(PMT-FCS=PMT-CARI)	0.062	0.081	0.530	0.194	0.553	0.005
p-val(PMT-FCS=Expert)	0.038	0.008	0.258	0.806	0.695	0.360
p-val(PMT-CARI=Expert)	0.845	0.396	0.579	0.291	0.300	0.069

Notes: PMT-Expenditure targeting serves as the reference group. The table reports regression results on household-level outcomes using survey data. Each column corresponds to a separate dependent variable, with the ITT treatment indicator as the variable of interest. Employment rate measures the share of working-age household members currently employed. Average work hours capture the mean weekly hours worked among employed household members. Income from work is total labour income, transformed using the inverse hyperbolic sine and winsorized at the 95th percentile. Debt is a dummy equal to one if the household holds any outstanding liabilities. The Social Cohesion Index incorporates measures of social tensions and collective actions following Bossuoy et al. (2022). The Trust Index aggregates both generalized and institutional trust. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A30: Impacts of Targeting on Downstream Outcomes with Weights (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	Life Satisf. Index (SD)	Worriedness Index (SD)	WHO-5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
PMT-FCS	-0.33 (0.89)	0.06 (0.63)	6.35 (4.52)	9.79 (11.82)	-0.04 (0.05)	-0.05 (0.05)	-0.09 (1.02)
PMT-CARI	0.96 (0.88)	0.33 (0.62)	5.41 (4.41)	15.50 (12.01)	-0.05 (0.05)	-0.07 (0.05)	-1.27 (1.05)
Expert	-0.00 (0.87)	0.56 (0.64)	5.65 (4.49)	10.48 (11.86)	-0.05 (0.05)	-0.02 (0.05)	-1.42 (1.02)
Mean PMT-Exp.	54.47	16.62	136.64	392.21	-0.00	0.02	23.00
Observations	6,358	6,358	6,358	6,358	6,358	6,357	6,300
p-val(PMT-FCS=PMT-CARI)	0.151	0.660	0.831	0.633	0.861	0.787	0.233
p-val(PMT-FCS=Expert)	0.713	0.434	0.876	0.953	0.825	0.484	0.165
p-val(PMT-CARI=Expert)	0.275	0.716	0.956	0.676	0.964	0.336	0.882

Notes: PMT-Expenditure targeting serves as the reference group. The table shows the regression results on pre-specified primary outcomes at the household level using survey data. Each column is its own dependent variable. The independent variable of interest is the ITT treatment indicator and includes beneficiaries and non-beneficiary households. Regressions use sampling weights; robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A31: Impacts of Targeting on Secondary Outcomes with Weights (ITT)

	Employment rate	Avg. work hours	Income from work (SD)	Has Debt	Social Cohesion Index (SD)	Trust Index (SD)
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Round 1						
PMT-FCS	-0.01 (0.02)	0.10 (0.72)	-0.01 (0.20)	0.03 (0.03)	-0.01 (0.07)	0.03 (0.05)
PMT-CARI	0.01 (0.02)	1.21 (0.79)	0.06 (0.19)	0.02 (0.03)	0.01 (0.06)	-0.03 (0.05)
Expert	-0.02 (0.02)	0.60 (0.76)	0.08 (0.20)	0.04 (0.03)	-0.08 (0.05)	0.03 (0.05)
Mean PMT-Exp.	0.29	12.07	3.48	0.44	0.00	0.02
Observations	5,792	5,792	6,358	6,325	6,358	6,356
p-val(PMT-FCS=PMT-CARI)	0.107	0.155	0.691	0.739	0.719	0.230
p-val(PMT-FCS=Expert)	0.784	0.508	0.622	0.567	0.269	0.933
p-val(PMT-CARI=Expert)	0.064	0.456	0.907	0.364	0.109	0.213
Panel B: Round 2						
PMT-FCS	-0.02 (0.02)	-0.97 (0.72)	-0.09 (0.17)	0.03 (0.03)	0.01 (0.07)	-0.01 (0.05)
PMT-CARI	-0.01 (0.02)	-0.13 (0.76)	-0.04 (0.17)	0.04 (0.03)	0.03 (0.06)	0.05 (0.05)
Expert	-0.02 (0.02)	-0.50 (0.74)	-0.13 (0.17)	0.04 (0.03)	-0.03 (0.06)	-0.01 (0.05)
Mean PMT-Exp.	0.30	11.91	3.61	0.50	-0.00	-0.00
Observations	5,938	5,938	6,465	6,450	6,465	6,465
p-val(PMT-FCS=PMT-CARI)	0.436	0.246	0.768	0.837	0.718	0.243
p-val(PMT-FCS=Expert)	0.826	0.504	0.813	0.862	0.648	0.974
p-val(PMT-CARI=Expert)	0.592	0.613	0.596	0.975	0.290	0.247

Notes: PMT-Expenditure targeting serves as the reference group. The table reports regression results on household-level outcomes using survey data. Each column corresponds to a separate dependent variable, with the ITT treatment indicator as the variable of interest. All regressions use sampling weights, with robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A32: Progressiveness regressions: selection on welfare proxies, by targeting arm

	Wealth index	Asset index
PMT-Exp	-1.554*** (0.064)	-1.450*** (0.061)
PMT-FCS	-2.042*** (0.079)	-2.015*** (0.078)
PMT-CARI	-1.790*** (0.070)	-1.756*** (0.069)
Expert	-1.052*** (0.054)	-1.080*** (0.053)
Joint test of equal slopes (p)	0.000	0.000

Notes: Each cell is the coefficient on the welfare index from a separate logit of an indicator for selection by that arm on the index (robust SE in parentheses); negative values mean poorer households are more likely selected. The joint test reshapes to household-by-arm and tests equality of the index slope across the four arms (logit with arm interactions, SE clustered by household). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A33: Progressivity: CGH index and targeting differential

Method	GT	CGH index			Targeting differential		
		10%	20%	43%	10%	20%	43%
PMT-Exp	Cons.	2.22	2.05	1.67	0.20	0.35	0.48
	FCS	1.53	1.38	1.29	0.09	0.13	0.21
	CARI	2.17	1.90	1.55	0.20	0.30	0.39
PMT-FCS	Cons.	2.03	1.83	1.60	0.17	0.27	0.43
	FCS	1.78	1.62	1.40	0.14	0.21	0.30
	CARI	1.98	1.79	1.52	0.16	0.26	0.38
PMT-CARI	Cons.	2.15	2.02	1.68	0.19	0.34	0.49
	FCS	1.35	1.42	1.34	0.06	0.14	0.25
	CARI	2.13	1.87	1.57	0.19	0.29	0.41
Expert	Cons.	1.23	1.32	1.32	0.04	0.11	0.23
	FCS	1.32	1.15	1.17	0.06	0.05	0.12
	CARI	1.20	1.19	1.24	0.03	0.06	0.17

Notes: CGH index (?) is the share of assistance reaching the deprived over their population share (equals 1 under random targeting; >1 is progressive). The targeting differential (?) is the difference between the share of the deprived and of the non-deprived reached (ranges -1 to 1 ; 0 is non-progressive). Both are shown at the 10/20/43% targeted shares, against each ground truth (GT: consumption, FCS, CARI).

Table A34: Who appeals? Correlates of appealing the discontinuation decision

	(1)	(2)	(3)
Panel A: No fixed effects			
Baseline wealth index (SD)	-0.018 (0.011)		
Distance to eligibility cutoff (SD)		-0.011 (0.010)	
Owns a phone or smartphone			0.009 (0.026)
Mean appeal rate	0.573	0.573	0.573
Observations	2,908	2,908	2,908
Panel B: Eligibility-space fixed effects			
Baseline wealth index (SD)	0.005 (0.013)		
Distance to eligibility cutoff (SD)		-0.007 (0.010)	
Owns a phone or smartphone			0.018 (0.026)
Mean appeal rate	0.573	0.573	0.573
Observations	2,908	2,908	2,908

Notes: The sample is households whose assistance was discontinued and for whom appeal status is recorded. The dependent variable is an indicator for having submitted an appeal to WFP. Each column reports a separate regression on a single covariate. The wealth index is the first principal component of baseline asset ownership, standardised. Distance to the eligibility cutoff is the gap between the household's score under its assigned targeting method and the 43rd-percentile threshold, signed so that higher values indicate households further from eligibility, and standardised within arm. Panel B adds eligibility-space fixed effects; because eligibility space is itself determined by the four vulnerability scores, these absorb much of the variation the covariates are meant to capture. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

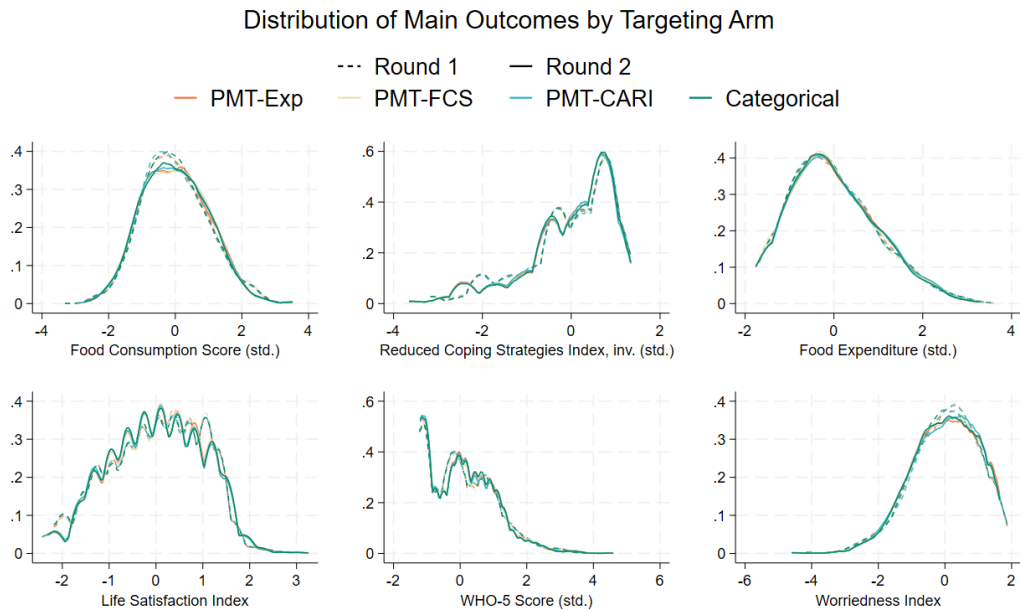


Figure A16: Distribution of outcomes across targeting arms

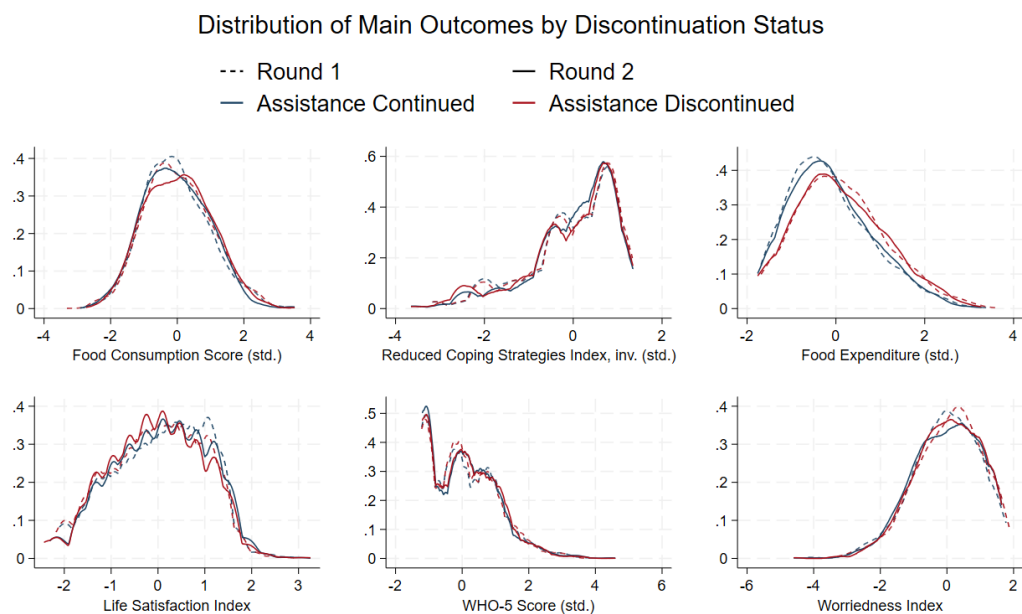


Figure A17: Baseline wealth and the probability of discontinuation, by targeting method

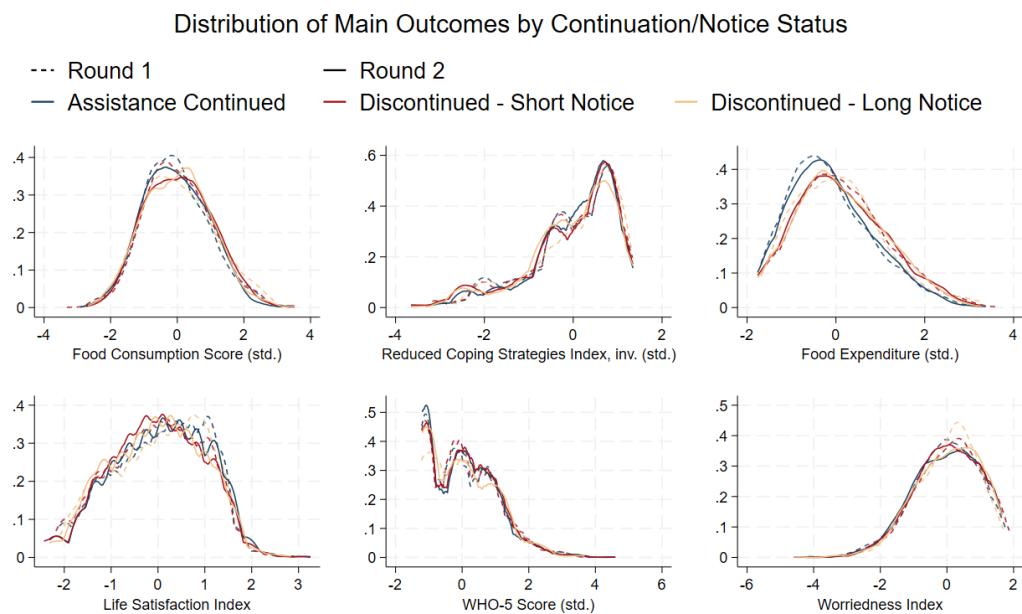


Figure A18: Distribution of outcomes, long-notice households

C.8 Eligibility Spaces and Marginal Households

Table A35: Impacts of Discontinuation by Eligibility on All Outcomes (ITT)

	FCS	rCSI	LCSI Stress	Food Exp.	Life Sat Index	WHO-5	Worry Index	No. Em HHMs	Avg. Work Hrs.	HH Income	HH Has Debt	SC Index	Trust Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: Round 1													
Disc. × Elig 2 Methods	0.08	0.07	0.00	0.19***	-0.06	-0.05	-0.01	0.03	-0.02	0.02	-0.03	0.19**	- 0.15*
	(0.08)	(0.08)	(0.02)	(0.07)	(0.08)	(0.08)	(0.08)	(0.07)	(0.08)	(0.03)	(0.04)	(0.07)	(0.08)
Disc. × Elig 3 Methods	0.07	0.04	-0.01	0.12*	0.08	0.02	-0.10	-0.02	-0.02	0.10*	-0.02	0.11*	- 0.16**
	(0.07)	(0.07)	(0.02)	(0.06)	(0.07)	(0.07)	(0.07)	(0.06)	(0.07)	(0.05)	(0.03)	(0.06)	(0.07)
Long Notice	0.07	0.09	-0.00	-0.07	0.07	0.03	-0.04	-0.03	-0.04	-0.02	0.01	0.04	0.09
	(0.06)	(0.06)	(0.01)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.04)	(0.03)	(0.06)	(0.06)
Mean Disc. Elig 1 Method	0.12	0.04	0.95	0.23	-0.04	0.01	0.11	0.13	0.10	-0.01	0.45	0.07	-0.10
Observations	6,358	6,358	6,358	6,358	6,358	6,300	6,357	6,358	5,792	6,358	6,325	6,358	6,356
Panel B: Round 2													
Disc. × Elig 2 Methods	0.06	0.04	-0.01	0.03	-0.03	0.02	0.05	0.09	0.03	-0.08	0.03	0.23***	0.02
	(0.08)	(0.08)	(0.04)	(0.07)	(0.08)	(0.08)	(0.08)	(0.07)	(0.08)	(0.11)	(0.04)	(0.07)	(0.08)
Disc. × Elig 3 Methods	0.01	0.06	-0.03	-0.02	0.03	-0.02	-0.03	0.04	0.01	-0.09	-0.00	0.00	-0.06
	(0.07)	(0.07)	(0.03)	(0.06)	(0.07)	(0.07)	(0.06)	(0.06)	(0.07)	(0.08)	(0.03)	(0.06)	(0.07)
Long Notice	-0.01	0.01	-0.01	0.01	0.01	-0.04	0.04	-0.02	-0.05	-0.01	0.00	-0.03	0.04
	(0.04)	(0.04)	(0.02)	(0.04)	(0.04)	(0.04)	(0.04)	(0.03)	(0.04)	(0.04)	(0.02)	(0.04)	(0.04)
Mean Disc. Elig 1 Method	0.09	-0.01	0.73	0.13	-0.03	0.08	0.05	0.13	0.12	0.11	0.51	0.12	-0.00
Observations	6,465	6,465	6,465	6,465	6,465	6,453	6,465	6,465	5,938	6,465	6,450	6,465	6,465

Notes: The interaction term is the product of the discontinued indicator and baseline eligibility. The reference group is discontinued households eligible under 1 method only. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A36: Impacts of Discontinuation by Eligibility on All Outcomes (2SLS)

	FCS	rCSI	LCSI Stress	Food Exp.	Life Sat Index	WHO-5	Worry Index	No. Em HHMs	Avg. Wk Hrs.	HH Income	HH Has Debt	SC Index	Trust Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Panel A: Round 1													
Disc. × Elig 2 Methods	0.39	0.34	0.02	0.85***	-0.24	-0.24	-0.08	0.14	-0.07	0.11	-0.11	0.80**	-
	(0.34)	(0.35)	(0.08)	(0.32)	(0.34)	(0.35)	(0.33)	(0.30)	(0.35)	(0.14)	(0.17)	(0.32)	0.65*
Disc. × Elig 3 Methods	0.42	0.26	-0.04	0.63*	0.42	0.13	-0.57	-0.10	-0.12	0.55*	-0.13	0.57*	-
	(0.35)	(0.37)	(0.08)	(0.33)	(0.35)	(0.36)	(0.36)	(0.32)	(0.37)	(0.29)	(0.18)	(0.34)	0.85**
Long Notice	-0.10	-0.02	0.00	-0.05	-0.08	-0.02	0.09	-0.09	-0.07	-0.06	0.04	-0.06	0.12
	(0.09)	(0.09)	(0.02)	(0.08)	(0.09)	(0.09)	(0.09)	(0.08)	(0.09)	(0.04)	(0.04)	(0.09)	(0.10)
Mean Disc. Elig 1 Method	-0.02	-0.01	0.95	-0.12	0.03	0.00	-0.03	-0.05	-0.05	-0.01	0.46	-0.02	0.06
Observations	6,358	6,358	6,358	6,358	6,358	6,300	6,357	6,358	5,792	6,358	6,325	6,358	6,356
Panel B: Round 2													
Disc. × Elig 2 Methods	0.34	0.42	-0.11	-0.14	0.18	0.16	0.24	0.56	0.17	-0.39	0.11	1.06***	0.11
	(0.39)	(0.40)	(0.18)	(0.37)	(0.41)	(0.40)	(0.38)	(0.35)	(0.42)	(0.52)	(0.19)	(0.35)	(0.40)
Disc. × Elig 3 Methods	0.13	0.42	-0.15	-0.15	0.27	-0.03	-0.14	0.27	0.02	-0.49	-0.02	0.07	-0.27
	(0.39)	(0.39)	(0.17)	(0.37)	(0.40)	(0.39)	(0.36)	(0.34)	(0.41)	(0.44)	(0.19)	(0.34)	(0.39)
Long Notice	-0.05	0.06	-0.03	0.03	0.09	-0.17	0.16	-0.10	-0.22	-0.02	-0.01	-0.15	0.20
	(0.18)	(0.18)	(0.08)	(0.17)	(0.18)	(0.18)	(0.17)	(0.15)	(0.18)	(0.19)	(0.09)	(0.18)	(0.18)
Mean Disc. Elig 1 Method	0.01	-0.03	0.73	-0.02	0.01	-0.04	0.02	-0.00	-0.00	-0.00	0.54	-0.02	-0.03
Observations	6,465	6,465	6,465	6,465	6,465	6,453	6,465	6,465	5,938	6,465	6,450	6,465	6,465

Notes: Interactions are formed by multiplying eligibility at baseline with each instrument for discontinued and long notice households. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A37: Impacts of Targeting on Downstream Outcomes by Eligibility (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	LCSI Stress (pp)	LCSI Crisis (pp)	LCSI Emer- gency (pp)	WHO-5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Reference method: PMT-Exp</i>								
Panel A: Eligible (continued under assigned method)								
PMT-FCS	-2.20*** (0.80)	-1.08* (0.63)	-4.83 (3.95)	-30.33*** (10.39)	0.37 (1.18)	2.12 (2.48)	-0.00 (1.40)	0.39 (0.97)
PMT-CARI	0.05 (0.81)	0.03 (0.65)	0.17 (4.06)	4.58 (10.85)	0.82 (1.16)	3.46 (2.48)	0.97 (1.43)	-0.61 (0.97)
Expert	-1.61* (0.83)	-1.64*** (0.62)	-19.41*** (3.90)	-40.22*** (10.80)	1.17 (1.14)	5.02** (2.46)	-1.32 (1.34)	-2.21** (0.93)
Mean PMT-Exp.	55.24	17.45	135.36	392.93	93.88	55.25	8.50	23.73
Observations	3,200	3,200	3,200	3,200	3,200	3,200	3,200	3,166
Panel B: Not eligible (discontinued under assigned method)								
PMT-FCS	2.54*** (0.84)	-0.01 (0.60)	18.89*** (4.32)	64.04*** (11.65)	2.22** (1.08)	4.09* (2.48)	-0.50 (1.38)	0.32 (0.95)
PMT-CARI	2.05** (0.84)	0.63 (0.60)	4.29 (4.31)	8.31 (11.34)	0.26 (1.17)	-0.83 (2.49)	-0.08 (1.39)	0.86 (0.97)
Expert	1.53* (0.83)	1.73*** (0.62)	28.70*** (4.22)	67.69*** (11.25)	1.92* (1.09)	-6.14** (2.50)	-1.29 (1.34)	2.02** (0.96)
Mean PMT-Exp.	53.69	15.78	137.93	391.48	94.06	56.89	8.34	22.26
Observations	3,158	3,158	3,158	3,158	3,158	3,158	3,158	3,134

Notes: ITT effects of each targeting arm relative to PMT-Expenditure, estimated from a single regression per outcome that interacts the targeting arm with an indicator for whether the household was *eligible* (continued) under its assigned method (Round 1). Panel A reports the arm effect among eligible households; Panel B among not-eligible (discontinued) households. Because eligibility is itself assigned by each method, this split blends targeting with the discontinuation margin. Food and total expenditure are winsorized at the 95th percentile; LCSI coefficients and means are in percentage points; WHO-5 ranges 0–100. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A38: Impacts of Targeting on Downstream Outcomes among Marginal Households (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	LCSI Stress (pp)	LCSI Crisis (pp)	LCSI Emer- gency (pp)	WHO- 5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Reference method: PMT-Exp</i>								
PMT-FCS	0.06 (0.62)	-0.69 (0.47)	7.18** (3.00)	17.03** (7.99)	1.12 (0.84)	3.78** (1.87)	-1.30 (1.05)	0.46 (0.72)
PMT-CARI	1.04* (0.62)	0.26 (0.48)	2.11 (3.04)	5.56 (8.02)	-0.08 (0.88)	1.52 (1.87)	0.00 (1.09)	0.49 (0.72)
Expert	-0.11 (0.62)	-0.22 (0.47)	4.54 (2.99)	14.75* (8.06)	1.05 (0.83)	-0.83 (1.87)	-1.84* (1.03)	0.18 (0.71)
Mean PMT-Exp.	54.60	16.82	135.61	391.22	94.33	56.50	8.97	22.63
Observations	5,566	5,566	5,566	5,566	5,566	5,566	5,566	5,517
p-val(PMT-FCS=PMT-CARI)	0.115	0.045	0.092	0.147	0.150	0.225	0.212	0.969
p-val(PMT-FCS=Expert)	0.783	0.316	0.372	0.774	0.924	0.013	0.586	0.694
p-val(PMT-CARI=Expert)	0.067	0.307	0.418	0.249	0.177	0.209	0.073	0.666

Notes: Sample restricted to *marginal* households—those selected by some but not all four targeting methods, excluding the always-eligible (selected by all four) and never-eligible (selected by none). PMT-Expenditure targeting serves as the reference group. All regressions include randomization-strata (eligibility-cell) fixed effects. Each column is its own dependent variable; the regressor of interest is the ITT treatment indicator (Round 1). Food and total expenditure are winsorized at the 95th percentile; LCSI coefficients and means are in percentage points; WHO-5 ranges 0–100. The p -values at the bottom test the null of equal treatment effects between targeting arms. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.